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# Optimization Principles and Trade-offs in Sensory Adaptation of Cells

Optimierungsprinzipien und Zielkonflikte in der  
sensorischen Adaption von Zellen

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# Abstract

Sensory adaptation is a process that is vital to all living organisms. In a mathematical framework, adaptation can be modelled as a response of a noisy biochemical feedback network. Recently, Lan et al. (2012 Nat. Phys. 8 422–8) proved that an adaptive system must dissipate, implying that the process is thermodynamically irreversible. They also established that the energy cost associated with adaptation is proportional to the adaptation accuracy. In such models for sensory adaptation, the physical limits of the underlying biochemical process are mathematically represented by the boundaries of the phase-space. However, the role of the phase-space limitations for this process is not well understood. An aim of this thesis is to shed some light on how the phase-space limitations impact the performance of such an adaptive system (e.g., a cell). It is shown that for a system in a fully adaptive state, the adaptation error is non-zero only at the boundaries. Also, numerical results show that phase-space boundaries reduce the steady state dissipation of the system and make the system lose its capacity to adapt accurately. Additionally, an empirical expression is conjectured, describing that in the vicinity of the domain boundaries the dissipation rate decays exponentially as the input level is raised. In addition, the adaptive performance for a system of two interacting subsystems is also studied. Two such cases of interacting systems are examined. In the first case, the interaction is a linear coupling between the outputs of the two subsystems. For the second case, the control elements of the two subsystems are coupled. Simulation results show that coupling the outputs deteriorates the adaptive performance of the system, however the energy cost associated with it also reduces. Whereas, it is observed that coupling the control elements reduces the energetic cost of adaptation without compromising on the adaptive performance.



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# Introduction

The surroundings of all living organisms constantly undergo changes. Diverse mechanical, electrical and chemical signaling pathways enable organisms to continuously respond to these changes [1]. The involved sensory systems can adjust their sensitivity to the external stimuli. This action of sensory systems is called adaptation [2]. The phenomenon of adaptation of a sensory system to changes in input signals is observed in a variety of biological systems [1–7]. An example of adaptation can be found in *E. coli*, which is a chemotactic bacterium that controls the direction of its flagellar rotation in response to its environment’s stimuli [3]. This operation allows them to sense and move along the gradients of a specific chemical. An adaptation to chemo-attractant concentrations is also observed in amoebae [4]. In fact, because bacterial chemotaxis is extremely well understood on the molecular and behavioral level, it is considered to be a benchmark problem for testing important notions in systems biology [8].

The motion of *E. coli* resembles a biased random-walk which is composed of runs, in which it moves along a certain direction and tumbles, in which the bacterium randomly changes its orientation. The cell reduces its tumbling frequency when it wants to move along the chemical gradient and increases it when it prefers to swim away [9]. The input to the sensory system of a cell is the concentration of a chemical (attractant or repellent) and the output is the tumbling rate. The molecules of this chemical are sensed by the receptors in a cell, which can be in two states i.e., active or inactive. When a receptor is active, it phosphorylates a receptor-regulator protein (CheY). Higher concentrations of this phosphorylated protein can alter the direction of flagellar rotation and hence increase the tumbling rate [10]. In its active state, the activity of a receptor is characterized by the concentration of the protein kinase that it is bound to. The attractant molecule reduces the receptor activity, which in turn reduces phosphorylation of CheY and thus, results in a reduced tumbling rate. However, each receptor also has methylation levels (number of methyl group molecules attached to the receptor) which can counter the effect of the attractant and increase the activity of the receptor [8]. At high methylation levels, the receptor is unable to bind to the attractant and therefore the activity of the receptor is increased. Moreover, the methylation of the receptors is catalyzed by a protein called CheR and reduced by protein CheB [11]. The action of CheB is directly related to the activity of the receptor. Consequently, when an attractant is added, the activity of receptor is reduced, making CheB less active. However, the action of CheR is unaffected. As a result,

the total methylation is increased and hence the receptor has a higher activity even though the attractant is present. Therefore, the tumbling rate increases [12]. In simple terms, at first, the activity of the receptor is reduced due to the attractant. The increased activity facilitates a steady increase in methylation level, which in-turn restores the activity [8].

Models of sensory adaptation have been studied extensively in the literature [1, 7, 13–15]. Sensory adaptation in living systems is frequently realized as responses generated by biochemical feedback loops [12, 15–17]. Such biochemical processes are usually influenced by thermal fluctuations if they involve a small number of molecules, making them inherently stochastic [1, 15]. Analysis of such noisy biochemical feedback loops can help deduce generic optimization and trade-off principles which are crucial in a variety of biological processes [1, 15, 18–21].

In Ref. [1], Sartori and Horowitz use a minimal feed-forward model to provide information-theoretic bounds to sensory adaptation. Studying this model enabled them to conclude that for any sensory adaptation system there is an energy cost associated with the measurement of any change in the surrounding, whereas no energy is required to erase the past information. G. Lan and Y. Tu in Ref. [15] present a noisy feedback control mechanism using which they establish that adaptation is inherently dissipative. They construct a minimal model for the biochemical network and prove that negative feedback results in broken detailed balance [22]. Additionally, they also present a cost-performance trade-off relation i.e., the “energy-speed-accuracy trade-off” (ESA), which relates the steady state dissipation for a fixed input with the error in adaptation and the adaptation speed, after a change in the input. They obtain this relation in the context of bacterial chemotaxis where their stochastic system involves a fast changing response variable to model changes in the output activity and a slow changing negative control element that represents methylation levels of the receptor. Here, the physical limits of the output and the control element (e.g., number of methylation sites) make up the phase-space of the model.

The work presented in Ref. [15] is fundamental, however, the analytical techniques used in calculations have their mathematical limitations. For instance, the validity of a few of the approximations used in Ref. [15] become doubtful near the phase-space boundaries. As a result, the soundness of the ESA at phase-space boundaries is ambiguous. Also, the role of effective temperatures in the adaptive performance of the model in Ref. [15] is not yet clearly understood. Moreover, it is known that the receptors in a cell reside in localized clusters. Therefore, a binding event in a receptor alters the state of other nearby receptors [23], which subsequently increases the overall sensitivity of the cell [24, 25]. Hence, it is natural to investigate the adaptive behavior of interacting sensory systems and ask whether any cost-performance trade-off dictates its adaptive performance. These are some of the questions that motivate the work in this thesis.

The research in this master’s thesis is inspired by the work in Ref. [15]. However, instead of using a finite difference approach to solve the Fokker-Planck equation, Brownian

dynamics simulations are used to treat a set of governing stochastic differential equations and a formalism from stochastic energetics [26] is employed to compute the steady-state dissipation in the system. This numerical framework enables us to study the adaptation accuracy and the energy cost of the system near the edges of the phase-space (i.e., when the change in chemo-attractant concentration is so large that even if all the methylation sites are occupied the activity of the receptor cannot be restored). After evaluating the numerical data, an empirical expression for the energy dissipation is conjectured which appears to hold true in a larger subset of the phase-space than the one suggested in Ref. [15]. Also, two examples of two interacting sensory systems are studied numerically and the effects of the interaction on the adaptive performance of the system are presented.





# Part I

## Theoretical Preliminaries



# Chapter 1

## Stochastic Dynamics of Particles

### 1.1 Introduction

Fluctuations are ubiquitous in nature, which makes all of physical studies essentially statistical. The fluctuations in physical systems are mainly due to two sources namely; small external perturbations and finite precision of measurements [27]. Physically, the fundamental source of external perturbations is due to thermal fluctuations. More intrinsically, thermal fluctuations are a consequence of the discrete nature of matter. If it weren't for that, all natural phenomena would be described using partial differential equations.

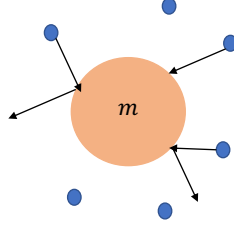
Various approaches have been developed to describe fluctuations in different fields of study; physics, chemistry, economics and more [28]. Out of the many ways to study fluctuations, in this chapter, I present a methodology which is a very appealing approach to non-equilibrium statistical mechanics and is also physically intuitive. This technique is a stochastic approach that helps describe the phase space dynamics of a single non-interacting particle suspended in a fluid at thermal equilibrium mesoscopically.

Moreover, this stochastic technique is mathematically equivalent to (but more descriptive than) a deterministic approach of time evolution of conditional probability distribution of the ensemble of instances via a convection-diffusion equation known as the Fokker-Planck equation [29].

### 1.2 Langevin Dynamics

#### 1.2.1 The Langevin Equation

Consider a macroscopic particle suspended in a fluid at thermal equilibrium. The motion of the particle in this setting is known as Brownian motion (named after botanist Robert Brown [30])



**Figure 1.1: Illustration of Brownian motion:** A macroscopic particle with mass  $m$ , in a fluid randomly colliding with the fluid molecules.

The macroscopic particle experiences two forces, the first force originates due to friction between the fluid and the particle. Being frictional in nature, this force is dependent on the velocity of the macroscopic particle and is intrinsically dissipative. The second force is a stochastic force and is a manifestation of granularity of the fluid in which the macroscopic particle is suspended (see Fig. 1.1). It encapsulates the force due to the collision between the fluid molecules and the particle [31]. If  $m \in \mathbb{R}_{\geq 0}^1$  is the mass of the particle, then the equation of motion of the macroscopic particle (in 1D) is given as,

$$m \frac{dv}{dt} = -\gamma v + \eta(t), \quad t > 0, v(t) \in \mathbb{R} \quad (1.1)$$

where  $\gamma \in \mathbb{R}_{\geq 0}$  is the friction coefficient between the fluid and the particle and  $\eta(t)$  is the random force. Moreover,

$$v = \frac{dx}{dt}, \quad x(t) \in \mathbb{R} \quad (1.2)$$

where  $x$  and  $v$  represent the position and velocity of the particle, respectively. Equation (1.1) was first proposed by French physicist Paul Langevin and is therefore, known as *Langevin equation*.

The random force  $\eta$  is modelled by means of a Gaussian white noise. Therefore, it has two key properties,

$$\langle \eta(t) \rangle = 0 \quad (1.3)$$

and

$$\langle \eta(t) \eta(t') \rangle = \Delta \delta(t - t'). \quad (1.4)$$

where  $\Delta \in \mathbb{R}_{\geq 0}$  characterizes the noise amplitude. The first property arises from the physical fact that the average force due to collisions with fluid molecules is zero [31] (provided that the fluid is homogeneous and isotropic). Whereas, the latter property is a mathematical idealization and arises because of the fact that the statistical correlation between forces at different time instances is short ranged. This statement signifies that the effect of the random force is only present for a short period. Here, one can make a mathematical approximation by taking a limit and assume the force to be instantaneous.

---

<sup>1</sup>  $\mathbb{R}_{\geq 0} = \{x \in \mathbb{R} | x \geq 0\}$

As per mathematical literature [32], (1.1) is a stochastic differential equation and can be re-written as,

$$m dv_t = -\gamma v_t + \sqrt{\Delta} dW_t. \quad (1.5)$$

Equation (1.5) informally represents a stochastic integral equation where  $W_t$  is a Wiener process. Wiener process  $W_t$ , is a continuous time stochastic process which is differentiable nowhere. Hence, its time derivative is mathematically ill-defined. However, since (1.5) is an integral equation, it is perfectly valid. The details of such a stochastic equation will be discussed in Chapter 2.

Langevin equation (1.1) can be treated as an inhomogeneous ordinary differential equation and can be solved by employing the method of variation of constants. Assuming  $v(0) = v_0$ , the solution to (1.1) is given by,

$$v(t) = v_0 e^{-\zeta t} + \frac{1}{m} \int_0^t d\tau e^{-\zeta(t-\tau)} \eta(\tau) \quad (1.6)$$

where  $\zeta = \frac{\gamma}{m}$  is the characteristic time of decay of deterministic motion [33]. Assuming the same initial condition for all trajectories, the statistical average of (1.6) is ,

$$\langle v(t) \rangle = v_0 e^{-\zeta t}. \quad (1.7)$$

This equation indicates that the contribution of initial condition becomes vanishingly small as time passes. This property is a consequence of the Markovian assumption (See Appendix A.1) which underlies the Langevin dynamics [27].

The correlation function of velocities is given as,

$$\langle v(t)v(t') \rangle = v_0^2 e^{-\zeta(t+t')} + \frac{1}{m^2} \int_0^t d\tau \int_0^{t'} d\tau' \langle \eta(\tau)\eta(\tau') \rangle e^{-\zeta(t-\tau)} e^{-\zeta(t'-\tau')}, \quad (1.8)$$

using (1.4) in (1.8) yields

$$\langle v(t)v(t') \rangle = v_0^2 e^{-\zeta(t+t')} + \frac{\Delta}{2\zeta m^2} (e^{-\zeta|t-t'|} - e^{-\zeta(t+t')}). \quad (1.9)$$

**Remark:** The integrals in (1.8) for an arbitrary (non-white) noise are separable only if the noise correlation falls faster than the characteristic time scale  $\zeta$  [34]. This characterization is called the underdamped limit. However, this property is automatically true for white noise, since the  $\delta$ -distribution falls instantaneously.

The second moment of the velocity can be obtained by setting  $t = t'$  in (1.9). Furthermore, for  $t \gg 1/\zeta$ , (1.9) becomes

$$\langle v^2(t) \rangle = \frac{\Delta}{2\zeta m^2}. \quad (1.10)$$

Since the system is in thermal equilibrium, the equipartition theorem holds i.e.,  $\frac{1}{2}m\langle v^2 \rangle = \frac{1}{2}k_B T$ , where  $T$  is the equilibrium temperature and  $k_B$  is the Boltzmann constant.

Employing the equipartition theorem in (1.10) leads to the following relation,

$$\Delta = 2\gamma k_B T. \quad (1.11)$$

This relation is known as the *Fluctuation Dissipation Theorem* (FDT). It relates the fluctuation amplitude  $\Delta$  with the friction coefficient  $\gamma$  via the thermal energy scale. This result allows one to connect the random force to the deterministic frictional force in (1.1).

### 1.2.2 Einstein-Stokes Relation

This problem was first studied by Albert Einstein in one of his seminal papers in 1905 [35]. There he also investigated the dynamics of particle position. To study at the dynamics of the position of the particle, consider (1.2),

$$x(t) = x_0 + \int_0^t v(t') dt',$$

where  $x_0 = x(0)$ . Using (1.6), one obtains the expression for the position of the particle

$$x(t) = x_0 + \frac{v_0}{\zeta}[1 - e^{\zeta t}] + \frac{1}{\zeta m} \int_0^t \eta(\tau)[1 - e^{-\zeta(\tau-t)}] d\tau. \quad (1.12)$$

Then the mean position is given by [31],

$$\langle x(t) \rangle = x_0 + \frac{v_0}{\zeta}(1 - e^{-\zeta t}), \quad (1.13)$$

and the variance is given by,

$$Var[x(t)] = \frac{\Delta}{\zeta^2 m^2} \left[ t - \frac{2}{\zeta}(1 - e^{-\zeta t}) + \frac{1}{2\zeta}(1 - e^{-2\zeta t}) \right].$$

In a long-time limit,

$$Var[x(t)] \approx 2Dt, \quad (1.14)$$

where  $D = \Delta/2(\zeta m)^2 = \frac{\Delta}{2\gamma^2}$  is called the diffusion coefficient.

From Stokes' law it is known that for a particle with radius  $r$  in a liquid with viscosity coefficient  $\mu$ , the friction coefficient is given by,

$$\gamma = 6\pi\mu r. \quad (1.15)$$

Using (1.15) in the expression for diffusion coefficient  $D$  results in the *Einstein-Stokes relation* [27],

$$D = \frac{k_B T}{6\pi\mu r}. \quad (1.16)$$

### 1.2.3 The Overdamped Limit

In addition to investigating the dynamics of a particle suspended in a fluid, it is possible to study a more general problem which investigates the time evolution of the trajectory of a Brownian particle subject to an arbitrary deterministic force  $\mathbf{F}$  (in three dimensions). To examine this problem, the deterministic force is assumed to be conservative i.e., there must exist a potential  $U(\mathbf{x}) \in \mathbb{R}$ , such that  $\mathbf{F} = -\nabla U(\mathbf{x})$ .

**Remark:** *In this case the position of the particle is a vector i.e  $\mathbf{x} \in \mathbb{R}^3$ .*

The equation of motion in this setting is given by,

$$m \frac{d^2 \mathbf{x}}{dt^2} = -\gamma \frac{d\mathbf{x}}{dt} - \nabla U(\mathbf{x}) + \boldsymbol{\eta}(t), \quad t > 0, \mathbf{x}(t) \in \mathbb{R}^3. \quad (1.17)$$

To make calculations easier in this case, the particle can be assumed to be very light, such that  $m/\gamma \rightarrow 0$ . This limit is known as the *overdamped limit*.

In the overdamped limit, the force due to inertia can be ignored, allowing one to obtain the following Langevin equation [28],

$$\gamma \frac{d\mathbf{x}}{dt} = -\nabla U(\mathbf{x}) + \boldsymbol{\eta}(t), \quad (1.18)$$

with

$$\langle \eta_i(t) \eta_j(t') \rangle = \Delta \delta_{ij} \delta(t - t').$$

where  $\delta_{ij}$  is the Kronecker delta and  $i, j = 1, 2, 3$ . Then defining  $\boldsymbol{\zeta}(t) \equiv \boldsymbol{\eta}(t)/\gamma$  and choosing  $\Delta = 2\gamma k_B T$  yields,

$$\langle \zeta_i(t) \zeta_j(t') \rangle = \frac{2k_B T}{\gamma} \delta_{ij} \delta(t - t') = 2D \delta_{ij} \delta(t - t'), \quad (1.19)$$

where the second equivalence is due to the definition of the diffusion coefficient and the FDT (1.11). Therefore, the resulting Langevin equation is [26],

$$\frac{d\mathbf{x}}{dt} = -\frac{1}{\gamma} \nabla U(\mathbf{x}) + \boldsymbol{\zeta}(t). \quad (1.20)$$

## 1.3 Fokker-Planck Equation

### 1.3.1 Continuous Markov Processes

Let  $Z(t)$  be a Markov process with sample space  $\Omega$  and let  $t_1 \leq t_2 \leq t_3$  be the time instances where measurements ( $Z(t_i) = z_i$ , for  $i \in \mathbb{N}$ ) are performed. Then  $Z(t)$  must satisfy the Chapman-Kolmogorov equation (see Appendix A.2) i.e.,

$$p(z_3, t_3 | z_1, t_1) = \int_{\Omega} dz_2 p(z_3, t_3 | z_2, t_2) \cdot p(z_2, t_2 | z_1, t_1). \quad (1.21)$$

If  $t_3 = t_2 + \Delta t$  where  $\Delta t \geq 0$  then,

$$\lim_{\Delta t \rightarrow 0} p(z_3, t_3 | z_2, t_2) = \delta(z_2 - z_3). \quad (1.22)$$

Additionally, if  $Z(t)$  is enforced to have a continuous state space then (1.22) is equivalent to,

$$\lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} \int_{|z_3 - z_2| > \epsilon} dz_3 p(z_3, t_2 + \Delta t | z_2, t_2) = 0, \quad \forall \epsilon > 0. \quad (1.23)$$

This is the Lindeberg condition, which states that a Markov process has a continuous sample path with probability 1, if (1.23) is true uniformly in  $z_2, t$  and  $\Delta t$  [36].

If the continuity assumption is violated, the trajectory is considered to be discontinuous. In that case, one can define “jumps” as,

$$w(z_3 | z_2, t_2) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} p(z_3, t_2 + \Delta t; z_2, t_2), \quad (1.24)$$

where  $\forall \epsilon > 0$  the limit is defined uniformly in  $z_2, z_3$  and  $t$  for  $|z_3 - z_2| \geq \epsilon$  [36].

### 1.3.2 Differential Chapman-Kolmogorov Equation

The Chapman-Kolmogorov equation (CKE) is an integral equation however, under certain assumptions it is possible to obtain a differential form of the CKE.

The CKE (1.21) is true irrelevant of the continuity of the sample paths. For obtaining the differential CKE, the differentiability constraints are divided into parts depending on its continuity properties;

- If the sample path is discontinuous, then the transition rates are given by (1.24).
- In the case when the sample path is continuous, the first two moments of the sample paths can be defined as (rename  $t_2 = t$ ,  $z_2 = z$  and  $z_3 = z'$ ),

$$A(z, t) + \mathcal{O}(\epsilon) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} \int_{|z - z'| < \epsilon} dz' (z' - z) p(z', t + \Delta t | z, t) \quad (1.25)$$

and

$$B(z, t) + \mathcal{O}(\epsilon) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} \int_{|z - z'| < \epsilon} dz' (z' - z)^2 p(z', t + \Delta t | z, t) \quad (1.26)$$

both are uniformly converging in  $z', \epsilon$  and  $t$  [36].

Here  $A, B$  are called the drift and diffusion coefficient, respectively.

For a higher dimensional system,  $\mathbf{Z}(t) \in \Omega \subseteq \mathbb{R}^n$  i.e.  $\mathbf{z}, \mathbf{z}' \in \Omega$ . Then,

$$A_i(\mathbf{z}, t) + \mathcal{O}(\epsilon) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} \int_{|\mathbf{z} - \mathbf{z}'| < \epsilon} d\mathbf{z}' (z'_i - z_i) p(\mathbf{z}', t + \Delta t | \mathbf{z}, t), \quad (1.27)$$



$$B_{ij}(\mathbf{z}, t) + \mathcal{O}(\epsilon) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} \int_{|\mathbf{z} - \mathbf{z}'| < \epsilon} d\mathbf{z}' (z'_i - z_i)(z'_j - z_j) p(\mathbf{z}', t + \Delta t | \mathbf{z}, t), \quad (1.28)$$

where  $\mathbf{A}(\mathbf{z}, t)$  is the drift vector and  $B_{ij}(\mathbf{z}, t)$  is the diffusion matrix.

**Remark:** In the limit of  $\epsilon \rightarrow 0$ , higher order moments of a trajectory do not contribute, as they vanish. See Ref. [36] for details.

The differential form of the CKE for  $t' > t$  subject to an initial condition at time  $t$ , can be derived using (1.27) and (1.28), and is given by (detailed derivation in Ref. [36])

$$\begin{aligned} \partial_t p(\mathbf{z}'', t' | \mathbf{z}, t) = & \\ & - \sum_i \frac{\partial}{\partial z''_i} [A_i(\mathbf{z}'', t) p(\mathbf{z}'', t' | \mathbf{z}, t)] + \frac{1}{2} \sum_{ij} \frac{\partial^2}{\partial z''_i \partial z''_j} [B_{ij}(\mathbf{z}'', t) p(\mathbf{z}'', t' | \mathbf{z}, t)] \\ & + \int d\mathbf{z}' [w(\mathbf{z}'' | \mathbf{z}', t) p(\mathbf{z}', t' | \mathbf{z}, t) - w(\mathbf{z}' | \mathbf{z}'', t) p(\mathbf{z}'', t' | \mathbf{z}, t)]. \end{aligned} \quad (1.29)$$

In literature, this equation is called the *forward differential Chapman-Kolmogorov equation* where the spatial derivatives in the right side are with respect to  $\mathbf{z}''$ , where  $\mathbf{z}''$  is a value the random variable  $\mathbf{Z}$  takes at a time instance between  $\mathbf{z}$  and  $\mathbf{z}'$ .

Similarly, it is possible to obtain a *backward differential Chapman-Kolmogorov equation* with  $t' > t$ ,

$$\begin{aligned} \partial_t p(\mathbf{z}'', t' | \mathbf{z}, t) = & \\ & - A_i(\mathbf{z}, t) \sum_i \frac{\partial p(\mathbf{z}'', t' | \mathbf{z}, t)}{\partial z_i} - \frac{1}{2} \sum_{ij} B_{ij}(\mathbf{z}, t) \frac{\partial^2 p(\mathbf{z}'', t' | \mathbf{z}, t)}{\partial z_i \partial z_j} \\ & + \int d\mathbf{z}' w(\mathbf{z}' | \mathbf{z}; t) [p(\mathbf{z}'', t' | \mathbf{z}, t) - p(\mathbf{z}'', t' | \mathbf{z}', t)], \end{aligned} \quad (1.30)$$

which is subject to a final condition at time  $t'$ .

### 1.3.3 Fokker-Planck Equation

For a Markov process  $\mathbf{Z}(t)$  with continuous sample paths, the forward differential Chapman-Kolmogorov equation is reduced from being an integro-differential equation to a partial differential equation (PDE), which is called the *Fokker-Planck equation* (FPE) [36],

$$\partial_t p(\mathbf{z}'', t' | \mathbf{z}, t) = - \sum_i \frac{\partial}{\partial z''_i} [A_i(\mathbf{z}'', t) p(\mathbf{z}'', t' | \mathbf{z}, t)] + \frac{1}{2} \sum_{ij} \frac{\partial^2}{\partial z''_i \partial z''_j} [B_{ij}(\mathbf{z}'', t) p(\mathbf{z}'', t' | \mathbf{z}, t)]. \quad (1.31)$$

The Fokker-Planck equation also represents the dynamics of a stochastic process (equivalent to a Langevin equation). The difference here is that, instead of looking at the time evolution of a trajectory, one focuses on the time evolution of the ensemble of trajectory by means of evolution of a probability density. A lot more on FPE's can be found in Refs. [37] and [36].

### 1.3.4 Smoluchowsky Equation

For the overdamped Langevin equation (1.20), the corresponding Fokker-Planck equation can be obtained by using (1.27) and (1.28),

$$A_i(\mathbf{x}) = \lim_{\Delta t \rightarrow 0} \frac{\langle x_i(t + \Delta t) - x_i(t) \rangle}{\Delta t} = -\frac{1}{\gamma} \partial_i U, \quad (1.32)$$

as  $\langle \boldsymbol{\eta}(t) \rangle = 0$  and<sup>2</sup>,

$$B_{ij}(\mathbf{x}) = \lim_{\Delta t \rightarrow 0} \frac{\langle (x_i(t + \Delta t) - x_i(t))(x_j(t + \Delta t) - x_j(t)) \rangle}{\Delta t} = 2D\delta_{ij}, \quad (1.33)$$

where  $i, j = 1, 2, 3$ .

Therefore, the Fokker-Planck equation corresponding to (1.20) is,

$$\partial_t p(\mathbf{x}, t) = \frac{1}{\gamma} \text{div}[p(\mathbf{x}, t) \nabla U(\mathbf{x})] + D \nabla^2 p(\mathbf{x}, t), \quad t > 0, \quad (1.34)$$

where  $p(\mathbf{x}, t) = p(\mathbf{x}, t | \mathbf{x}_0, t_0)$  with  $\mathbf{x}_0 = \mathbf{x}(t_0)$ ,  $t_0$  being the initial time point. This FPE for the overdamped system is also called the *Smoluchowsky Equation*.

The Fokker-Planck equation is a probability conservation equation and resembles the hydrodynamic continuity equation, this resemblance can be seen by defining the probability current,

$$\mathbf{J}(\mathbf{x}, t) = -\frac{1}{\gamma} \nabla U(\mathbf{x}) p(\mathbf{x}, t) - D \nabla p(\mathbf{x}, t). \quad (1.35)$$

Then FPE (1.34) becomes,

$$\partial_t p(\mathbf{x}, t) = -\text{div}(\mathbf{J}(\mathbf{x}, t)), \quad (1.36)$$

which is the continuity equation and dictates that the probability is conserved [37].

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2

$$\lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} \left\langle \left[ -\frac{1}{\gamma} \partial_i U \Delta t + dW_i(t) \right] \left[ -\frac{1}{\gamma} \partial_j U \Delta t + dW_j(t) \right] \right\rangle = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} 2\Delta t D \delta_{ij}$$

# Chapter 2

## Stochastic Differential Equations

### 2.1 Wiener Process

In the previous chapter, the Langevin equation (1.1) was introduced and it was briefly mentioned that (1.1) is a stochastic differential equation (SDE) which is to be understood as an integral equation and can be re-written as (1.5).

Similarly, the overdamped Langevin equation (1.19) can also be written as an integral equation (in one-dimension),

$$x(t) = x(0) - \frac{1}{\gamma} \int_0^t a(x) d\tau + \int_0^t \zeta(s) ds. \quad (2.1)$$

where  $a(x) \equiv \partial_x U(x)$ .

Equation (2.1) can also be written as,

$$dx_t = -\frac{1}{\gamma} a(x) dt + dW_t, \quad t > 0, \quad (2.2)$$

where  $W_t \equiv W(t) = \int_0^t \zeta(\tau) d\tau$  is called the *Wiener process* and  $dW_t = \zeta(t) dt = W_{t+dt} - W_t$  is called the *Wiener increment* [36].

**Remark:** Here, the diffusion constant  $D$  is chosen to be  $1/2$  such that  $\zeta(t)$  is a unit white noise.

**Definition (Wiener Process)** A one-dimensional Wiener Process (Mathematical Brownian Motion) is a real-valued stochastic process,  $W_t$ ,  $t \geq 0$ , if [38]

- $W_0 = 0$ .
- $W$  has independent increments; for  $0 < t_1 < \dots < t_n$ ,  $dW_0 = W_1 - W_0$ ,  $dW_1 = W_2 - W_1$ , ...,  $dW_{n-1} = W_n - W_{n-1}$  are all statistically independent.

- Increments of  $W$  are Gaussian; if  $u, t \geq 0$ ,

$$P(W_{t+u} - W_t) = \frac{1}{\sqrt{2\pi u}} \int_{\mathbb{R}} e^{-x^2/2u} dx, \quad x \in \mathbb{R}.$$

- $W$  has continuous paths;  $t \mapsto W_t$  is continuous  $\forall t \geq 0$ .

The third requirement in this definition has three implications,

- $\langle W_t \rangle = 0, \forall t \geq 0$ .
- $\langle W_{\Delta t} W_{\Delta t} \rangle = \Delta t$ , where  $W_{\Delta t} \equiv W_{t+\Delta t} - W_t$  and  $\Delta t \geq 0$ .
- The Fokker-Planck equation for a Wiener process,  $W_t$ , is the heat (diffusion) equation,

$$\partial_t p(w, t) + \partial_x^2 p(w, t) = 0, \quad w = W_t, \quad t > 0. \quad (2.3)$$

**Remark:** Analogously, the above-mentioned properties hold true for any higher-dimensional Wiener process,  $\mathbf{W}_t \in \mathbb{R}^n$ , with the additional property that all components of  $\mathbf{W}_t$  are statistically independent of each other [38].

## 2.2 Stochastic Differential Equations

This section is based on the discussion presented in Chapter 4 of the book by Gardiner (Ref. [36]). Here, I briefly summarize parts of it that are relevant in the framework of this thesis.

The SDE in (2.2) is of the form,

$$dX_t = a(X_t) dt + b dW_t, \quad t \geq 0, \quad (2.4)$$

where  $dt > 0$  and  $X_t$  is the stochastic process with sample space  $\Omega \subseteq \mathbb{R}$ , that solves this SDE. In (2.4),  $a : \Omega \rightarrow \mathbb{R}$  and  $b \in \mathbb{R}$  is a constant independent of  $X_t$  and  $t$ .

One can generalize (2.4) to the case where  $b = b(X_t, t)$ , then this equation is said to carry *multiplicative noise*, as the last term becomes a product of two stochastic variables. Then, (2.4) in integral form is [39],

$$X_t = X_0 + \int_0^t a(X_\tau, \tau) d\tau + \int_0^t b(X_s, s) dW_s. \quad (2.5)$$

The first integral in (2.5) equation is trivial, however, the integral on the right is a *stochastic integral* and requires some explaining.

### 2.2.1 Stochastic Integrals

Consider the following stochastic integral,

$$I = \int_0^t G(X_s, s) dW_s, \quad (2.6)$$

where  $G : \Omega \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$  is an arbitrary function of time  $t$  and real-valued stochastic process  $X_t$ . Then the integral (2.6) can be defined as the limit of partial sums  $S_n$ ,

$$I(\alpha) = \lim_{n \rightarrow \infty} S_n(\alpha) = \lim_{n \rightarrow \infty} \sum_{i=1}^n [G(\alpha X_{\tau_i} + (1 - \alpha)X_{\tau_{i-1}}, \tau_{i-1})][W(t_i) - W(t_{i-1})], \quad (2.7)$$

where  $\alpha \in [0, 1]$  and the domain  $[0, t]$  is divided into a partition

$$0 \leq t_1 \leq t_2 \leq \dots \leq t_{n-1} \leq t, \quad \text{and} \quad t_{i-1} \leq \tau_i \leq t_i,$$

for all  $i \in \{1, 2, \dots, n\}$ .

- If  $\alpha = 0$  then,

$$I = \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \sum_{i=1}^n G(X_{t_{i-1}}, t_{i-1})[W(t_i) - W(t_{i-1})]. \quad (2.8)$$

This definition of stochastic integral is termed as *Itô stochastic integral* or *Itô interpretation* of stochastic integration. In this interpretation,  $G_{t-1} \equiv G(X_{t_{i-1}}, t_{i-1})$  is independent of  $\Delta W_i = W(t_i) - W(t_{i-1})$ , since  $X_{t_{i-1}}$  does not depend on  $\Delta W_{i-1}$ .

- If  $\alpha = 1/2$  then,

$$I = \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \sum_{i=1}^n G\left(\frac{X_{t_i} + X_{t_{i-1}}}{2}, t_{i-1}\right)[W(t_i) - W(t_{i-1})] \quad (2.9)$$

Note that in this definition, the average is only over  $X_t$  and not for  $t$ . Consequently, the two terms in the product are not independent of each other. Equation (2.9) is called the *Stratonovich integral* or *Stratonovich interpretation* of stochastic integration [36].

**Remark:** *The existence of above limits is subject to convergence with respect to the mean square limit.*

Both the interpretations are crucial in the theory of stochastic processes. Itô's interpretation is more prevalent in mathematics and finance, whereas Stratonovich's approach is the preferred choice in physics. The reason for this choice is that using Itô's interpretation introduces a slight change to the already known rules of calculus (discussed later).

Itô and Stratonovich integrals are crucial when working with stochastic differential equations with multiplicative noise. Moreover, they are also useful when one has to interpret product of two stochastic processes.

### 2.2.2 Itô's Formula

Let  $X_t$  be a stochastic process which obey's *Itô stochastic differential equation* i.e.,

$$dX_t = a(X_t, t) dt + b(X_t, t) dW_t. \quad (2.10)$$

The solution of Itô SDE is only guaranteed to exist (and is also unique) if both  $a$  and  $b$  are Lipschitz continuous and satisfy the linear growth condition (Appendix B) in their domain.

Consider an arbitrary function  $f : \Omega \subseteq \mathbb{R} \rightarrow \mathbb{R}$  of stochastic process  $X(t) \equiv X_t$  with sample space  $\Omega$ , where  $X(t) = x$  at time  $t$  satisfies (2.10). Then the SDE obeyed by  $f$  is,

$$\begin{aligned} df[X(t)] &= f[X(t) + dX(t)] - f[X(t)] \\ &= f'[X(t)]\{a[X(t), t]dt + b[X(t), t]dW_t\} \\ &\quad + \frac{1}{2}f''[X(t)]\{a[X(t), t]dt + b[X(t), t]dW_t\}^2 \dots, \end{aligned} \quad (2.11)$$

where, first,  $f[X(t) + dX(t)]$  was Taylor expanded and then, (2.10) was used. Using  $[dW_t]^2 = dt$  and ignoring the terms with order higher than  $\mathcal{O}(dt)$  yields,

$$df[X(t)] = a[X(t), t] \frac{\partial f}{\partial x} dt + \frac{1}{2}b^2[X(t), t] \frac{\partial^2 f}{\partial x^2} dt + b[X(t), t] \frac{\partial f}{\partial x} dW_t. \quad (2.12)$$

This is known as *Itô's formula*, which shows that under Itô's interpretation, the change of variables is not the same as in regular calculus (unless  $f[X(t)]$  is linear in  $X(t)$ ).

### 2.2.3 Connection with Fokker-Planck Equation

Additionally,  $f$  must satisfy,

$$\frac{d}{dt} \langle f[X(t)] \rangle = \left\langle \frac{df[X(t)]}{dt} \right\rangle. \quad (2.13)$$

The right side of (2.13) is the statistical average of time derivative of  $f$ , which can be evaluated using (2.12) as,

$$\left\langle \frac{df[X(t)]}{dt} \right\rangle = \left\langle a[X(t), t] \frac{\partial f}{\partial x} \right\rangle + \left\langle \frac{1}{2}b^2[X(t), t] \frac{\partial^2 f}{\partial x^2} \right\rangle + \left\langle b[X(t), t] \frac{\partial f}{\partial x} dW_t \right\rangle \frac{1}{dt}, \quad (2.14)$$

under Itô's interpretation,  $\langle b[X(t), t] \partial_x f dW_t \rangle = \langle b[X(t), t] \partial_x f \rangle \langle dW_t \rangle = 0$ .

$X(t)$  has a conditional probability density  $p(x, t|x_0, t_0)$  where  $X(t_0) = x_0$ , hence the left side of (2.13) is,

$$\frac{d}{dt} \langle f[X(t)] \rangle = \int_{\Omega} dx f(x) \partial_t p(x, t|x_0, t_0). \quad (2.15)$$

Similarly at time  $t$ , (2.14) can be written as,

$$\left\langle \frac{df[x(t)]}{dt} \right\rangle = \int_{\Omega} dx [a(x, t) \partial_x f + \frac{1}{2} b(x, t)^2 \partial_x^2 f] p(x, t | x_0, t_0). \quad (2.16)$$

Since (2.15) and (2.16) are equal,

$$\begin{aligned} \int_{\Omega} dx f(x) \partial_t p(x, t | x_0, t_0) = \\ \int_{\Omega} dx f(x) \left( -\partial_x [a(x, t) p(x, t | x_0, t_0)] + \frac{1}{2} \partial_x^2 [p(x, t | x_0, t_0) b(x, t)^2] \right), \end{aligned} \quad (2.17)$$

where, partial integration is used for the right-side. Since  $f$  is arbitrary and the integrals in (2.17) are over the same domain, it yields

$$\partial_t p(x, t | x_0, t_0) = -\partial_x [a(x, t) p(x, t | x_0, t_0)] + \frac{1}{2} \partial_x^2 [p(x, t | x_0, t_0) b(x, t)^2], \quad (2.18)$$

which is the Fokker-Planck equation.

### 2.2.4 Itô & Stratonovich Dilemma

The stochastic process  $X_t$  governed by Itô SDE (2.10) also obeys

$$dX_t = \alpha(X_t, t) dt + \beta(X_t, t) \circ dW_t, \quad (2.19)$$

which is the *Stratonovich stochastic differential equation*. Here  $\circ$  indicates that the stochastic integral is to be taken in the Stratonovich sense. Rewriting (2.19) as an integral equation,

$$X(t) = X(t_0) + \int_{t_0}^t \alpha(X_{\tau}, \tau) d\tau + \int_{t_0}^t \beta(X_s, s) \circ dW_s. \quad (2.20)$$

Since  $X_t$  in (2.10) and (2.19) is the same stochastic process, it is possible to relate the two SDE's [36].

To obtain this relation, first consider the definition of the Stratonovich integral

$$\int_{t_0}^t \beta(X_s, s) \circ dW_s \simeq \sum_i \beta \left[ \frac{X_i + X_{i-1}}{2}, t_{i-1} \right] [W(t_i) - W(t_{i-1})], \quad (2.21)$$

where  $X_i \equiv X(t_i)$  and  $X_i = X_{i-1} + dX_{i-1}$ . Since  $X_t$  obeys Itô's SDE for all  $t$ ,

$$dX_i = a(X_{i-1}, t_{i-1})[t_i - t_{i-1}] + b(X_{i-1}, t_{i-1})[W_i - W_{i-1}]. \quad (2.22)$$

The following term in (2.21) can be re-written as,

$$\begin{aligned}
\beta \left[ \frac{X_i + X_{i-1}}{2}, t_{i-1} \right] &= \beta[X_{i-1} + \frac{1}{2}dX_{i-1}, t_{i-1}] \\
&= \beta[X_{i-1}, t_{i-1}] + d\beta \\
&= \beta[X_{i-1}, t_{i-1}] + a[X_{i-1}, t_{i-1}] \partial_x \beta[X_{i-1}, t_{i-1}] \frac{dt_{i-1}}{2} \\
&\quad + \frac{1}{2} b^2[X_{i-1}, t_{i-1}] \partial_x^2 \beta[X_{i-1}, t_{i-1}] \frac{dt_{i-1}}{2} \\
&\quad + \frac{1}{2} b[X_{i-1}, t_{i-1}] \partial_x \beta[X_{i-1}, t_{i-1}] dW_{i-1},
\end{aligned} \tag{2.23}$$

where for the last equivalence, Itô's formula was used with  $dt_{i-1} \equiv t_i - t_{i-1}$  and  $dW_{i-1} \equiv W_i - W_{i-1}$ . Substituting (2.23) in (2.21), and ignoring terms with order higher than  $\mathcal{O}(dt)$ ,

$$\begin{aligned}
\int_{t_0}^t \beta(X_s, s) \circ dW_s &\simeq \sum_i \beta[X_{i-1}, t_{i-1}] dW_{i-1} + \frac{1}{2} \sum_i b[X_{i-1}, t_{i-1}] \partial_x \beta[X_{i-1}, t_{i-1}] dt_{i-1} \\
&\quad + \mathcal{O}(dt dW) + \mathcal{O}(dt^2).
\end{aligned} \tag{2.24}$$

Therefore,

$$\int_{t_0}^t \beta(X_s, s) \circ dW_s = \int_{t_0}^t \beta[X_s, s] dW_s + \frac{1}{2} \int_{t_0}^t b[X_\tau, \tau] \partial_x \beta[X_\tau, \tau] d\tau, \tag{2.25}$$

where the first term on the right is Itô's stochastic integral. Equation (2.25) connects Stratonovich's stochastic integral with Itô's stochastic integral.

Equation (2.25) allows one to directly relate Itô SDE (2.10) with Stratonovich's SDE (2.19) by making the following choice for the functions  $\alpha(X_t, t)$  and  $\beta(X_t, t)$ ,

$$\alpha(X_t, t) = a(X_t, t) - \frac{1}{2} b(X_t, t) \partial_x b(X_t, t), \tag{2.26a}$$

$$\beta(X_t, t) = b(X_t, t). \tag{2.26b}$$

Then, Itô SDE (2.10) is the same as a Stratonovich SDE of the form,

$$dX_t = [a(X_t, t) - \frac{1}{2} b(X_t, t) \partial_x b(X_t, t)] dt + b(X_t, t) \circ dW_t. \tag{2.27}$$

The additional term in (2.27), gives rise to an additional drift term in the corresponding Fokker-Planck equation. This additional drift term is called the *noise-induced drift* or *spurious drift*. See Ref. [40] for more on noise-induced drift.



### 2.2.5 Multivariate Systems

For an  $n$ -dimensional real-valued stochastic process  $\mathbf{X}(t)$  obeying the following Itô SDE,

$$d\mathbf{X} = \mathbf{A}(\mathbf{X}, t) dt + \mathbf{B}(\mathbf{X}, t) d\mathbf{W}(t), \quad (2.28)$$

Itô's formula is given by,

$$\begin{aligned} df(\mathbf{X}(t)) = & \sum_i A_i(\mathbf{X}, t) \frac{\partial f(\mathbf{X})}{\partial x_i} dt + \frac{1}{2} \sum_{i,j} [\mathbf{B}(\mathbf{X}, t) \mathbf{B}^T(\mathbf{X}, t)]_{ij} \frac{\partial^2 f(\mathbf{X})}{\partial x_i \partial x_j} dt \\ & + \sum_{i,j} B_{ij}(\mathbf{X}, t) \frac{\partial f(\mathbf{X})}{\partial x_i} dW_j(t) \end{aligned} \quad (2.29)$$

where  $\mathbf{W}(t)$  is an  $n$ -dimensional Wiener process,  $\mathbf{A} \in \mathbb{R}^n$  is the drift vector and  $\mathbf{B} \in \mathbb{R}^{n \times n}$  is the diffusion matrix. The Fokker-Planck equation corresponding to (2.27) is,

$$\partial_t p(\mathbf{x}, t) = - \sum_i \partial_i [A_i(\mathbf{x}, t) p(\mathbf{x}, t)] + \frac{1}{2} \sum_{i,j} \partial_i \partial_j \{ [\mathbf{B}(\mathbf{x}, t) \mathbf{B}^T(\mathbf{x}, t)]_{ij} p(\mathbf{x}, t) \}, \quad (2.30)$$

where  $p(\mathbf{x}, t) \equiv p(\mathbf{x}, t | \mathbf{x}_0, t_0)$ .

Lastly, the Stratonovich SDE that  $\mathbf{X}(t)$  obeys and is equivalent to 2.28 is,

$$d\mathbf{X} = \mathbf{A}^s(\mathbf{X}, t) dt + \mathbf{B}^s(\mathbf{X}, t) \circ d\mathbf{W}(t), \quad (2.31)$$

where,

$$A_i^s = A_i - \frac{1}{2} \sum_{j,k} B_{kj} \partial_k B_{ij}, \quad (2.32a)$$

$$B_{ij}^s = B_{ij}. \quad (2.32b)$$



# Chapter 3

## Stochastic Thermodynamics

### 3.1 Introduction

Classical dynamical systems can be described on at least three distinct levels. The first one, which is completely deterministic, is by means of microscopic Hamiltonian dynamics [26]. In this description, if the phase space coordinates are known at an initial time, then the phase space coordinates at any other time point are uniquely determined [41]. Coarse graining certain details on the microscopic level yields a mesoscopic view of the dynamical system. In this view, the description is a semi-phenomenological one. The dynamics in this setting are no longer deterministic and can be characterized via stochastic dynamics, which was developed by likes of Einstein, Langevin and Itô. And lastly, a macroscopic description to classical systems is given by the framework of equilibrium thermodynamics. This formalism does not look at the dynamics of a system, but one studies the energetics under the influence of change in external parameters. In this framework, the system is particularly studied in a long-time limit with the assumption that the system has completely relaxed to an equilibrium state [26].

Various methods have been developed to relate these three levels of descriptions. For instance, Hamilton dynamics can be related to a generalized Langevin approach via the Zwanzig-Mori projection operator formalism [42]. Additionally, the framework of equilibrium statistical mechanics forms a bridge between the microscopic description and macroscopic thermodynamical approach [28].

Similarly, stochastic thermodynamics provides a conceptual framework that allows one to determine thermodynamic quantities like heat exchange and entropy production at the trajectory level by employing stochastic dynamics and thereby linking the mesoscopic and the macroscopic description of a system [43].

## 3.2 Recap of Classical Thermodynamics

Before looking at stochastic thermodynamics, in this section I present a brief outlook of some important aspects of Classical (Reversible) Thermodynamics for systems in equilibrium. Here, the focus is on certain physical aspects of equilibrium thermodynamics and not on the important equations. Therefore, in this section, I first introduce some relevant terminologies, which is followed by a brief discussion of the zeroth, first and second law of thermodynamics. A lot more on this subject is discussed in Refs. [28, 44–46].

### 3.2.1 Important Terminologies

In thermodynamics, one way to categorize the system is by means of the boundary it shares with its surroundings [47],

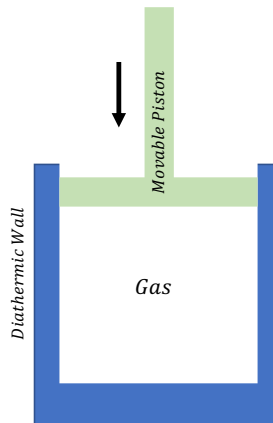
- **Isolated systems:** System cannot exchange mass and energy with its surroundings.
- **Closed systems:** System can exchange energy with the surrounding but the exchange of mass is forbidden.
- **Open systems:** Both energy and mass exchange with the surrounding is possible.

The boundaries of a thermodynamic system can be considered either *diathermic* or *adiabatic*. In case of a diathermic wall, the exchange of heat between the system and its surrounding is possible. In the latter case, no heat exchange is allowed.

The surroundings of a system can be considered to have two contributions: the *external agent* and the *thermal environment*. The latter is defined as the physical environment that surrounds a system and shares its boundary with it. The thermal environment is assumed to contain a large number of particles (in comparison to the system) such that it can maintain its own temperature irrespective of the amount of energy exchange between the system and thermal environment. The external agent is an external system that can control at least one macroscopic parameter (degree of freedom) of the system [47].

An illustrative example in classical thermodynamics is that of a gas in a cylinder with diathermic walls and a movable piston (see Fig. 3.1). The cylinder makes up the boundary of the closed system, where the system is the gas, the thermal environment is the physical surrounding outside the cylinder and the external agent is an experimenter that controls pressure applied on the gas [47].

The state of a thermodynamic system is said to be in *equilibrium* if it is in a state where all of its macroscopic quantities do not change [28]. In thermodynamics, the evolution of a system is controlled by an external agent, which can be modeled by the evolution of a *control parameter*  $a(t)$ . A thermodynamic process can either be considered *reversible* or *irreversible* depending on the rate of change of control parameter  $a(t)$ ;



**Figure 3.1: Example for thermodynamic system and thermal environment:** A container made with diathermic walls containing gas with a movable piston is an example of a closed system with the physical surrounding of the piston as the thermal environment.

- **Reversible processes:** If the control parameter is changed quasi-statically, such that the system stays infinitesimally close to the equilibrium, then this process is considered to be reversible.
- **Irreversible processes:** If the control parameter is changed at a finite (non-zero) rate, the system can drift away from the equilibrium. The process is then considered to be irreversible [47].

Physically, all processes in practice are irreversible as it is not possible to change a parameter at an infinitesimal rate.

### 3.2.2 Laws of Thermodynamics

*Zeroth Law:* If a thermodynamic system is left isolated from its surrounding for a long enough time, then it will reach an equilibrium state and will stay in that state unless perturbed by its surroundings [28].

A thermodynamic process, in general, alters the total energy of a system. The total energy of a system is given by its internal energy,  $U$ , which is the sum of the microscopic kinetic and potential energy of the particles in the system.

The total change in the internal energy of a system is due to the energy exchange with the thermal bath and as a response to the change in the control parameter induced by the action of an external agent. The energy transferred to the system from the thermal environment is termed as *heat*,  $Q$ . Whereas, the internal energy change caused by the change in control parameter is called *work*,  $W$ , i.e.,

$$\Delta U = Q + W. \quad (3.1)$$

This is the content of the *First Law of Thermodynamics* [47].

For a closed system, when the control parameter is changed infinitesimally, the change in internal energy is also infinitesimal, which is denoted by  $dU$ . Then (3.1) yields,

$$dU = \delta Q + \delta W, \quad (3.2)$$

where,  $dU$  is an exact differential, whereas  $\delta Q$  and  $\delta W$  are inexact differentials. That is, if a process moves from an equilibrium state A to another equilibrium state B reversibly, then the total change in heat is given by,

$$\Delta Q_{rev} = \int_A^B \delta Q = Q_B - Q_A. \quad (3.3)$$

However, if the process happens irreversibly, then

$$\Delta Q_{irr} = \int_A^B \delta Q \neq Q_B - Q_A. \quad (3.4)$$

In the irreversible case, the change in heat becomes dependent on the path between the two end points A and B. However, when the process is reversible the total change in heat depends only on the end points A and B.

This functionality does not hold for internal energy, since an infinitesimal change in internal energy is represented by an exact differential,

$$\Delta U = \int_A^B dU = U_B - U_A. \quad (3.5)$$

This is true irrespective of the reversibility of the process. In (3.2), even though both quantities on the right are inexact differentials, their sum is always an exact differential [45].

Furthermore, for a reversible process at fixed temperature  $T$

$$\frac{\Delta Q_{rev}}{T} = \int_A^B \frac{\delta Q}{T} = S_B - S_A \equiv \Delta S, \quad (3.6)$$

where  $S_A$  and  $S_B$  is the *entropy* of the system in state A and B, respectively.

If the above-mentioned process is done irreversibly, then (3.6) does not hold anymore, instead the following inequality holds true [44],

$$\frac{\Delta Q_{irr}}{T} \leq \frac{\Delta Q_{rev}}{T} = \Delta S, \quad (3.7)$$

where the equality only holds if the process is reversible. This relation is known as the *Second Law of Thermodynamics* [47].

Equation (3.7) implies the existence of a non-negative quantity  $S_{\text{prod}}$  such that,

$$S_{\text{prod}} + \frac{\Delta Q_{\text{irr}}}{T} = \Delta S. \quad (3.8)$$

Here,  $S_{\text{prod}}$  is known as the *entropy production* in the system. It is the increase in entropy of the system due to the change in its internal degrees of freedom.

**Remark:** *The above statement is only true for an ensemble average. In general, the entropy production of an individual trajectory can be negative. However, the average entropy production of a collection of trajectories must always be non-negative.*

Equation (3.8) also entails that

$$\Delta S - \frac{\Delta Q_{\text{irr}}}{T} \geq 0 \iff -\Delta Q_{\text{irr}} \geq -T\Delta S.$$

Then (3.1) for an irreversible process yields,

$$W \geq \Delta U - T\Delta S. \quad (3.9)$$

Moreover, in macroscopic thermodynamics, the change in *Helmholtz Free Energy* ( $F$ ) is defined as,

$$\Delta F = \Delta U - T\Delta S. \quad (3.10)$$

This definition implies that,

$$W \geq \Delta F, \quad (3.11)$$

where the equality sign holds only if the process is reversible. Equation (3.11) indicates that  $\Delta F$  is the minimum work that the external agent needs to do to drive the system from one equilibrium state ( $A$ ) to another ( $B$ ) [47].

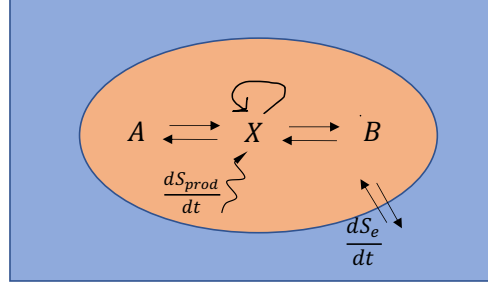
This relation also implies that for an irreversible process, there exists some excess work, (called *dissipation*)  $W_{\text{diss}}$ , that is inherent to the irreversible nature of the process [48],

$$\begin{aligned} W_{\text{diss}} &= W - \Delta F \\ &= \Delta Q_{\text{irr}} - T\Delta S \\ &= TS_{\text{prod}} \geq 0, \end{aligned} \quad (3.12)$$

where the dissipation vanishes only for a reversible process.

### Sign Convention

The sign convention in this thesis is in accordance with Ref. [28] – anything that the system receives is assigned a positive sign, whereas anything that the surrounding gains (system loses) is assigned a negative sign.



**Figure 3.2: Entropy production and exchange rate:** Fluxes A, B drive the state of system X out-of-equilibrium, which induces the rate of entropy production  $\frac{dS_{\text{prod}}}{dt}$  and entropy exchange rate  $\frac{dS_e}{dt}$ .

### 3.3 Entropy Production in Stochastic Processes

Entropy production is intrinsic to systems in non-equilibrium and is therefore considered a hallmark for irreversibility [49]. It is a quantity that can be used as a measure to determine if a stochastic process is in equilibrium or not. This quantity is especially useful when there are no visible gradients in a stochastic process. Moreover, in cases when an effective temperature is known, the knowledge of entropy production can help complete the energetic characterization of a process [49].

For stochastic processes, the time derivative of the total entropy of a system is determined by two terms. The first is the rate of change of entropy production, which measures how far the system is away from equilibrium. The second contribution characterizes the entropy exchange due to interaction between the system and its surroundings (see Fig. 3.2).

For a discrete stochastic process governed by a master equation, the rate of change of entropy is given by [49],

$$\frac{dS(t)}{dt} = \underbrace{\frac{1}{2} \sum_{i,j=1}^n J_{ji} \ln \frac{\omega_{ji} p_i}{\omega_{ij} p_j}}_{\frac{dS_{\text{prod}}}{dt} \geq 0} - \underbrace{\frac{1}{2} \sum_{i,j=1}^n J_{ji} \ln \frac{\omega_{ji}}{\omega_{ij}}}_{\frac{dS_e}{dt}} \quad (3.13)$$

where  $J_{ji}$  is the probability current of transition from state  $i$  to state  $j$ ,  $\omega_{ji}$  is the corresponding rate of the process and  $p_i$  is the steady-state probability of being in state  $i$ , where  $i, j \in \mathbb{Z}_{\geq 0}$ <sup>1</sup> [50].

Since the focus in this thesis is on continuous time stochastic processes, one can derive an analogous equation for the rate of entropy. This derivation has also been discussed in

---

<sup>1</sup>  $\mathbb{Z}_{\geq 0} = \{x \in \mathbb{Z} | x \geq 0\}$ .



some detail in Refs. [51] and [52].

To determine the entropy production rate for continuous time stochastic processes, consider the Fokker-Planck equation for the overdamped Langevin system (1.19),

$$\begin{aligned}\partial_t p(\mathbf{x}, t) &= \frac{1}{\gamma} \operatorname{div}[(\nabla U(\mathbf{x}))p(\mathbf{x}, t)] + D \nabla^2 p(\mathbf{x}, t), \quad t > 0 \\ &= -\operatorname{div}(\mathbf{J}(\mathbf{x}, t))\end{aligned}\tag{1.33}$$

where the probability current is,

$$J_i(\mathbf{x}, t) = -\frac{1}{\gamma} \partial_i U(\mathbf{x}) p(\mathbf{x}, t) - D \partial_i p(\mathbf{x}, t).\tag{1.34}$$

Then using the von-Neumann entropy,

$$S(t) = - \int_{\mathbb{R}^3} p(\mathbf{x}, t) \ln p(\mathbf{x}, t) \, d\mathbf{x},\tag{3.14}$$

and differentiating it with respect to  $t$  yields,

$$\frac{dS(t)}{dt} = - \int_{\mathbb{R}^3} \partial_t p(\mathbf{x}, t) \ln p(\mathbf{x}, t) \, d\mathbf{x} - \partial_t \int_{\mathbb{R}^3} p(\mathbf{x}, t) \, d\mathbf{x}.$$

Using that  $\int_{\mathbb{R}^3} p(\mathbf{x}, t) \, d\mathbf{x} = 1$  must be true at all times eliminates the second term. Then using (1.33) yields,

$$\frac{dS(t)}{dt} = \int_{\mathbb{R}^3} \operatorname{div}(\mathbf{J}(\mathbf{x}, t)) \ln p(\mathbf{x}, t) \, d\mathbf{x}.\tag{3.15}$$

Then employing integration by parts, where the boundary terms vanish as  $p(\mathbf{x}, t)|_{\mathbf{x}=\pm\infty} = 0$ ,

$$\frac{dS(t)}{dt} = - \int_{\mathbb{R}^3} \frac{1}{p(\mathbf{x}, t)} \mathbf{J}(\mathbf{x}, t) \cdot \nabla p(\mathbf{x}, t) \, d\mathbf{x}.\tag{3.16}$$

Moreover, equation (1.34) implies  $\partial_i p(\mathbf{x}, t) = -\frac{1}{\gamma D} \partial_i U(\mathbf{x}) p(\mathbf{x}, t) - \frac{1}{D} J_i(\mathbf{x}, t)$  and thus,

$$\frac{dS(t)}{dt} = \sum_{i \in \{x, y, z\}} \left[ \int_{\mathbb{R}^3} \frac{J_i^2(\mathbf{x}, t)}{D p(\mathbf{x}, t)} \, d\mathbf{x} + \int_{\mathbb{R}^3} \frac{\partial_i U(\mathbf{x})}{D \gamma} J_i(\mathbf{x}, t) \, d\mathbf{x} \right],\tag{3.17}$$

where the first term corresponds to the rate of entropy production and the second corresponds to the entropy exchange rate. A justification of this claim can be seen as follows.

The probability current can also be written as,

$$J_i(\mathbf{x}, t) = p(\mathbf{x}, t) v_i(\mathbf{x}, t),\tag{3.18}$$

where  $\mathbf{v}(\mathbf{x}, t)$  is the mean local velocity [51]. Then,

$$\int_{\mathbb{R}^3} \frac{J_i^2(\mathbf{x}, t)}{Dp(\mathbf{x}, t)} d\mathbf{x} = \langle v_i^2(\mathbf{x}, t) \rangle / D \geq 0, \quad (3.19)$$

hence,

$$\frac{dS_{\text{prod}}}{dt} = \sum_{i \in \{x, y, z\}} \langle v_i^2(\mathbf{x}, t) \rangle / D \geq 0. \quad (3.20)$$

Additionally, in a steady-state  $\dot{S}(t) = 0$ , hence

$$\frac{dS_{\text{prod}}}{dt} = \sum_{i \in \{x, y, z\}} \int_{\mathbb{R}^3} \frac{J_i^2(\mathbf{x}, t)}{Dp(\mathbf{x}, t)} d\mathbf{x} = \sum_{i \in \{x, y, z\}} \int_{\mathbb{R}^3} \frac{-\partial_i U(\mathbf{x})}{D\gamma} J_i(\mathbf{x}, t) d\mathbf{x} = -\frac{dS_e}{dt}. \quad (3.21)$$

In this case, if the (effective) temperature  $T$  of the system is known, then the steady-state dissipation rate is determined as [48]

$$-\dot{Q} = T \frac{dS_e}{dt} = T \frac{dS_{\text{prod}}}{dt} = \dot{W}_{\text{diss}}. \quad (3.22)$$

## 3.4 Stochastic Energetics

The discussion in the previous section presents an approach to compute the energetics of a continuous time stochastic process, provided that the governing probability distribution is known. However, in many cases one does not have access to the probability distribution. Furthermore, the above formalism cannot be used to determine the energetics of an individual realization of the stochastic system. In such cases the Fokker-Planck equation is not directly helpful and one must rely on the stochastic dynamics of the system. Sekimoto's framework does exactly that [26, 28]. It establishes a connection between the Langevin's description of stochastic systems to the thermodynamics of the system. And hence, can be used to determine the thermodynamic quantities of a trajectory.

**Remark:** One must note that the definition of heat (sign) in Ref. [26] differs from the one in Ref. [28]. This document consistently follows the one presented in the book by Sekimoto (Ref. [28]) – to keep the sign convention the same as in macroscopic thermodynamics.

### 3.4.1 Sekimoto's Framework

Since the Langevin equation is a force-balance equation, the energy of the system and its surroundings must be conserved.

To understand the above statement, consider the overdamped Langevin equation (1.17) for a one-dimensional system,

$$0 = -\gamma \frac{dx}{dt} + \eta(t) - \frac{\partial U(x)}{\partial x}, \quad (3.23)$$

where the first two terms on the right side are due to the interaction of the system (Brownian particle) with the heat bath.

Looking at this system mechanically, the system and the environment must satisfy Newton's 3rd law of motion. That is, if thermal bath exerts a force  $-\gamma \frac{dx}{dt} + \eta(t)$ , then the particle also exerts equal and opposite force (reaction force) on the bath.

Since the variable  $x(t)$  characterizes the state of the system,  $dx(t)$  describes the change of state of system in a time interval  $dt$ . Then the work done by the system on its environment is the product of the change of state  $dx$  with the reaction force i.e.,  $-(-\gamma \frac{dx}{dt} + \eta) \circ dx(t)$ , where  $\circ$  represents that the product is to be taken in the Stratonovich sense. Alternatively, the negative of this expression determines the energy lost by the thermal bath to the particle as heat

$$\delta Q \equiv \left( -\gamma \frac{dx}{dt} + \eta(t) \right) \circ dx(t). \quad (3.24)$$

Similarly, taking the product of  $dx$  with the Langevin equation (3.23) and using (3.24) yields,

$$0 = \delta Q - \frac{\partial U(x)}{\partial x} \circ dx. \quad (3.25)$$

Since  $dU$  is an exact differential,

$$0 = -\delta Q + dU. \quad (3.26)$$

Equation (3.26) is an energy balance equation [26]. Equation (3.26) is the first law of thermodynamics for a single realization of the stochastic process  $x(t)$ , where there is no work done  $dW$  as no external agent is involved.

If a system were to contain an external agent controlling the parameter  $a(t)$ , the potential becomes a function of both the control parameter and the state of the system. Then

$$dU(x, a) = \frac{\partial U}{\partial x} \circ dx + \frac{\partial U}{\partial a} \circ da. \quad (3.27)$$

Adding zero to (3.25) and using (3.27)

$$\begin{aligned} \frac{\partial U}{\partial a} \circ da &= -\delta Q + \frac{\partial U}{\partial x} \circ dx + \frac{\partial U}{\partial a} \circ da \\ &= -\delta Q + dU(x, a). \end{aligned} \quad (3.28)$$

Since, the work done by the system is the response of the system to change introduced to the control parameter,  $\delta W = \frac{\partial U}{\partial a} \circ da$ . Therefore,

$$\delta W = -\delta Q + dU, \quad (3.29)$$

which is the first law of thermodynamics as in (3.2), but for an individual realization of a stochastic process. Equation (3.29) is the main result of this formalism as it holds at the level of individual trajectories and is derived directly from the Langevin dynamics of the system [28].



# Chapter 4

## Numerical Methods

Stochastic differential equations are essential when studying the physics of noisy biological systems. In addition to biological processes, they are also crucial in the field of engineering, economics and finance. However, even though they are extremely useful, they are most often not easy to handle analytically [39]. Therefore, the best way to understand them is by means of computer simulations i.e., numerically.

In this chapter, I briefly introduce some of the most used methods for simulating systems that are governed by SDE's. To do that, first the notion of weak and strong convergence of a numerical method is introduced. This definition is followed by a brief discussion of some of the most popular numerical techniques such as the Euler-Maruyama method. Lastly, three examples are presented to motivate the choice of the method used for the research work discussed in this thesis.

### 4.1 Convergence

Consider the following one-dimensional stochastic differential equation,

$$\begin{aligned} dX(t) &= a[X(t), t] dt + b[X(t), t] dW_t, \quad t \in [0, T], \quad X(t) \in \Omega, \\ X(0) &= X_0, \end{aligned} \tag{4.1}$$

where  $\Omega \subseteq \mathbb{R}$  is the state space,  $X_0$  is the given initial value and  $W_t$  is the Wiener process.

Let  $X_n$  be the numerical approximation of  $X(t_n)$  where  $t_n = n\tau$  and time-step  $\tau = T/N$ , with  $N \in \mathbb{N}$ .

**Definition (Strong and Weak Convergence)** For  $n > 0$  and  $T > 0$ , approximation  $X_n \sim X(t_n)$  converges

- **strongly** with order  $\alpha > 0$ , if there exists a constant  $C > 0$ , independent of  $\tau$  such that

$$\max_{n=0, \dots, N} \mathbb{E}(|X(t_n) - X_n|) \leq C\tau^\alpha$$

for all sufficiently small  $\tau$ , and

- **weakly** with order  $\alpha > 0$ , with respect to a function  $f : \Omega \rightarrow \mathbb{R}$ , if there exists a constant  $C > 0$ , independent of  $\tau$  such that

$$\max_{n=0,\dots,N} \mathbb{E}(|f[X(t_n)] - f(X_n)|) \leq C\tau^\alpha$$

for all sufficiently small  $\tau$  [39].

In simple terms, this definition means that a numerical solver converges to the solution strongly with order  $\alpha$ , if the maximum error between the numerical and the actual solution is of the order of  $\mathcal{O}(\tau^\alpha)$ . Alternatively, the weak convergence is defined similarly with respect to an arbitrary function  $f$  of the stochastic process  $X(t)$ .

## 4.2 Numerical Methods for SDE

### 4.2.1 Itô-Taylor Expansion

Consider the SDE (4.1), where it is assumed that  $a[X(t), t]$  and  $b[X(t), t]$  are sufficiently smooth and satisfy a linear-growth condition (see Appendix B). An additional assumption that is required is that  $a$  and  $b$  must not explicitly depend on  $t$ . That is,  $a[X(t), t] = a[X(t)]$  and  $b[X(t), t] = b[X(t)]$ .

Then the Itô-Taylor expansion for (4.1) is [53],

$$X(t) = X(t_0) + a[X(t_0)] \int_{t_0}^t d\tau' + b[X(t_0)] \int_{t_0}^t dW_{\tau'} + R \quad (4.2)$$

with  $t_0 < t$  and

$$\begin{aligned} R = & \int_{t_0}^t \int_{t_0}^{\tau'} L_0[a[X(\tau)]] d\tau d\tau' + \int_{t_0}^t \int_{t_0}^{\tau'} L_1[a[X(\tau)]] d\tau d\tau' \\ & + \int_{t_0}^t \int_{t_0}^{\tau'} L_0[b[X(\tau)]] dW_\tau dW_{\tau'} + \int_{t_0}^t \int_{t_0}^{\tau'} L_1[b[X(\tau)]] dW_\tau dW_{\tau'}, \end{aligned} \quad (4.3)$$

where the operators  $L_0$  and  $L_1$  are defined as,

$$L_0 := a[X(t)] \frac{\partial}{\partial X} + \frac{1}{2} b^2[X(t)] \frac{\partial^2}{\partial X^2}, \quad L_1 := b[X(t)] \frac{\partial}{\partial X}.$$

The Itô-Taylor expansion is the result of employing Itô's Formula in the SDE (4.1) and is crucial to obtain the numerical methods below. A detailed derivation of the expansion is discussed in Appendix B.

### 4.2.2 Euler-Maruyama Method

The Euler-Maruyama method is one of the most popular methods for solving SDE's in finance and economics and is also used for Brownian dynamics simulations in physics and biology [54].

For (4.1), the Euler-Maruyama discretization is [54],

$$\begin{aligned} X_{i+1} &= X_i + a(X_i, t_i)\Delta t_i + b(X_i, t_i)\Delta W_i, \\ X_0 &= X(0), \end{aligned} \quad (4.4)$$

where  $\Delta t_i = t_{i+1} - t_i$  with  $t_{i+1} > t_i \forall i \in \mathbb{N}$  and  $\Delta W_i = N_i \sqrt{\Delta t_i}$  with  $N_i \in \mathcal{N}(0, 1)$ <sup>1</sup>.

This method can be obtained from (4.2) by truncating the expansion and setting  $R \rightarrow 0$ . The time integral evaluates to  $(t - t_0)$ , then choosing  $t = t_{i+1}$  and  $t_0 = t_i$  results in (4.4).

It has been shown in the literature that Euler-Maruyama method converges strongly with order  $\mathcal{O}(\Delta t^{1/2})$  and converges weakly with order  $\mathcal{O}(\Delta t)$  [39].

### 4.2.3 Weak Order 2.0 Taylor Method

The Itô-Taylor expansion is also useful to obtain higher order methods such as Milstein Method [54], which is both strongly and weakly convergent with order  $\mathcal{O}(\Delta t)$  and the Weak Order 2.0 Taylor Method which converges strongly with order  $\mathcal{O}(\Delta t^{3/2})$  and weakly with order  $\mathcal{O}(\Delta t^2)$  [39].

In order to derive a higher order method, the integral form of Itô's formula (see Appendix B, equation (B.4)) can be used in Itô-Taylor expansion, for the terms in the error term  $R$  (see Appendix B for details). Following this approach gives rise to integrals of the form [36],

$$\int_{t_i}^{t_{i+1}} \int_{t_i}^{\tau} dW_{\tau'} dW_{\tau} = \frac{1}{2}[W(t_{i+1}) - W(t_i)]^2 - \frac{1}{2}(t_{i+1} - t_i) = \frac{1}{2}[(\Delta W_i^2 - \Delta t_i)], \quad (4.5)$$

where  $i > 0$ , and similarly the double Itô integral is,

$$\int_{t_i}^{t_{i+1}} \int_{t_i}^{\tau'} dW_{\tau'} d\tau = \Delta Z_i, \quad (4.6)$$

where  $\Delta Z_i$  is normally distributed with zero mean, variance  $\langle \Delta Z_i^2 \rangle = \frac{1}{3}\Delta t_i^3$  and co-variance  $\langle \Delta Z_i \Delta W_i \rangle = \frac{1}{2}\Delta t_i^2$ .

---

<sup>1</sup>Here,  $\mathcal{N}(\mu, \sigma)$  is a normal distribution with mean  $\mu$  and standard deviation  $\sigma$ .

Using the integrals results from (4.5) and (4.6), and the Itô-Taylor expansion [53][39], the Weak Order 2.0 Taylor method (also called the Itô-Taylor method [54]) can be obtained,

$$\begin{aligned}
X_{i+1} &= X_i + a\Delta t_i + b\Delta W_i + \frac{1}{2}bb_X(\Delta W_i^2 - \Delta t_i) \\
&\quad + a_Xb\Delta Z_i + \frac{1}{2}(aa_X + \frac{1}{2}a_{XX}b^2)\Delta t_i^2 \\
&\quad + (ab_X + \frac{1}{2}b_{XX}b^2)(\Delta W_i\Delta t_i - \Delta Z_i), \\
X_0 &= X(0)
\end{aligned} \tag{4.7}$$

where  $a_X = \partial_X a$ ,  $b_X = \partial_X b$ ,  $a_{XX} = \partial_X^2 a$ ,  $b_{XX} = \partial_X^2 b$  and  $\Delta Z_i = \frac{1}{2}\Delta t_i(\Delta W_i + \Delta V_i/\sqrt{3})$ , with  $\Delta V_i$  chosen independently from  $N_i\sqrt{\Delta t_i}$  with  $N_i \in \mathcal{N}(0, 1)$  [54].

### 4.3 Examples

In this section, three examples are discussed for which the above-described numerical methods are employed to solve the system of stochastic equations. The energetics for these examples are then also computed numerically. The first example deals with an undamped stochastic harmonic oscillator problem (as in Ref. [55]) in which the energetics are computed using both Itô and Stratonovich pictures, displaying the importance of correct interpretation. The second and third example treat the problem of a damped harmonic oscillator and its energetics. The underlying potential in the second example is taken to be independent of time, whereas in the third example the potential has explicit time dependence.

#### 4.3.1 Undamped Stochastic Harmonic Oscillator

The behavior of an undamped stochastic harmonic oscillator is governed by the following set of equations [55]:

$$\frac{dx(t)}{dt} = v(t), \tag{4.8a}$$

$$\frac{dv(t)}{dt} = -\omega^2 x(t) - \gamma v(t) + \sqrt{D}\eta(t), \quad t > 0, \quad x(t), v(t) \in \mathbb{R}, \tag{4.8b}$$

where  $\omega = \sqrt{k/m}$  is the frequency of the oscillator,  $\eta(t)$  is a unit Gaussian white noise with  $\langle \eta(t)\eta(s) \rangle = \delta(t-s)$  and  $D \in \mathbb{R}_{\geq 0}$  is used to scale the amplitude of the noise. In (4.8),  $k \in \mathbb{R}_{\geq 0}$  is the spring constant,  $\gamma \in \mathbb{R}_{\geq 0}$  is the friction (damping) coefficient and  $m \in \mathbb{R}_{\geq 0}$  is the mass attached to the spring.

To solve this system, (??) and (4.8b) can be expressed in form of a matrix-vector equation,

$$\frac{dX(t)}{dt} = AX(t) + Z(t), \tag{4.9}$$



where

$$X(t) = \begin{pmatrix} x(t) \\ v(t) \end{pmatrix}, \quad A = \begin{pmatrix} 0 & 1 \\ -\omega^2 & -\gamma \end{pmatrix}, \quad Z(t) = \begin{pmatrix} 0 \\ \sqrt{D}\eta(t) \end{pmatrix}.$$

Following the calculation in Ref. [31], the solution to (4.8) is given by,

$$x(t) = \frac{\sqrt{D}}{\lambda_1 - \lambda_2} \int_0^t (e^{\lambda_1(t-t')} - e^{\lambda_2(t-t')}) \eta(t') dt', \quad (4.10)$$

$$v(t) = \frac{\sqrt{D}}{\lambda_1 - \lambda_2} \int_0^t (\lambda_1 e^{\lambda_1(t-t')} - \lambda_2 e^{\lambda_2(t-t')}) \eta(t') dt', \quad (4.11)$$

where  $\lambda_1$  and  $\lambda_2$  are the eigenvalues of the matrix  $A$ . Using (4.10) and (4.11), it is possible to compute the correlation between  $x$  and  $v$ . The correlation functions can then be used to obtain the expression for potential energy,

$$\frac{d}{dt} \langle E_p(t) \rangle = k \langle v(t) \circ x(t) \rangle = \frac{Dk}{2\omega^2} \sin^2(\omega t), \quad (4.12)$$

where  $E_p$  is the potential energy of the harmonic oscillator and  $\circ$  indicates that the product is to be taken in the Stratonovich sense. Similarly for kinetic energy  $E_k$ ,

$$\frac{d}{dt} \langle E_k(t) \rangle = -k \langle x(t) \circ v(t) \rangle + m\sqrt{D} \langle \eta(t) \circ v(t) \rangle = \frac{Dk}{2\omega^2} \sin^2(\omega t) + \frac{k}{\omega^2} D. \quad (4.13)$$

Equation (4.12) was obtained by multiplying (4.8a) with  $kx$  and (4.13) is obtained by multiplying (4.8b) with  $mv$  and then taking ensemble averages [55].

Integrating (4.12) and (4.13) with zero initial conditions yields

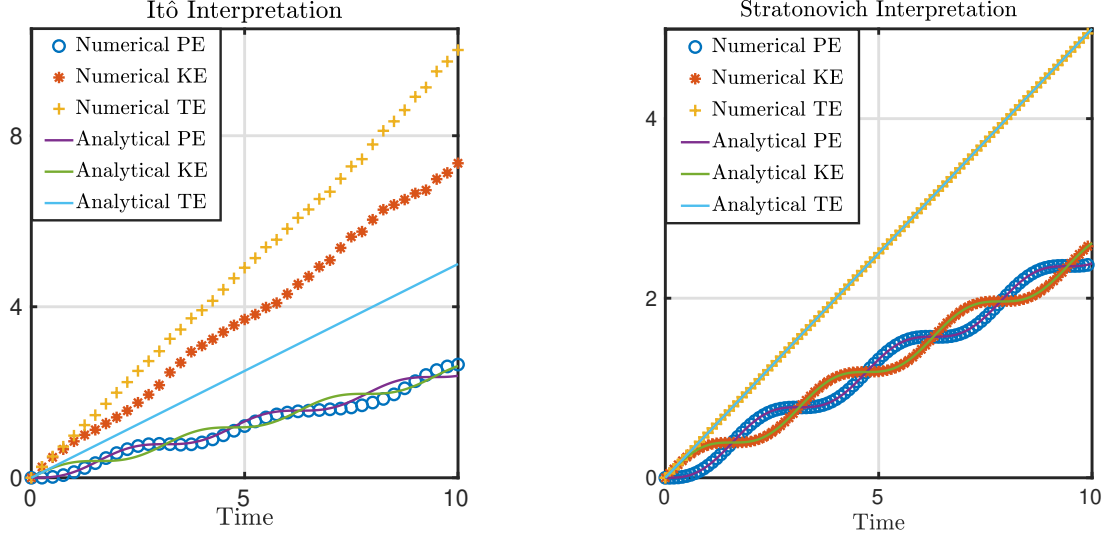
$$\langle E_k(t) \rangle = h \frac{2\omega t + \sin(2\omega t)}{8\omega}, \quad (4.14a)$$

$$\langle E_p(t) \rangle = h \frac{2\omega t - \sin(2\omega t)}{8\omega}, \quad (4.14b)$$

$$\langle E(t) \rangle = \langle E_k(t) \rangle + \langle E_p(t) \rangle = h \frac{t}{2}, \quad (4.14c)$$

where  $h \equiv \frac{Dk}{\omega^2} = Dm$  and  $E(t)$  is the total energy.

Figure 4.1 presents the results of numerical simulations and compares them to the analytical results. The numerical simulations were performed using the Euler-Maruyama method for stochastic integration and the explicit Euler method to solve the ODE's (4.12) and (4.13). These results indicate the importance of choosing the correct interpretation for the correlation, since the numerical results obtained using Itô interpretation disagree with the analytical results.



(a) Energies obtained analytically and numerically using the Itô interpretation for  $\circ$  in correlations.

(b) Energies obtained analytically and numerically using the Stratonovich interpretation for  $\circ$  in correlations.

**Figure 4.1: Simulation results for undamped harmonic oscillator:** Kinetic (KE), potential (PE) and total (TE) energies obtained numerically and analytically for undamped harmonic potential with  $h = 1$  and  $\omega = 1$ . The numerical solution of the SDE's (4.8) were used to compute correlations (numerically) which were used in (4.12) and (4.13). These equations were then integrated numerically to obtain the energies. (a) Simulation results obtained using Itô interpretation do not agree with the results of analytical calculations. (b) Simulation results obtained using Stratonovich interpretation completely agree with the analytical results.

### 4.3.2 Damped Harmonic Oscillator

#### Case 1: Time independent potential

The Langevin equation for a one-dimensional damped harmonic oscillator is,

$$\begin{aligned} \frac{dx}{dt} &= -\partial_x V(x) + \eta(t) \\ &= -\lambda_1(x - \lambda_2) + \eta(t), \quad t > 0, \quad x(t) \in \mathbb{R}, \end{aligned} \quad (4.15)$$

where  $V(x) = \frac{1}{2}\lambda_1(x - \lambda_2)^2$ , with  $\lambda_1 > 0, \lambda_2 \in \mathbb{R}$ . In the (4.15), the time correlation of noise is defined differently, as the causality plays a crucial role in this example. Hence,  $\eta(t)$  is a Gaussian white noise with,

$$\langle \eta(t)\eta(t') \rangle = D\delta(t - t')\Theta(t - t') + D\delta(t' - t)\Theta(t' - t), \quad (4.16)$$

where  $\Theta(x)$  is the Heaviside function,  $\delta(x)$  is the delta distribution and  $D \in \mathbb{R}_{>0}$  is the strength of the noise.

With  $x(0) = x_0$  as the initial condition, the solution to (4.15) is,

$$x(t) = x_0 e^{-\lambda_1 t} + \lambda_2 (1 - e^{-\lambda_1 t}) + \int_0^t e^{-\lambda_1(t-\tau)} \eta(\tau) d\tau. \quad (4.17)$$

The first moment of  $x(t)$ , can be determined easily as  $\langle \eta(t) \rangle = 0$ ,

$$\langle x(t) \rangle = x_0 e^{-\lambda_1 t} + \lambda_2 (1 - e^{-\lambda_1 t}), \quad (4.18)$$

however, the computation for second moment requires certain care,

$$\begin{aligned} \langle x^2(t) \rangle &= \lim_{\omega \rightarrow 0} \langle x(t+\omega)x(t) \rangle \\ &= \langle x \rangle^2 + \lim_{\omega \rightarrow 0} \int_0^{t+\omega} d\tau \int_0^t d\tau' e^{-\lambda_1(t+\omega-\tau)} e^{-\lambda_1(t-\tau')} \langle \eta(\tau)\eta(\tau') \rangle \\ &= \langle x \rangle^2 + \lim_{\omega \rightarrow 0} \int_0^{t+\omega} d\tau \int_0^t d\tau' e^{-\lambda_1(t+\omega-\tau)} e^{-\lambda_1(t-\tau')} \\ &\quad [D\delta(\tau - \tau')\Theta(\tau - \tau') + D\delta(\tau' - \tau)\Theta(\tau' - \tau)] \\ &= \langle x \rangle^2 + \lim_{\omega \rightarrow 0} \int_0^t d\tau' \int_0^{t+\omega} d\tau e^{-\lambda_1(t+\omega-\tau)} e^{-\lambda_1(t-\tau')} D\delta(\tau - \tau')\Theta(\tau - \tau') \\ &\quad + \lim_{\omega \rightarrow 0} \int_{[0,t]} d\tau \int_0^t d\tau' e^{-\lambda_1(t+\omega-\tau)} e^{-\lambda_1(t-\tau')} D\delta(\tau' - \tau)\Theta(\tau' - \tau) \\ &\quad + \lim_{\omega \rightarrow 0} \int_{(t,t+\omega]} d\tau \int_0^t d\tau' e^{-\lambda_1(t+\omega-\tau)} e^{-\lambda_1(t-\tau')} D\delta(\tau' - \tau)\Theta(\tau' - \tau) \\ &= \langle x \rangle^2 + \lim_{\omega \rightarrow 0} \int_0^t d\tau' \int_0^{t+\omega} d\tau e^{-\lambda_1(t+\omega-\tau)} e^{-\lambda_1(t-\tau')} D\delta(\tau - \tau')\Theta(\tau - \tau') \\ &\quad + \lim_{\omega \rightarrow 0} \int_0^t d\tau \int_0^t d\tau' e^{-\lambda_1(t+\omega-\tau)} e^{-\lambda_1(t-\tau')} D\delta(\tau' - \tau)\Theta(\tau' - \tau) + 0 \\ &= \langle x \rangle^2 + \lim_{\omega \rightarrow 0} \int_0^t d\tau' e^{-\lambda_1(t-\tau')} e^{-\lambda_1\omega} D + \lim_{\omega \rightarrow 0} \int_0^t d\tau e^{-\lambda_1(t-\tau)} e^{-\lambda_1\omega} D \\ &= \langle x \rangle^2 + \frac{D}{\lambda_1} - \frac{D}{\lambda_1} e^{-2\lambda_1 t}, \end{aligned}$$

where in the second line, (4.16) is used. Then for the next equivalence, the order of integrals is swapped for the first term and the integral is split into two domains for the second term. The last term then seems to vanish as the domains of  $\tau$  and  $\tau'$  do not overlap. Since the expectation value of  $x$  is already known,

$$\langle x^2(t) \rangle = (x_0 e^{-\lambda_1 t} + \lambda_2 (1 - e^{-\lambda_1 t}))^2 + \frac{D}{\lambda_1} - \frac{D}{\lambda_1} e^{-2\lambda_1 t}. \quad (4.19)$$

The correlation function for the noise and position can be obtained in a similar manner,

$$\begin{aligned}
\langle x(t)\eta(t) \rangle &= \lim_{\omega \rightarrow 0} \langle x(t+\omega)\eta(t) \rangle \\
&= \lim_{\omega \rightarrow 0} \left\langle \left( x_0 e^{-\lambda_1(t+\omega)} + \lambda_2(1 - e^{-\lambda_1(t+\omega)}) + \int_0^{t+\omega} e^{-\lambda_1(t+\omega-\tau)} \eta(\tau) d\tau \right) \eta(t) \right\rangle \\
&= \lim_{\omega \rightarrow 0} \int_0^{t+\omega} e^{-\lambda_1(t+\omega-\tau)} \left[ D\delta(t-\tau)\Theta(t-\tau) + D\delta(\tau-t)\Theta(\tau-t) \right] d\tau \\
&= \lim_{\omega \rightarrow 0} \int_0^t e^{-\lambda_1(t+\omega-\tau)} D\delta(t-\tau) d\tau + \lim_{\omega \rightarrow 0} \int_{(t, t+\omega]} e^{-\lambda_1(t+\omega-\tau)} D\delta(\tau-t) d\tau \quad (4.20) \\
&= \lim_{\omega \rightarrow 0} \int_0^t e^{-\lambda_1(t+\omega-\tau)} D\delta(t-\tau) d\tau + 0 = D.
\end{aligned}$$

In the third last step the Heaviside function in the first integral becomes 1, since  $t \geq \tau$  for  $\tau \in [0, t]$ . Whereas, for the second term  $\tau \in (t, t+\omega]$ , hence  $\tau > t$ , as a result  $\delta$  distribution vanishes in the following step.

#### *Energetics:*

To compute the dissipation in this system, Sekimoto's formalism (see Section 3.4) is used. Since in this case no external agent is involved, there is no associated work done. Therefore, (3.26) can be used with  $U = V(x)$ , which implies that

$$\begin{aligned}
\frac{dU}{dt} &= \frac{dV(x)}{dt} = \lambda_1(x - \lambda_2) \circ \frac{dx}{dt} \\
&= \lambda_1(x - \lambda_2) \circ (-\lambda_1(x - \lambda_2) + \eta(t)) = \frac{dQ}{dt}.
\end{aligned} \quad (4.21)$$

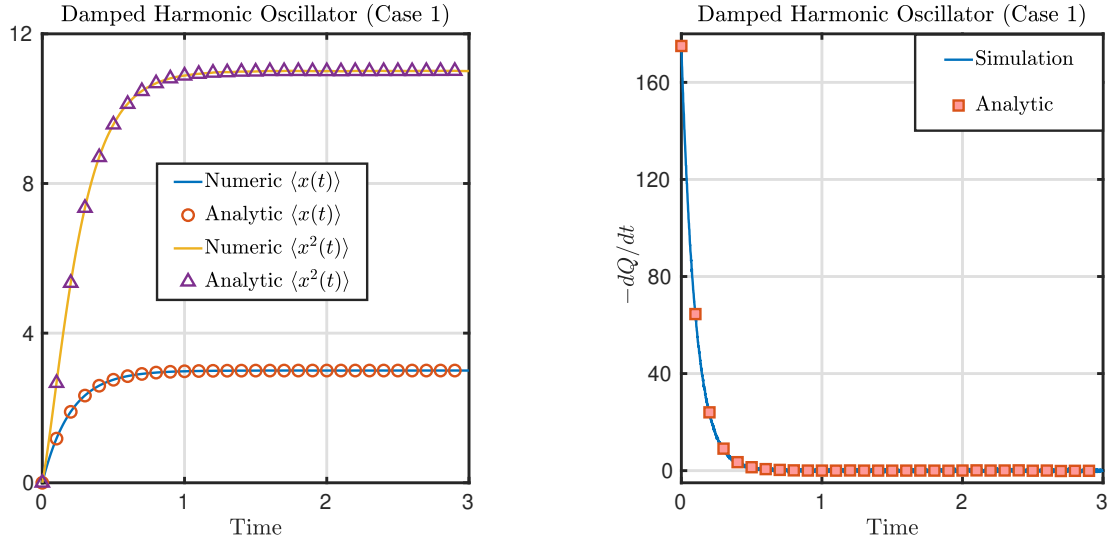
Therefore, in steady-state,

$$\begin{aligned}
-\langle \dot{Q} \rangle &= \langle \dot{W}_{\text{diss}} \rangle = -\lambda_1 \langle x \circ \dot{x} \rangle + \lambda_1 \lambda_2 \langle \dot{x} \rangle \\
&= \lambda_1^2 \langle x^2(t) \rangle + (\lambda_1 \lambda_2)^2 - 2\lambda_1^2 \lambda_2 \langle x \rangle + \lambda_1 \langle x(t) \circ \eta(t) \rangle,
\end{aligned} \quad (4.22)$$

where  $\circ$  indicates that the product is to be taken in the Stratonovich sense. Since all the correlations in the right-hand side are known, the dissipation is analytically determined.

#### *Numerical Results:*

Figure 4.2 shows the results of simulation performed using Euler-Maruyama method. The method successfully reproduces both, the moments of the solution and the dissipation in the system (see Fig. 4.2b). These results could also have been produced using the Itô-Taylor method (4.7). The results of the convergence test in Fig. 4.4a confirms this claim. The convergence test displays the error in  $\langle x \rangle$  on a log-log plot as the time-stepping is halved. It can be seen from the convergence test that Itô-Taylor method has a weak convergence of  $\mathcal{O}(\Delta t^2)$ , whereas Euler-Maruyama method only has a weak convergence of order  $\mathcal{O}(\Delta t)$ .



(a) First and second moment of  $x(t)$  obtained numerically and analytically

(b) Average rate of heat exchange as a function of time obtained numerically and analytically.

**Figure 4.2: Simulation results using Euler-Maruyama method:** (a, b) The moments and energy dissipation computed numerically using the Euler-Maruyama method agrees well with the analytical results in equations (4.18), (4.19) and (4.22). *Parameters:*  $D = 20$ ,  $\lambda_1 = 5$ ,  $\lambda_2 = 3$  and  $x_0 = 0$ .

### Case 2: Time Dependent Potential

For this case,  $V(x, t) = \frac{1}{2}ktx^2$ . Therefore, the overdamped Langevin equation is,

$$\frac{dx}{dt} = -ktx + \eta(t), \quad t > 0, \quad x(t) \in \mathbb{R} \quad (4.23)$$

where  $\eta(t)$  is as in case 1 and  $k > 0$ . The solution to this equation can also be computed using variation of parameters, with  $x(0) = x_0$ ,

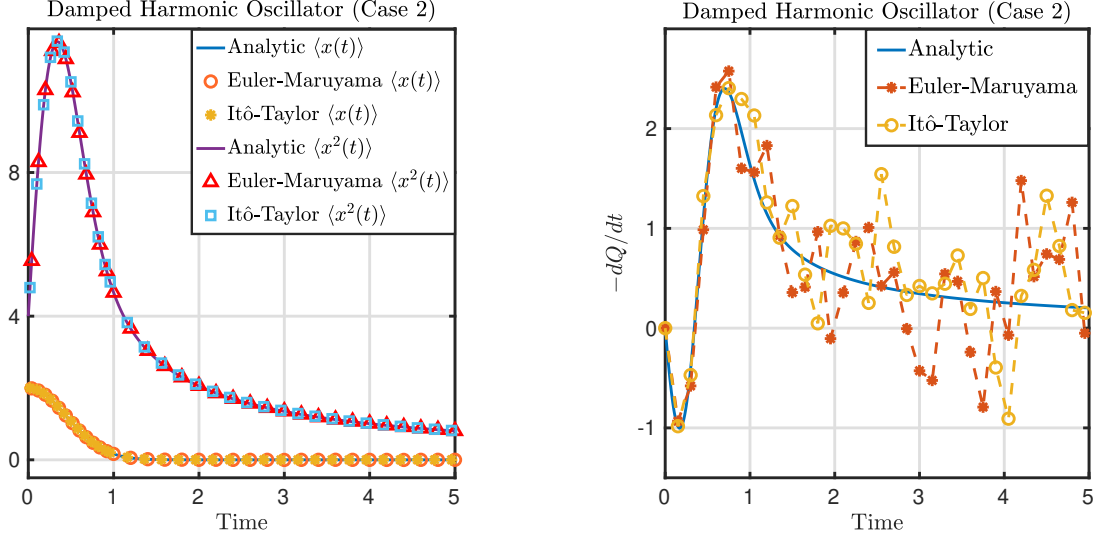
$$x(t) = x_0 e^{-k\frac{t^2}{2}} + \int_0^t \eta(\tau) e^{-\frac{k}{2}(t^2 - \tau^2)} d\tau. \quad (4.24)$$

The first moment, second moment of  $x$  and its correlation with noise  $\eta(t)$  is given by,

$$\langle x(t) \rangle = x_0 e^{-k\frac{t^2}{2}}, \quad (4.25)$$

$$\langle x^2(t) \rangle = \lim_{\omega \rightarrow 0} \langle x(t + \omega)x(t) \rangle = x_0^2 e^{-kt^2} + D \sqrt{\frac{\pi}{k}} e^{-kt^2} \text{Erfi}(\sqrt{k}t), \quad (4.26)$$

$$\langle x(t)\eta(t) \rangle = D. \quad (4.27)$$



(a) First and second moment of  $x(t)$  using both the numerical methods

(b) Dissipation for case 2 computed using both the methods become unstable as  $t$  increases

**Figure 4.3: Simulation results for case 2:** (a) The first and second moment of the numerical solution of (4.23) using both numerical methods agree with the analytical calculations. (b) The energetic calculations using both numerical methods become unstable. *Parameters:*  $D = 20, k = 5$ , and  $x_0 = 2$ .

Figure 4.3a shows the first and second moment of  $x$ , computed using both Euler-Maruyama and Itô-Taylor method. Both the methods successfully solve (4.23).

*Energetics:*

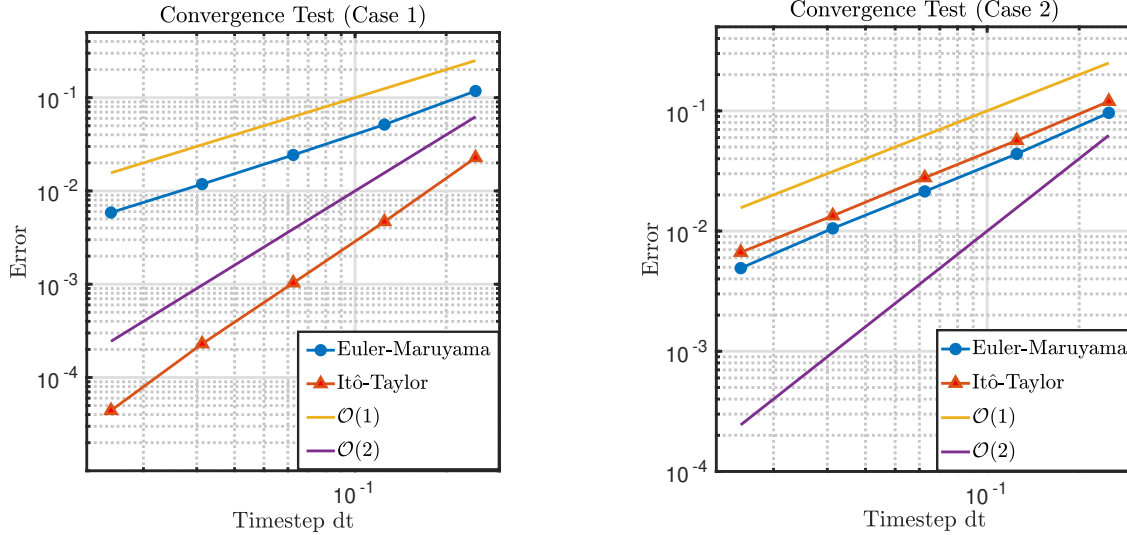
Again, employing Sekimoto's formalism to determine dissipation with  $U(x, t) = V(x, t) = \frac{1}{2}tkx^2$ .

$$-\left\langle \frac{dQ}{dt} \right\rangle = -kt\langle x \circ \dot{x} \rangle = k^2t^2\langle x^2 \rangle - kt\langle x \circ \eta \rangle \quad (4.28)$$

*Numerical Results:*

Numerical computation of dissipation is presented in Fig. 4.3b, where the dissipation is computed using both the methods. It is clear from the figure that computation for the dissipation becomes unstable as simulation time increases. This unstable behavior is expected when using Euler-Maruyama method but not for a higher order method like Itô-Taylor method.

In case of Euler-Maruyama method, the instability in the dissipation arises from the fact that the weak convergence of the method is of order  $\mathcal{O}(\Delta t)$  and (4.28) contains a term where the second moment of  $x(t)$  is multiplied with  $t^2$ , which carries a factor of  $N^2$  – where



(a) Convergence test for Euler-Maruyama and Itô-Taylor method for case 1.

(b) Convergence Test for Euler-Maruyama and Itô-Taylor method for case 2.

**Figure 4.4: Convergence test results:** (a) For case 1, Euler-Maruyama method has a weak  $\mathcal{O}(\Delta t)$  and Itô-Taylor method has weak  $\mathcal{O}(\Delta t^2)$  convergence. (b) For case 2, both the methods i.e., Euler-Maruyama method and Itô-Taylor method have weak  $\mathcal{O}(\Delta t)$  convergence.

$N$  is the number of time steps. Therefore, the resulting computation does not converge. However, this unstable behavior is not expected when using a method with weak convergence order of  $\mathcal{O}(\Delta t^2)$ .

The reason for this unstable behavior is illustrated in Fig. 4.4b, which shows the convergence test for both the methods for this case. The convergence test shows that Itô-Taylor method is only able to achieve weak  $\mathcal{O}(\Delta t)$  instead of  $\mathcal{O}(\Delta t^2)$ . At this point it is not clear as to why this behavior is observed. However, it is believed that the cause of this behavior arises from the fact that the potential  $V(x, t)$  has an explicit dependence on  $t$ . But it is still not clear why the convergence of Euler-Maruyama method in this case is not spoiled, as the assumptions for both the methods are same.

### Conclusion:

Even though the convergence and stability of the higher order methods is well established in the literature of numerical methods of SDE [39], the results of the examples above indicate that Euler-Maruyama method seems to be more reliable. Since investigating the convergence of higher order methods is not the focus of the work presented in this thesis, Euler-Maruyama method will be used for the simulating the stochastic equations discussed in the later chapters of this thesis.





## Part II

# Sensory Adaptation in Cells



# Chapter 5

## The Stochastic System

### 5.1 Feedback Network

All living systems, from single-celled to complex multi-cellular organisms dissipate, i.e., they consume energy to perform essential actions. One crucial yet dissipative function is sensory adaptation, which is performed by all biological systems [56]. A way to model sensory adaptation in living system is by using a biochemical feedback loop [57]. Sensory adaptation in cells is usually mathematically represented as a response of a noisy biochemical feedback loop, where the noise is due to the small size of the cell.

In Ref. [15], Lan and Tu use a three-node negative feedback loop (see Fig. 5.1) as a minimal model for achieving accurate adaptation in cells. The model consists of an input *stimulus* ( $s \in \mathbb{R}_{\geq 0}$ ), a slowly changing dynamic negative *control* element (methylation level  $m \in [0, m_0]$ , where  $m_0 \in \mathbb{R}_{\geq 0}$ ) and a fast response to the output *activity* ( $a \in [0, 1]$ ). Any change in the input stimulus  $s$ , brings-forth a change in the activity level  $a$ . The change in  $a$  triggers a change in  $m$ , which counters the effect of  $s$ , bringing  $a$  to a *stimulus-independent level*  $a_0 \in [0, 1]$ . The dynamics of the system are governed by a system of two coupled Langevin equations,

$$\frac{da}{dt} = F_a(a, m, s) + \eta_a(t) \quad (5.1a)$$

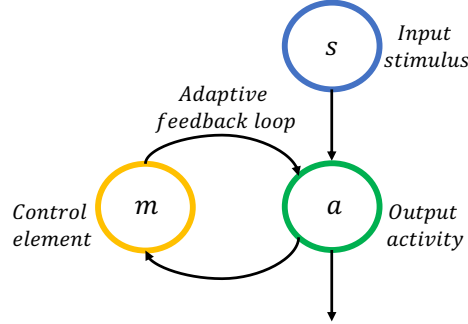
$$\frac{dm}{dt} = F_m(a, m, s) + \eta_m(t) \quad (5.1b)$$

where  $\eta_a$  and  $\eta_m$  are the white-noise terms arise due to the small size of a cell. The noise terms satisfy the following correlations,

$$\langle \eta_a(t) \eta_a(t') \rangle = 2\Delta_a \delta(t - t'), \quad (5.2a)$$

$$\langle \eta_m(t) \eta_m(t') \rangle = 2\Delta_m \delta(t - t'), \quad (5.2b)$$

where  $\Delta_a, \Delta_m \in \mathbb{R}_{\geq 0}$  are the corresponding noise amplitudes.



**Figure 5.1: Three-node system with an adaptive feedback loop:** Consisting of an input stimulus  $s$ , a control element  $m$  (methylation level) and output activity  $a$ .

The coupled system (5.1) corresponds to the following Fokker-Planck equation for the probability distribution  $P \equiv P(a, m, t)$ ,

$$\partial_t P = -\frac{\partial}{\partial a}(F_a P) - \frac{\partial}{\partial m}(F_m P) + \Delta_m \frac{\partial^2 P}{\partial m^2} + \Delta_a \frac{\partial^2 P}{\partial a^2}, \quad (a, m)^T \in \Omega \quad (5.3)$$

where  $\Omega = (0, 1) \times (0, m_0)$  is the interior of the phase space  $\bar{\Omega}$ . Moreover, it can be shown that when the system is in equilibrium, it must satisfy

$$\Delta_a \partial_a F_m = \Delta_m \partial_m F_a \quad (5.4)$$

to preserve detailed balance (derivation in Appendix C.1). Since the biochemical network consists of a negative feedback loop, it is required that  $\partial_a F_m$  and  $\partial_m F_a$  must be of opposite signs (this claim can be proved by using the time-scale separation between adaptation and response, and the nullclines of (5.1). Detailed calculations can be found in Ref. [56]). Therefore, it implies that all systems with negative feedback loops must break detailed balance, which entails that adaptation is a thermodynamically irreversible process.

The functions  $F_a$  and  $F_m$  provide a coarse-grained description of the biochemical processes inside the cell. Their functional form as prescribed in Ref. [15] are,

$$F_a(a, m, s) = -\omega_a[a - G(m, s)], \quad (5.5a)$$

$$F_m(a, m, s) = -\omega_m(a - a_0)[\beta - (1 - \beta)C\partial_m G(s, m)], \quad (5.5b)$$

where,  $F_a$  is chosen such that activity  $a$  relaxes to its mean value  $G(s, m)$  with a fast response rate  $\omega_a > 0$ . Whereas,  $F_m$  describes the slow adaptation dynamics controlled by rate  $\omega_m > 0$ . To ensure that the time scale of response is much slower than that of adaptation, it is assumed that  $\omega_a \gg \omega_m$ .

**Remark:** As proved in Ref. [56], it is required for adaptation that  $G(s, m)$  must have opposite dependence on  $s$  and  $m$ .

To obtain an equilibrium interaction  $F_m^{eq}$ , the detailed balance condition (5.4) can be integrated with respect to  $a$ ,

$$\frac{\Delta_m}{\Delta_a} \int_{a_0}^a \partial_m(-\omega_a[a - G(s, m)]) da = F_m^{eq}(a) - F_m^{eq}(a_0).$$

In equilibrium, the system fails to adapt, therefore it can be assumed that  $F_m^{eq}(a_0) = 0$ . Hence,

$$\omega_a \frac{\Delta_m}{\Delta_a} (a - a_0) \partial_m G(s, m) = F_m^{eq}(a). \quad (5.6)$$

For processes that break detailed balance (non-equilibrium),

$$F_m^{neq} = -\omega_m(a - a_0), \quad (5.7)$$

such that the system still attains  $a_0$  as its steady-state value, where factor  $(a - a_0)$  ensures accurate adaptation. The linear combination,  $\beta F_m^{eq} + (1 - \beta) F_m^{neq}$ , yields (5.5b), where  $C = \frac{\Delta_m/\omega_m}{\Delta_a/\omega_a}$  is a constant and  $\beta \in [0, 1]$  is a parameter that controls whether the detailed balance is broken or not i.e., it allows the system to move in and out-of-equilibrium.

When  $\beta = 0$ ,  $F_a$  and  $F_m$  satisfy (5.4) and therefore, the system is in equilibrium. In case when  $\beta = 1$ , the detailed balance condition is violated, hence the system becomes dissipative and adaptation is possible. For any other  $\beta \neq 0$ , even though the system is away from equilibrium it still may or may not adapt.

Additionally, when the system is in equilibrium, it satisfies the fluctuation dissipation theorem (1.11). Therefore, using the FDT, the *effective temperature* for the activity  $a$  and the control element  $m$  can be defined as,

$$T_{\text{eff}(a)} = \frac{\Delta_a}{\omega_a}; \quad T_{\text{eff}(m)} = \frac{\Delta_m}{\omega_m}, \quad (5.8)$$

where  $k_B \equiv 1$  and  $1/\gamma_i = \omega_i$  for  $i = \{a, m\}$ . Equation (5.8) implies that the constant  $C$  in (5.5b) is the ratio of effective temperatures.

**Remark:** For simplification, it is assumed throughout this thesis that the effective temperatures for both  $a$  and  $m$  are equal i.e.,  $T_{\text{eff}(a)} = T_{\text{eff}(m)} = T_{\text{eff}}$ . This assumption entails that  $C = 1$ , unless stated otherwise. In Ref. [15], no such assumption was made.

**Remark:** For simulations in Ref. [15],  $G(s, m)$  was chosen to be a sigmoidal of the form,

$$G(s, m) = \frac{1}{1 + (s/\exp(2m))^H}, \quad (5.9)$$

where the Hill coefficient  $H = 1$ . This choice agrees with experimental findings [56]. The same functional form for  $G(s, m)$  with  $H = 1$  is used here, unless stated otherwise.

### 5.1.1 Effective Potential & Phase Portraits

To further investigate this model analytically, it is possible to obtain an approximate steady-state solution to the FPE (5.3). Since (5.3) is a PDE, it is important that the boundary conditions associated to the problem are known.

The boundary conditions imposed here are no flux (or reflective) boundary conditions i.e., for any stimulus level  $s$  and  $\beta$ ,

$$\mathbf{n} \cdot \mathbf{J} = 0, \quad \text{for } (a, m)^T \in \partial\Omega, \quad (5.10)$$

where  $\partial\Omega$  is the boundary of the phase space  $\bar{\Omega} = [0, 1] \times [0, m_0]$ ,  $\mathbf{n}$  is the normal vector and  $\mathbf{J} = (J_a, J_m)^T$  is the probability current vector where,

$$J_i(a, m, t) \equiv F_i P - \Delta_i \partial_i P \quad (5.11)$$

with  $i = \{a, m\}$ . Equation (5.10) is equivalent to  $J_a(a = 0, m, t) = J_a(a = 1, m, t) = J_m(a, m = 0, t) = J_m(a, m = m_0, t) = 0$ , for all  $t$ .

In steady-state,  $\partial_t P^s = 0$ , where  $P^s$  is the steady-state solution<sup>1</sup> of (5.3). Also, since  $\omega_a \gg \omega_m$ , the timescales for response and adaptation are separated, hence adiabatic approximation holds. This approximation implies that  $\partial_m J_m \approx 0$ . Moreover, as in Ref. [15], the probability distribution can be written as

$$P(a, m) = P_a(a/m) P_m(m). \quad (5.12)$$

Then, (5.3) entails that  $\partial_a J_a \approx 0$  everywhere in  $\Omega$ . With these approximations, the system is decomposed into two one-dimensional systems, which entails that  $J_a \approx 0$ . Using the definition of probability current in (5.11) and integrating along  $a$  yields,

$$\begin{aligned} \ln P_a(a/m) &= -\frac{\omega_a}{\Delta_a} \int (a - G(s, m)) da \\ P_a(a/m) &= \exp \left\{ -\frac{\omega_a}{2\Delta_a} (a - G(s, m))^2 \right\} \underbrace{\exp \left( C + \frac{\omega_a}{\Delta_a} G^2(s, m) \right)}_{\text{const in } a}, \end{aligned} \quad (5.13)$$

where  $C$  is the constant of integration. The part constant in  $a$ , in (5.13), can be absorbed in  $P_m(m)$ . Hence,

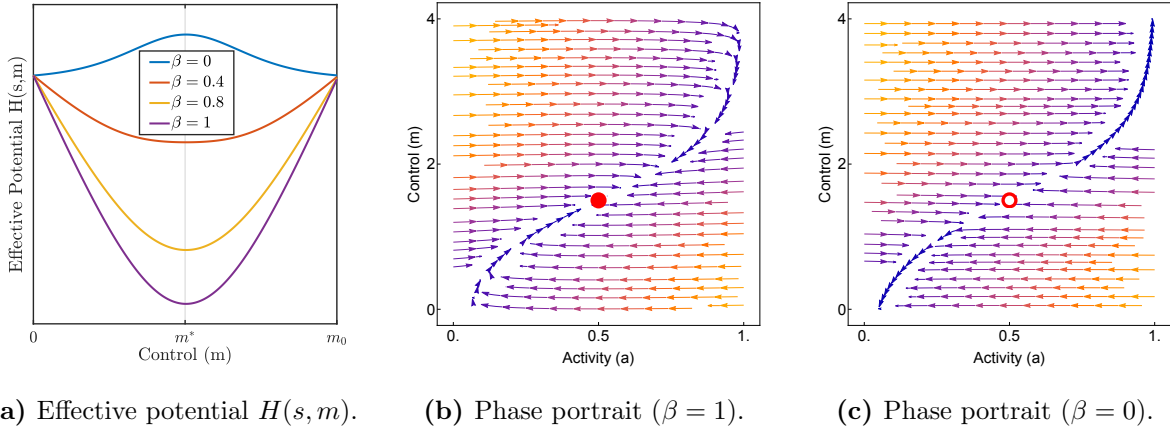
$$P_a(a/m) = \exp \left( -\frac{\omega_a}{2\Delta_a} (a - G(s, m))^2 \right). \quad (5.14)$$

Additionally, flux conservation also entails,

$$0 = \int_0^1 J_m(a, m) da,$$

---

<sup>1</sup>dropping the superscript  $s$  notation from here on



**Figure 5.2: Effective potential and phase portraits:** (a) Effective potential  $H(s, m)$  (with fixed  $s$ ) for varying  $\beta$ , where at  $m = m^*$  the system changes from being unstable to stable as  $\beta$  is increased from 0 to 1. (b) Phase portrait for a fully-adaptive system has a stable fixed point at  $(a, m) = (a_0, m^* = -\frac{1}{2} \log(\frac{1}{s}(\frac{1}{a} - 1)))$ , which is the adapted state. (c) Phase portrait for the system in equilibrium has an unstable fixed point, hence the system is unable to adapt.

$$0 = \int_0^1 (F_m P - \Delta_m P_a \partial_m P_m - \Delta_m P_m \partial_a \ln P_a) da, \quad (5.15)$$

where to obtain (5.15), (5.12) is used. Then, using (5.14) and (5.5b) in (5.15) yields,

$$0 = \int_0^1 e^{-F} (-\omega_m [a - a_0] [\beta - (1 - \beta) \partial_m G] P_m - \Delta_m \partial_m P_m + \Delta_m P_m \frac{\omega_a}{\Delta_a} (a - G) \partial_m G) da, \quad (5.16)$$

where

$$F \equiv \frac{\omega_a}{2\Delta_a} (a - G(s, m))^2. \quad (5.17)$$

Since the response time is much faster than the adaptation time, the following approximation holds [58],

$$\lim_{\omega_m \ll \omega_a} \exp \left( -\frac{\omega_a}{2\Delta_a} (a - G(s, m))^2 \right) \approx \sqrt{\frac{2\pi\Delta_a}{\omega_a}} \delta(a - G). \quad (5.18)$$

Using (5.18) in (5.16) and performing the integration yields,

$$\sqrt{\frac{2\pi\Delta_a}{\omega_a}} \left( -\underbrace{\omega_m [G - a_0] [\beta - (1 - \beta) \partial_m G]}_{\Delta_m h(s, m)} P_m - \Delta_m \partial_m P_m + 0 \right) \approx 0,$$

which entails that

$$h(s, m) P_m(m) = -\partial_m P_m(m), \quad (5.19)$$

must be true. The general solution of (5.19) is,

$$P_m(m) = \frac{1}{Z} \exp \left( -\int h(s, m) dm \right) = \frac{1}{Z} \exp(-H(s, m)), \quad (5.20)$$

where  $Z = \sqrt{2\pi\Delta_a/\omega_a} \int_0^{m_0} e^{-H(s,m)} dm$  is the normalization constant (partition function) and

$$H(s, m) = \frac{\omega_m}{\Delta_m} \int (G(s, m) - a_0) [\beta - (1 - \beta) \partial_m G(s, m)] dm, \quad (5.21)$$

is the *effective potential*.

Figure 5.2a displays the effective potential as a function of  $m$  for different values  $\beta$ . For  $\beta = 0$ , the system is unstable, whereas the system is stable when  $\beta = 1$ . This behavior can also be observed in the phase portraits in Figs. 5.2b and 5.2c. The nullclines of these phase portraits are obtained by setting  $F_a(a, m) = 0$  and  $F_m(a, m) = 0$ , for fixed  $s$  and  $a_0$ . The intersection of these two nullclines yield the fixed point  $(a_0, m^*)$  for fixed  $s$  and  $a_0$  and also implies that  $G(s, m^*) = a_0$ .

This change in behavior of the system due to variation in the parameter  $\beta$  indicates that there must exist a critical value  $\beta = \beta_c$ . At this critical value  $\partial_m^2 H(s, m)|_{m=m^*} = 0$ . Using this condition  $\beta_c$  can be determined as,

$$\beta_c - (1 - \beta_c) C \partial_m G(s, m^*) = 0 \implies \beta_c = \frac{C \partial_m G(s, m^*)}{1 + C \partial_m G(s, m^*)} \in (0, 1), \quad (5.22)$$

where, adaptation is possible only for  $\beta > \beta_c$ .

### 5.1.2 Numerical Simulation

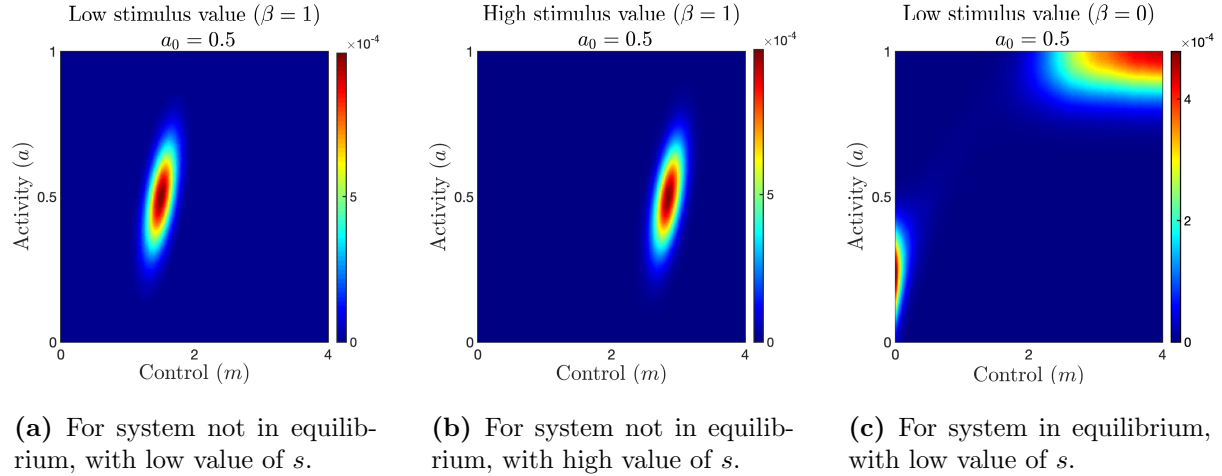
To further investigate the properties of the stochastic system (5.1), numerical simulations are performed using MATLAB. In Ref. [15], simulations were performed by using a grid-based finite difference approach to numerically solve the Fokker-Planck equation (5.3). However, the numerical approach used in this thesis solves the set of Langevin equations instead of the FPE.

To solve the set of SDE's (5.1), Euler-Maruyama method (Section 4.2.2) is used. However, one must note that even though FPE (5.3) and the stochastic system (5.1) are different descriptions for the same stochastic process, when the boundary conditions (5.10) are imposed on (5.3), the correspondence breaks down. This correspondence can be restored by introducing an additional term in Langevin equations [59].

In order to ensure that the equivalence holds, it is essential to impose reflective boundary conditions when numerically treating the Langevin equations. The boundary conditions can be imposed by reflecting the result of every iteration of numerical method (resulting from time-stepping) which tends to go outside the state space  $\bar{\Omega}$ . Since  $\bar{\Omega}$  in this case is a rectangular domain, the *reflection operator* is defined using a “central modulus” like mapping [60].

The histogram for the numerical solution of the stochastic system is presented in Fig.





**Figure 5.3: Steady-state probability distribution: (a, b)** The system can only adapt if it is away from equilibrium. The steady-state probability distribution in non-equilibrium is a vortex around the stable fixed point  $(a_0, m^*)$ . As the value of stimulus  $s$  is increased, the location of minima  $m^*$  shifts towards the right. **(c)** In equilibrium, the system has one unstable fixed point and hence it is always driven to the corners of the phase-space as in Fig. 5.2c. Therefore, the steady-state probability distribution in equilibrium is always localised at either one of the two or both the corners of the domain. *Simulation parameters:*  $\omega_a = 50$ ,  $\omega_m = 5$ ,  $\Delta_a = 0.01\omega_a$ ,  $\Delta_m = 0.01\omega_m$ ,  $m_0 = 4$ .

5.3. Figures 5.3a and 5.3b show the case in which the system is able to adapt ( $\beta = 1$ ). Whereas, Fig. 5.3c exhibits the steady-state probability distribution when the system is in equilibrium ( $\beta = 0$ ) and is unable to achieve adaptation.

## 5.2 Energetic Cost for Adaptation

In the previous section, the dynamical properties and the steady-state probability distribution of the stochastic system were discussed. Moreover, it was also established that adaptation requires the detailed balance to be broken i.e., it is accompanied with an energy cost. That is, since adaptation is a thermodynamically irreversible process, an adaptive system must dissipate some of its energy as heat to the environment. Therefore, it is only natural to investigate the amount of energy dissipated during adaptation.

Since the set of Langevin equations (5.1) govern a stochastic process in a continuous state-space, the energetics of the system can be explored using the formalism discussed in Section 3.3. Using (3.17) with  $F_i = -\frac{1}{\gamma}\partial_i U$ , the equation for total rate of entropy of this system can be computed using the equation,

$$\dot{S} = \iint_{\Omega} \left[ \frac{J_a^2}{\Delta_a} + \frac{J_m^2}{\Delta_m} \right] \frac{1}{P} da dm - \iint_{\Omega} \left[ \frac{J_a F_a}{\Delta_a} + \frac{J_m F_m}{\Delta_m} \right] da dm, \quad (5.23)$$

where the first term corresponds to the rate of entropy production and the second term is associated with the entropy exchange rate of the system with its surroundings.

When studying the thermodynamics of a system, one is interested in the steady-state behavior of energetic quantities. In steady-state,  $\dot{S} = 0$ , which implies that in steady-state the rate of entropy production becomes equal to the rate of entropy exchange (as in (3.21)). Then using the effective temperatures of the system, the steady-state dissipation (which is the same as the heat exchange in steady-state) can be determined as,

$$\dot{W}_{\text{diss}} = T_{\text{eff}} \int_0^1 da \int_0^{m_0} dm \left[ \frac{J_a^2}{\Delta_a P} + \frac{J_m^2}{\Delta_m P} \right]. \quad (5.24)$$

By employing the adiabatic approximation and the functional form of steady-state probability distribution  $P(a, m)$  (as obtained in (5.14) and (5.20)), it is possible to analytically approximate the integrals in the (5.24). The calculation yields the following expression for dissipation (see Appendix C.2),

$$\dot{W}_{\text{diss}} \approx T_{\text{eff}} \omega_m \beta^2 \frac{\Delta_a / \omega_a}{\Delta_m / \omega_m} \left( 1 + \frac{\Delta_m / \omega_m}{\Delta_a / \omega_a} G'(s, m^*) \right)^2. \quad (5.25)$$

**Remark:** Since it is assumed that  $\Delta_m / \omega_m = \Delta_a / \omega_a$ , (5.25) becomes

$$\dot{W}_{\text{diss}} \approx T_{\text{eff}} \omega_m \beta^2 (1 + G'(s, m^*))^2. \quad (5.26)$$

For  $\beta = 0$ , (5.26) vanishes identically. This result agrees with the understanding that systems in equilibrium do not dissipate.

### 5.3 Adaptation Accuracy

In addition to the energy cost of adaptation, another quantity of interest is the adaptation accuracy, which can be quantified by the *relative adaptation error*  $\epsilon$  (or simply *adaptation error*),

$$\epsilon \equiv \left| 1 - \frac{\langle a \rangle}{a_0} \right| \quad \text{where,} \quad \langle a \rangle = \iint_{\Omega} a P da dm \quad (5.27)$$

is the mean activity.

To further investigate the properties of the adaptation error, one can integrate the steady-state Fokker-Planck equation for  $m$ ,

$$\int_0^1 da \int_0^{m_0} dm J_m = 0.$$

Using (5.11) and integrating out the term on the right,

$$\begin{aligned} \int_0^1 da \int_0^{m_0} dm [F_m P - \Delta_m \partial_m P] &= 0, \\ \int_0^1 da \int_0^{m_0} dm F_m P &= \int_0^1 da \Delta_m [P(a, m_0) - P(a, 0)], \end{aligned}$$

then utilizing (5.5b) yields,

$$\begin{aligned} - \int_0^1 da \int_0^{m_0} dm \omega_m (a - a_0) \beta P + \int_0^1 da \int_0^{m_0} dm \omega_m P (1 - \beta) C(a - a_0) \partial_m G \\ = \int_0^1 da \Delta_m [P(a, m_0) - P(a, 0)], \end{aligned}$$

definition of  $\langle a \rangle$  from (5.27) and normalization of  $P$  entails,

$$\begin{aligned} -\omega_m (\langle a \rangle - a_0) \beta + \int_0^1 da \int_0^{m_0} dm \omega_m P(a, m) (1 - \beta) C(a - a_0) \partial_m G \\ = \int_0^1 da \Delta_m [P(a, m_0) - P(a, 0)], \end{aligned}$$

lastly, taking absolute value on both sides

$$\begin{aligned} \left| 1 - \frac{\langle a \rangle}{a_0} \right| &= \left| \frac{1}{\beta \omega_m a_0} \int_0^1 da \Delta_m P(a, 0) da - \frac{1}{\beta \omega_m a_0} \int_0^1 da \Delta_m P(a, m_0) da \right. \\ &\quad \left. + \frac{1 - \beta}{\beta a_0} \int_0^1 da \int_0^{m_0} dm P(a, m) C \partial_m G(s, m) (a - a_0) \right|. \end{aligned}$$

Using the definition of adaptation error yields,

$$\epsilon = |\epsilon_1 - \epsilon_2 + \epsilon_3|, \quad (5.28)$$

with

$$\epsilon_1 = \frac{1}{\beta \omega_m a_0} \int_0^1 da \Delta_m P(a, 0) da, \quad (5.29a)$$

$$\epsilon_2 = \frac{1}{\beta \omega_m a_0} \int_0^1 da \Delta_m P(a, m_0) da, \quad (5.29b)$$

$$\epsilon_3 = \frac{1 - \beta}{\beta a_0} \int_0^1 da \int_0^{m_0} dm P(a, m) C \partial_m G(s, m) (a - a_0). \quad (5.29c)$$

Equation (5.28) implies that the total adaptation error contains contributions due to three terms. The first two ( $\epsilon_1$  and  $\epsilon_2$ ) terms are the contribution to the error due to the boundary of the system, where as the remaining third term is the error in the bulk. Equation (5.29c) also indicates that when the system is in a fully adaptive state i.e. when  $\beta = 1$ , the total adaptation error vanishes if the system is away from the boundaries.

### 5.3.1 The ESA Relation

Since an analytical approximation for the dissipation in the system has been obtained, it is possible to relate it to the adaptation accuracy as done in Ref. [15]. The relation presented in Ref. [15] is,

$$\dot{S} \approx c_0 \frac{\Delta_a}{\omega_a} \times \omega_m \times \ln(\epsilon_0/\epsilon). \quad (5.30)$$

Equation (5.30) is the *Energy-Speed-Accuracy trade-off relation* where  $c_0$  and  $\epsilon_0$  are constants that depend on system parameters and the choice of  $G(s, m)$ . Equation (5.30) implies that the system requires higher dissipation rate to achieve higher adaptation accuracy. Moreover, it also shows that higher dissipation is needed if the system wants to adapt faster.

To obtain (5.30), one needs to first consider (5.28) and then consider the problem in different cases,

- Case 1:  $\beta = 1$

This case deals with the system when it is in fully adaptive state. Hence in this case the bulk error  $\epsilon_3$  is identically zero. Therefore, the adaptation error in this case is due to the contribution of the boundaries i.e.  $\epsilon_1$  and  $\epsilon_2$ . Moreover, if it is assumed that  $\Delta_a/\omega_a \gg \Delta_m/\omega_m$  i.e.  $C \ll 1$ , then (5.25) becomes,

$$\dot{S} \approx \omega_m \frac{\Delta_a/\omega_a}{\Delta_m/\omega_m} = \frac{\omega_a}{C}. \quad (5.31)$$

Furthermore, if  $m^*$  is closer to  $m_0$  then,  $\epsilon_1 \gg \epsilon_2$ . Then steepest descent method can be used to integrate along  $a$ , which results in the following expression for adaptation error,

$$\epsilon \approx \epsilon_0 \exp\left(-\frac{d}{\Delta_m/\omega_m}\right), \quad (5.32)$$

where,  $\epsilon_0 \equiv \sqrt{\Delta_m G'/2\pi\omega_m a_0^2}$  and  $d \equiv \int_{m_0}^{m^*} G(s, m) dm$ . Then using (5.32) in (5.31) yields (5.30).

- Case 2:  $\beta_c < \beta < 1$

In this case, Lan and Tu focus on  $\epsilon_3$  and use Taylor expansion and steepest descent method to obtain (see Appendix C.3 for details),

$$\epsilon \approx \frac{1-\beta}{\beta} e_0 (\beta - (1-\beta) \frac{\Delta_m/\omega_m}{\Delta_a/\omega_a} G'(s, m^*))^{-1}, \quad (5.33)$$

where  $e_0 \equiv 3 \frac{(\Delta_m/\omega_m)^2}{\Delta_a/\omega_a} G''/2a_0$ . If  $\beta$  is assumed to be very close to 1, then (5.33) yields,

$$\beta \approx 1/(1 + \epsilon_3/e_0) \quad (5.34)$$

Furthermore, they employ (5.25) and (5.34) to get,

$$\frac{\dot{S}}{C_0\omega_m} \approx \frac{1}{(1 + \epsilon_3/e_0)^2}, \quad (5.35)$$

where  $C_0 = \frac{1}{\bar{C}}(1+CG')^2$ . To proceed further, one can choose  $\epsilon_3 = e^x$  and  $e_0 = e^{x_0}$  and use that  $\partial_x^2 \dot{S} = 0$  at  $x = x_0 - \ln(2)$  to obtain the ESA relation, where  $\epsilon_0 = e^{3/2}e_0/2$  and  $c_0 = 8C_0\omega_a/27\Delta_a$ .



# Chapter 6

## Boundary Potential & Energetics

### 6.1 Introduction

In the previous chapter, the stochastic system (5.1) that models sensory adaptation in cells was introduced. In addition, an approach to simulate the model, the definition of adaptation error and a way to analytically approximate the energetic cost of adaptation was discussed.

One way to determine the energetic cost is by means of an approximation of integrals presented in Ref. [15] (using (5.25)). The other approach to compute the dissipation is a more mechanical one and is numerically more accurate. This approach utilizes the fact that in steady-state, the dissipation equals the heat exchange between the system and its surroundings (as in (3.22)). Furthermore, from the point of view of classical mechanics, the rate of change of heat is given by the product of the force and the velocity of the particle. Therefore,

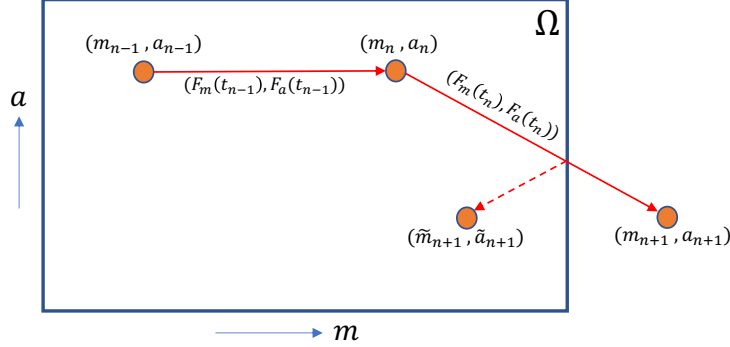
$$\langle \dot{W}_{\text{diss}} \rangle = -\langle \dot{Q} \rangle \sim \left\langle F_a \circ \frac{da}{dt} \right\rangle + \left\langle F_m \circ \frac{dm}{dt} \right\rangle, \quad (6.1)$$

where  $\circ$  indicates that the product of the random variables is to be taken in the Stratonovich sense. The choice of Stratonovich interpretation in (6.1) is made such that the standard rules of calculus can be used. The above expression is numerically computed using the following discretization,

$$-\langle \dot{Q}_i \rangle \sim \left\langle F_a(a_{i+1/2}, m_{i+1/2}) \cdot \frac{a_{i+1} - a_i}{\Delta t} \right\rangle + \left\langle F_m(a_{i+1/2}, m_{i+1/2}) \cdot \frac{m_{i+1} - m_i}{\Delta t} \right\rangle, \quad (6.2)$$

where  $x_{i+1/2} \equiv \frac{x_{i+1} + x_i}{2}$  for  $x \in \{a, m\}$ . On examination of (6.2), it can be seen that when the state of the system (determined by coordinates  $(a, m)$ ) is near the boundary of the phase-space, this expression is prone to boundary effects, hence, fails to determine the dissipation correctly.

Figure 6.1 shows how the phase-space boundary affects the computation of dissipation.



**Figure 6.1: Effect of reflective boundaries:** Phase-space interior  $\Omega$  with coordinates  $(m_i, a_i) = (m(t_i), a(t_i))$  and forces  $(F_m, F_a)$ . The phase-space coordinates at time  $t_n$  are completely determined by the coordinates and forces at  $t_{n-1}$  (ignoring the noise for now). The forces at time  $t_n$  push the state of the system outside the phase-space, however the boundaries reflect the motion and hence the new coordinates at time  $t_{n+1}$  are determined as  $(\mathcal{R}(m_{n+1}), \mathcal{R}(a_{n+1})) = (\tilde{m}_{n+1}, \tilde{a}_{n+1})$ , where  $\mathcal{R}$  is the reflection operator.

Due to the explicit nature of the used numerical method, the coordinates at time  $t = t_{n+1}$  i.e.  $(m_{n+1}, a_{n+1})$  are determined by the forces  $(F_m(t_n), F_a(t_n))$ . However, since these coordinates lie outside the phase-space, they are reflected back into the domain  $\bar{\Omega}$  due to reflective action of the boundaries. Hence, the resulting coordinates at time  $t = t_{n+1}$  are  $(\tilde{m}_{n+1}, \tilde{a}_{n+1}) \neq (m_{n+1}, a_{n+1})$ . Therefore, the coordinates determined by the force at time  $t_n$  are not equal to the actual coordinates. Furthermore, the actual coordinates can be considered as a result of an unknown force, which accounts for the effect of the boundary and the actual force at  $t = t_n$ . This unknown force results in an inconsistency in the computation of the heat exchange when using the reflection operators.

## 6.2 Boundary Potential

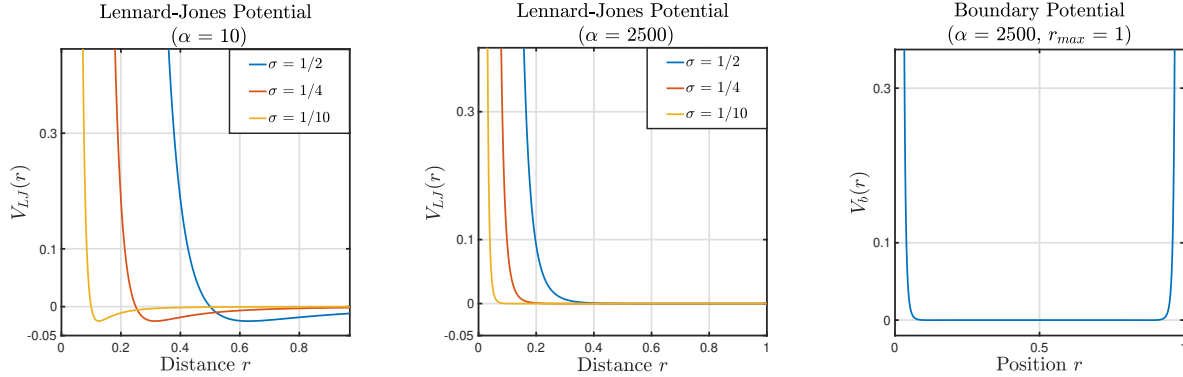
The inconsistency in the computation of the heat exchange is due to the reflective nature of the boundaries. This problem can be resolved by modelling the reflectivity of the boundaries by means of a repulsive potential at the boundaries instead of using the reflection operators.

### 6.2.1 Using Lennard-Jones Potential

The reflective nature of the walls of the domain can be modeled using a potential similar to the Lennard-Jones potential [61], and called here Lennard-Jones or LJ potential for simplicity,

$$V_{LJ}(r) = 4\epsilon \left( \left( \frac{\sigma}{r} \right)^6 - \left( \frac{\sigma}{r} \right)^3 \right), \quad (6.3)$$



(a) Lennard-Jones potential for small  $\alpha$ .(b) Lennard-Jones potential for large  $\alpha$ (c) Boundary potential for large  $\alpha$  and  $\sigma = 1/10$ 

**Figure 6.2: Lennard-Jones and boundary potential:** (a, b) As  $\sigma$  becomes smaller, the potential shifts towards left i.e., the minima of the potential becomes smaller. Also, for small  $\alpha$  the minimum value is large enough (with negative sign) to be visible clearly. However, for large  $\alpha$ , the amplitude of minimum value becomes vanishingly small. (c) The boundary potential is constructed by superimposing the LJ potential and its mirror image.

where  $r$  is the distance of the particle from the wall. Figure 6.2a presents the LJ potential for different values of  $\sigma$  and for a fixed value of  $\alpha \equiv 1/4\epsilon$ . As the distance  $r$  becomes smaller, the potential first becomes negative, reaches a minimum and then grows to infinity. The distance at which the function takes negative values depends on the parameter  $\sigma$  and the amplitude of the minimum value can be scaled by changing  $\alpha$  (see Fig. 6.2b).

Figure 6.2b indicates that as  $\sigma \rightarrow 0$  and  $\alpha \rightarrow \infty$ , the LJ potential becomes an infinite wall potential,

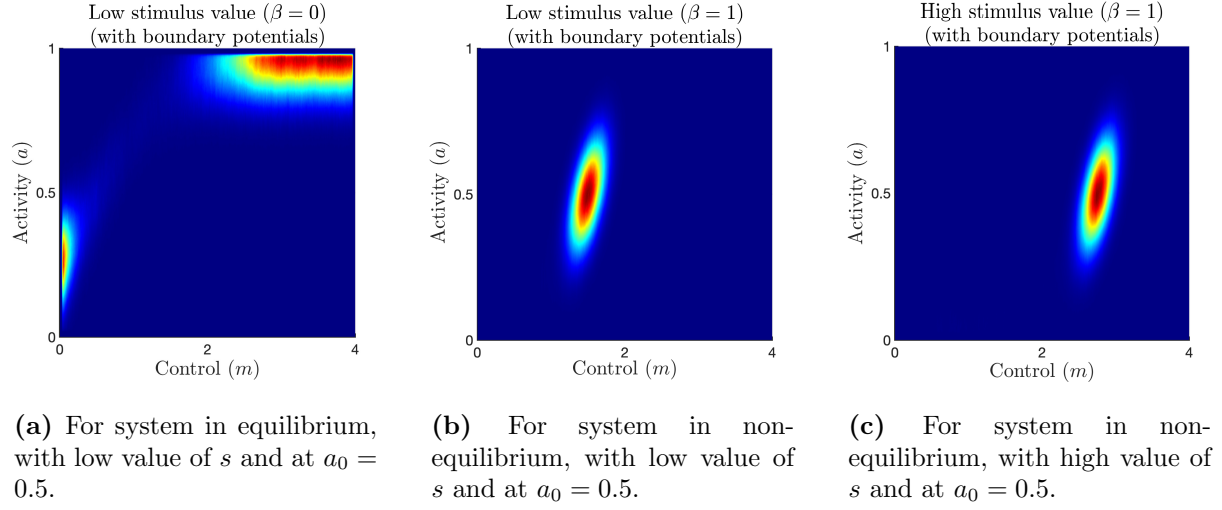
$$\lim_{\substack{\alpha \rightarrow \infty \\ \sigma \rightarrow 0}} V_{LJ}(r) = \begin{cases} \infty, & r = 0 \\ 0, & r > 0. \end{cases}$$

To have a potential wall on both the sides of the domain, a mirrored LJ potential can be super-imposed with the LJ potential [62],

$$\begin{aligned} V_b(r, r_{\max}) &\equiv V_{LJ}(r) - V_{LJ}(r_{\max} - r) \\ &= \frac{1}{\alpha} \left( \left( \frac{\sigma}{r} \right)^6 - \left( \frac{\sigma}{r} \right)^3 \right) - \frac{1}{\alpha} \left( \left( \frac{\sigma}{r_{\max} - r} \right)^6 - \left( \frac{\sigma}{r_{\max} - r} \right)^3 \right), \end{aligned} \quad (6.4)$$

where  $r_{\max}$  is the maximum value of  $r$  and  $V_b$  is the boundary potential which is a function of position  $r$  in a one-dimensional system. Figure 6.2c presents  $V_b(r, r_{\max})$  for  $r \in [0, 1]$  and  $r_{\max} = 1$ .

**Remark:** For any finite  $\alpha > 0$ ,  $V_b$  will always become negative before it starts to grow. However, the amplitude of the minimum value approaches zero as  $\alpha \rightarrow \infty$ . For the simulations results presented in this thesis,  $\alpha$  is chosen to be large enough, such that even though



**Figure 6.3: Probability distribution using boundary potentials:** The results of numerical solution using boundary potentials nearly perfectly reproduce the results in Fig. 5.3. The system parameters are the same as in Fig. 5.3. *Parameters for boundary potentials:*  $\sigma = 1/15, \alpha = 2500$ .

*the minimum is negative, the amplitude of the minimum is inconsiderable in comparison to the other potentials.*

For the stochastic system (5.1), the boundary potential can be constructed using a linear combination of the boundary potential for the activity and the control element,

$$V(a, m) = V_b(a, a_{\max}) + V_b(m, m_{\max}), \quad (6.5)$$

where  $a_{\max} \equiv 1$  and  $m_{\max} \equiv m_0$ .

Figure 6.3 presents the simulation results when the reflection operators are replaced by the boundary potentials. The results in Fig. 6.3 agree very well with the ones in Fig. 5.3. The only difference is in the equilibrium case, where there is a small gap in the boundary and the probability distribution, which is because  $\sigma \neq 0$ . One could choose  $\sigma$  to be closer to zero, however, this increases the computation time since a smaller  $\Delta t$  is also required.

### 6.3 Stochastic Energetics

In Section 6.1, an expression for the dissipation was discussed. A more precise equation for energy dissipation in the stochastic system can be obtained using Sekimoto's formalism (see Section 3.4). To make use of Sekimoto's framework, it is required to have a functional form for the potential that governs the dynamics of the system in equilibrium. Using this approach also allows one to include the boundary potential in the expression for heat exchange.

### 6.3.1 Equilibrium Potential

The *equilibrium potential*  $\Phi(a, m, s)$  can be constructed by integrating  $F_a$  and  $F_m$ ,

$$\begin{aligned}\Phi &= \frac{-1}{\omega_a} \int F_a da = \int (a - G(s, m)) da \\ &= \frac{a^2}{2} - G(s, m)a + c(s, m),\end{aligned}\tag{6.6}$$

where  $c(s, m)$  is a constant in  $a$  and results from indefinite integration. Also, since the system is in equilibrium,  $\beta = 0$ ,

$$\begin{aligned}\Phi &= \frac{-1}{\omega_m} \int F_m dm = - \int (a - a_0) \partial_m G(s, m) dm \\ &= a_0 G(s, m) - a G(s, m) + d(a, s),\end{aligned}\tag{6.7}$$

where  $d(a, s)$  is the constant of integration. Since both (6.6) and (6.7) must be equal, it can be concluded that  $c(m, s) = a_0 G(s, m)$  and  $d(a, s) = a^2/2$ . Therefore,

$$\Phi(a, m, s) = \frac{a^2}{2} - (a - a_0)G(s, m),\tag{6.8}$$

which satisfies the following relations,

$$-\omega_a \frac{\partial \Phi}{\partial a} = F_a,\tag{6.9a}$$

$$-\omega_m \frac{\partial \Phi}{\partial m} = F_m.\tag{6.9b}$$

To account for the reflective nature of the boundary, the boundary potential can be reconstructed using the LJ potential and the adaptation rates, and added to the equilibrium potential to obtain the total equilibrium potential,

$$\tilde{\Phi}(a, m, s) = \frac{a^2}{2} - (a - a_0)G(s, m) + \tilde{V}(a, m)\tag{6.10}$$

where

$$\tilde{V}(a, m) = \frac{1}{\omega_a} V_b(a, a_{\max}) + \frac{1}{\omega_m} V_b(m, m_{\max}).\tag{6.11}$$

Hence, the stochastic system (in equilibrium) that takes into account the boundary potential is,

$$\frac{da}{dt} = -\omega_a \frac{\partial \tilde{\Phi}}{\partial a} + \eta_a(t) = F_a + f_a + \eta_a(t),\tag{6.12a}$$

$$\frac{dm}{dt} = -\omega_m \frac{\partial \tilde{\Phi}}{\partial m} + \eta_m(t) = F_m + f_m + \eta_m(t),\tag{6.12b}$$

where,  $f_i \equiv -\partial_i V_b(i, i_{\max}) = -\omega_i \partial_i \tilde{V}(a, m)$  for  $i \in \{a, m\}$ .

**Remark:** The boundary potential is a conservative potential, introducing it in (5.1) does not violate or change the detailed balance condition (5.4).

### 6.3.2 Energetics

To compute the steady-state dissipation in the system, the stochastic system (6.12) is written as a stochastic differential equation (Langevin equation) for a two-dimensional system

$$\frac{d\mathbf{x}}{dt} = \mathbf{F} + \mathbf{f} + \boldsymbol{\eta}(t) \quad (6.13)$$

where,

$$\mathbf{x} = (a, m)^T, \quad (6.14a)$$

$$\mathbf{F} = (F_a, F_m)^T, \quad (6.14b)$$

$$\mathbf{f} = (f_a, f_m)^T, \quad (6.14c)$$

$$\boldsymbol{\eta}(t) = (\eta_a(t), \eta_m(t))^T. \quad (6.14d)$$

The total force vector  $\mathcal{F}$  can be defined as,

$$\mathcal{F} \equiv \mathbf{F} + \mathbf{f} = \mathbf{F}_{nc} + \mathbf{F}_c \quad (6.15)$$

where  $\mathbf{F}_c$  and  $\mathbf{F}_{nc}$  are the conservative and non-conservative parts of the total force  $\mathcal{F}$ , respectively,

$$\mathbf{F}_{nc} = \begin{pmatrix} 0 \\ F_{m(nc)} \end{pmatrix} = \begin{pmatrix} 0 \\ \beta F_m^{neq} \end{pmatrix}, \quad (6.16a)$$

$$\mathbf{F}_c = \begin{pmatrix} f_a + F_a \\ f_m + F_{m(c)} \end{pmatrix} = \begin{pmatrix} f_a + F_a \\ f_m + (1 - \beta)F_m^{eq} \end{pmatrix}, \quad (6.16b)$$

where  $F_{m(c)}$  and  $F_{m(nc)}$  are the conservative and non-conservative parts of  $F_m$ . In (6.16),  $F_m^{neq}$  and  $F_m^{eq}$  are as in Section 5.1.

For this two-dimensional system, the rate matrix can be defined as

$$M = \begin{pmatrix} \omega_a & 0 \\ 0 & \omega_m \end{pmatrix}. \quad (6.17)$$

Since there is no inertia, the change in internal energy equals the change in potential energy. Therefore,

$$\begin{aligned} dU = d\Phi &= \frac{\partial \Phi}{\partial s} ds + \nabla_{\mathbf{x}} \Phi \cdot d\mathbf{x} \\ &= \frac{\partial \Phi}{\partial s} ds + (M^{-1} M \nabla_{\mathbf{x}} \Phi) \cdot d\mathbf{x} \\ &= \frac{\partial \Phi}{\partial s} ds + M^{-1} (M \nabla_{\mathbf{x}} \Phi \cdot d\mathbf{x} + \mathbf{F}_{nc} \cdot d\mathbf{x} - \mathbf{F}_{nc} \cdot d\mathbf{x}) \\ &= \frac{\partial \Phi}{\partial s} ds + M^{-1} \mathbf{F}_{nc} \cdot d\mathbf{x} - M^{-1} \underbrace{(-M \nabla_{\mathbf{x}} \Phi + \mathbf{F}_{nc})}_{=\mathbf{F}_c} \cdot d\mathbf{x} \\ &= \underbrace{\frac{\partial \Phi}{\partial s} ds + M^{-1} \mathbf{F}_{nc} \cdot d\mathbf{x}}_{dW} - \underbrace{M^{-1} \mathcal{F} \cdot d\mathbf{x}}_{dQ}. \end{aligned} \quad (6.18)$$

The last line of (6.18) is the First law of thermodynamics as in Sekimoto's formalism (as in 3.29). Hence, the heat exchange and therefore the dissipation is given by,

$$-\dot{Q} = \frac{1}{\omega_a}(F_a + f_a) \circ \frac{da}{dt} + \frac{1}{\omega_m}(F_m + f_m) \circ \frac{dm}{dt} = \dot{W}_{\text{diss}} \quad (6.19)$$

where  $\circ$  represents the product in the Stratonovich sense.

It should be noted that (6.19) is the dissipation for an individual trajectory. The statistical average of (6.19) is equivalent to the right-most term in (5.23). This equivalence can be observed by using (3.18) and (3.22) where  $T = T_{\text{eff}}$  and  $dx/dt = v_x$  for  $x \in \{a, m\}$ .



# Chapter 7

## Role of Phase-Space Boundaries in Adaptation

### 7.1 Introduction

In Ref. [15], Lan and Tu compute an energy-speed-accuracy (ESA) trade-off relation (5.30) for sensory adaptation in cells. They do this by means of a minimal three-node feedback network (see Fig. 5.1). To compute the energy dissipation they use (5.24) and approximate the integrals using the timescale separation assumption to obtain (5.25). Additionally, they show that the adaptation error  $\epsilon$  has three key contributions (5.28). Two of which are due to the boundaries of the phase-space and the third contribution arises in the bulk of the domain, which is zero only if the system is in a fully-adaptive state. With the functional form of dissipation and adaptation error known, Lan and Tu were able to relate the two quantities to obtain the ESA relation.

In this chapter, I present how the steady-state dissipation and adaptation error of the model in Ref. [15] behave in a slightly different setting – when the effective temperatures of the system are equal. That is, when  $C = 1$ .

Moreover, the numerical results presented in this chapter are used to explain the significance of the phase-space boundaries in this model. It is shown that the domain boundaries are essential in quantifying the adaptation capability of the system. Specifically, it will be seen that as the boundaries becomes relevant, the system is unable to adapt and concomitantly it loses its capacity to dissipate energy.

Additionally, it is shown that as the system gets closer to the phase-space boundaries the analytical approximation for dissipation suggested in Ref. [15] i.e.,

$$\dot{W}_{\text{diss}} \approx \dot{W}_{\text{diss(a)}} \equiv T_{\text{eff}} \frac{\omega_m}{C} \beta^2 (1 + CG'(s, m^*))^2, \quad (5.22)$$

(where ‘a’ in  $\text{diss}(a)$  stands for analytical) disagrees with the numerical results obtained using the statistical mean of

$$\dot{W}_{\text{diss}} = \frac{1}{\omega_a}(F_a + f_a) \circ \frac{da}{dt} + \frac{1}{\omega_m}(F_m + f_m) \circ \frac{dm}{dt}. \quad (6.19)$$

This inconsistency arises due to the limitations of the steepest descent method, which is used to compute (5.25) i.e., the validity of this approximation method becomes questionable near the phase-space boundaries. Therefore the ESA, (5.30) does not hold when the system variables are driven towards the boundaries of the phase-space.

Lastly, the insights gained throughout this chapter are used to conjecture a functional form for the numerically computed dissipation (given by (6.19)), which appears to hold true for a larger subset of the phase-space than the one suggested in Ref. [15] (provided the boundary at  $m = 0$  is ignored).

## 7.2 Significance of Domain Boundaries

Consider the stochastic system (5.1) in a fully-adaptive state ( $\beta = 1$ ). In this state, the system is away from equilibrium therefore, the probability currents of the system are non-zero. Furthermore, the current vectors for the system form a vortex which encloses the adapted state of the system (see Fig. 6.3b). The location of the vortex in the phase-space  $\bar{\Omega}$  is governed by the input  $s$  and the setpoint  $a_0$ . That is, provided that  $a_0$  and  $s$  are known,  $m^*$  is completely determined.

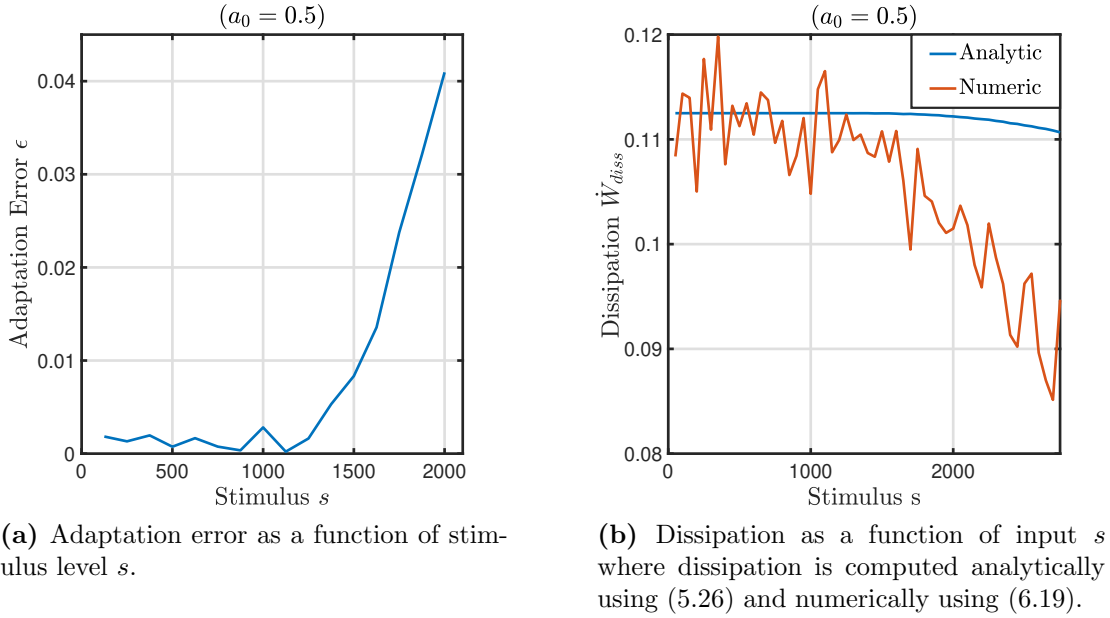
**Remark:** *In the language of control theory, an  $a_0$  like parameter, which the output of the system/plant tries to track, is called the setpoint.*

For a fixed setpoint, the *adapted state*  $(a, m) = (\langle a \rangle = a_0, m^*)$  moves along the positive  $m$ -axis as  $s$  is increased (see Figs. 6.3b and 6.3c). As long as the probability distribution does not touch the phase-space boundaries, the stochastic dynamics for a fully-adaptive system always relaxes to the adapted state i.e.,  $\langle a \rangle = a_0$  and as a result  $\epsilon = 0$ .

**Remark:** *This discussion assumes the system does not interact with the boundaries at  $a = 0, 1$ . This assumption is largely only true when  $a_0 = 1/2$  (i.e.,  $a$  can reach the boundary but it is very unlikely). For any other value of the setpoint, the probability distribution tends to touch the phase-space boundaries at  $a = 0, 1$  irrespective of the value of the input  $s$ .*

However, when  $s$  is large enough such that the boundary at  $m = m_0$  becomes relevant, the shape of the probability distribution deforms due to the reflective nature of the boundaries. This deformation in the probability distribution disrupts the flow of the probability fluxes, which changes the dynamics of the system. In the case when  $a_0 = 1/2$ , the distribution is exactly in the middle along the  $a$ -axis. Therefore, for a large enough  $s$ , only the boundary





**Figure 7.1: System behavior for varying stimulus level:** (a) For low stimulus level the adaptation error is zero (up to numerical precision). However, as  $s$  increases beyond a threshold value the adaptation accuracy decreases with increase in  $s$ . (b) As stimulus level exceeds a threshold value the capacity of the system to dissipate gets lowered. This behavior is true for both analytically and numerically computed dissipation. Moreover, computation using both the approaches agree for small values of  $s$ , however, the numerical results falls much faster than the analytical ones for large stimulus levels. *Simulation parameters: as in Fig. 6.3.*

at  $m = m_0$  is relevant. When the system interacts with this boundary, the deformation in the probability distribution alters the location of  $\langle a \rangle$ , making  $\langle a \rangle \neq a_0$ , which results in a non-zero adaptation error.

Figure 7.1a shows that the adaptation error is vanishingly small for small values of  $s$  and then starts to grow as  $s$  becomes larger than a threshold value ( $s_0 \approx 1200$  in Fig. 7.1a). The *input threshold*  $s_0$  can be defined as the value of  $s$  at which the probability distribution of the system first touches the phase-space boundary at  $m = m_0$ .

In other terms, for a fixed  $a_0$ , when the input  $s$  is changed the system tries to set the value of  $m$  such that, as  $m$  relaxes to  $m^*$  (on average), the mean output activity  $\langle a \rangle$  relaxes to setpoint  $a_0$ . However, in the case where  $s$  becomes larger than  $s_0$ , the required value of  $m$  for accurate adaptation tends to lie outside the bulk of  $\bar{\Omega}$  (for very large  $s$ , the required value of  $m^*$  can also be outside  $\bar{\Omega}$ ). That is, the change in the concentration of the chemo-attractant becomes so large that the required methylation level for adaptation is larger than the number of methylation sites in the receptor.

At first, the adaptation error becomes non-zero due to the reflective nature of the boundary at  $m = m_0$ , which deforms the distribution making it impossible for the system to get to the required value of  $m^*$ . Consequently, the output activity of the system is unable to reach the setpoint  $a_0$ . As a result, the system starts to lose its capacity to adapt perfectly. Moreover, if  $s$  is increased further such that the required  $m^*$  value lies outside the phase-space, the best the system can do is to set  $m^*$  to the boundary value  $m_0$  (which is smaller than the required control level). Hence, causing  $\langle a \rangle$  to deviate further away from  $a_0$ . As a result, increasing the adaptation error.

Figure 7.1a shows that beyond a certain value of  $s$ , the adaptation error  $\epsilon$  increases for an increase in the input value. Additionally, from Fig. 7.1b it can be observed that, as the system loses its capacity to adapt, the amount of energy it dissipates also reduces. Therefore, the adaptive performance of the system appears to be directly related to the steady-state dissipation in the system.

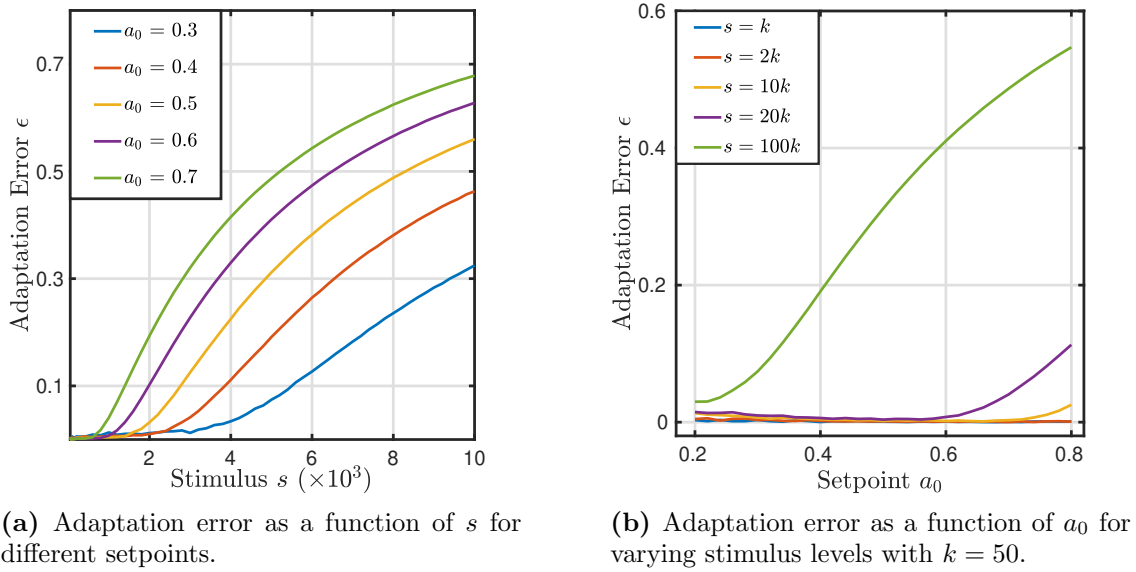
However, for  $s < s_0$ , the behavior in Figs. 7.1a and 7.1b suggest that even if the external stimulus to the system changes, the system is able to adapt (nearly perfectly). This adaptation occurs without any changes in the energy cost, as long as the phase-space boundaries do not intervene.

**Remark:** *Another parameter that is crucial in this model is the effective temperature. The effective temperature determines the variance of the Gaussian noise which characterizes the size of the probability distribution. For simplicity reasons, the effective temperature in this thesis is kept fixed.*

### 7.3 Different Setpoints

In the previous section, the behavior of adaptation accuracy and energy dissipation was discussed for varying stimulus levels. The results of numerical simulations showed that for a fully adaptive system, the system adapted nearly perfectly for small input values (i.e.,  $s < s_0$ ). However, for value of  $s$  larger than a threshold value ( $s_0$ ), the adaptive performance and the steady-state dissipation decreased dramatically with the increase in  $s$ . Therefore, implying a possible trade-off between the adaptation accuracy and the steady-state dissipation at the phase-space boundaries.

In this discussion, it was required that one ignores the presence of the boundaries at  $a = 0, 1$  i.e., the boundaries at the top and the bottom of the rectangular domain  $\bar{\Omega} = [0, 1] \times [0, m_0]$ , which was achieved by fixing  $a_0$  to  $1/2$ . By setting  $a_0 = 1/2$ , the probability distribution becomes localized exactly at the middle of the  $a$ -axis, which minimizes the effect of the domain boundaries at the top and bottom. However, relaxing this constraint by allowing the setpoint to change brings up some interesting features of this system which will be discussed further in this section.



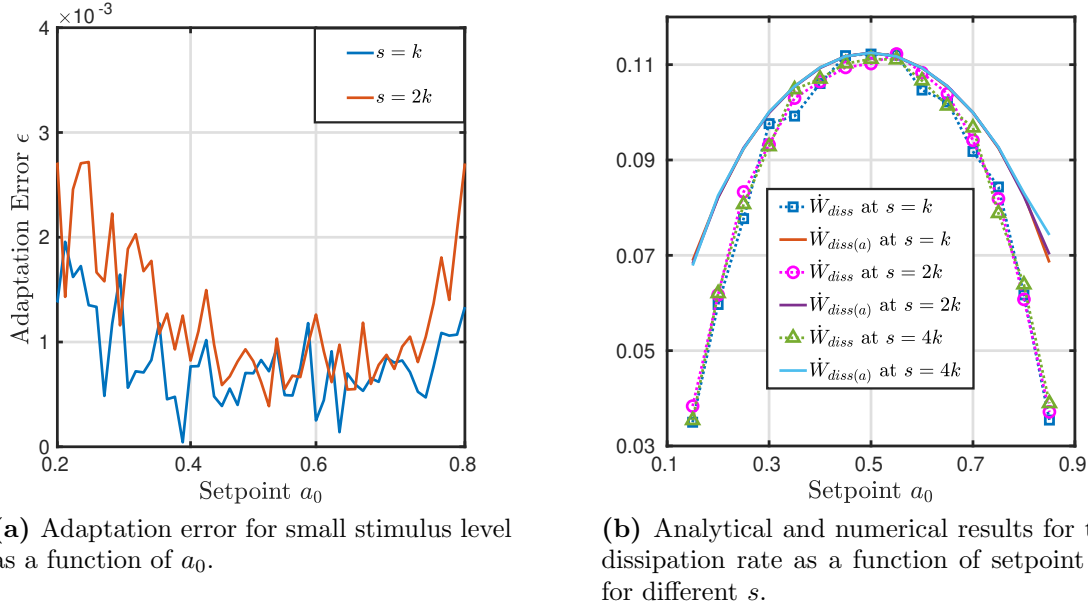
**Figure 7.2: Adaptation for different setpoints and stimulus levels:** (a) For a fixed  $s$ , the adapted state moves along positive  $m$ -axis, when  $a_0$  is increased. As a result, the adaptation accuracy for a fixed input decreases as  $a_0$  increases. (b) For high stimulus levels ( $s = 100k$ ) the adaptation error is non-zero for all  $a_0$ . However, for smaller  $s$  the error corresponding to each  $a_0$  is smaller. *Simulation parameters: as in Fig. 6.3.*

### 7.3.1 Error in Adaptation

Based on Figs. 6.3b and 6.3c it was argued that the adapted state of the system moves along the  $m$ -axis as the stimulus level is increased. However, if  $s$  is fixed,  $m^*$  can also be changed by altering  $a_0$ . Therefore, the adapted state (and the vortex) moves by changing the setpoint, even if the input is fixed.

It is trivially true that the setpoint determines the location of adapted state along the  $a$ -axis. But for a fixed input, if  $a_0$  is increased then  $m^*$  also increases, which translates the adapted state diagonally (like a sigmoidal) towards the top-right corner of the phase-space. This behavior is directly determined from the phase-space dynamics of the system (cf. Fig. 5.2b) and is only true as long as the system is away from the domain boundaries at  $m = 0, m_0$  (i.e., in the *bulk* of the phase-space).

Figure 7.2a shows that for larger setpoints, the input at which the error starts to grow is smaller i.e., if  $a_0$  becomes larger the adapted state reaches the boundary at  $m = m_0$  faster. Implying that, for a larger setpoint, a smaller input is required to make the adaptation error non-zero. Whereas, Fig. 7.2b exhibits that if  $s$  is large, the adaptation error is non-zero for all  $a_0$ . However, if the input is small ( $< 10k$ ), the adaptation error is vanishingly small as long as  $a_0$  is small ( $< 0.7$ ).



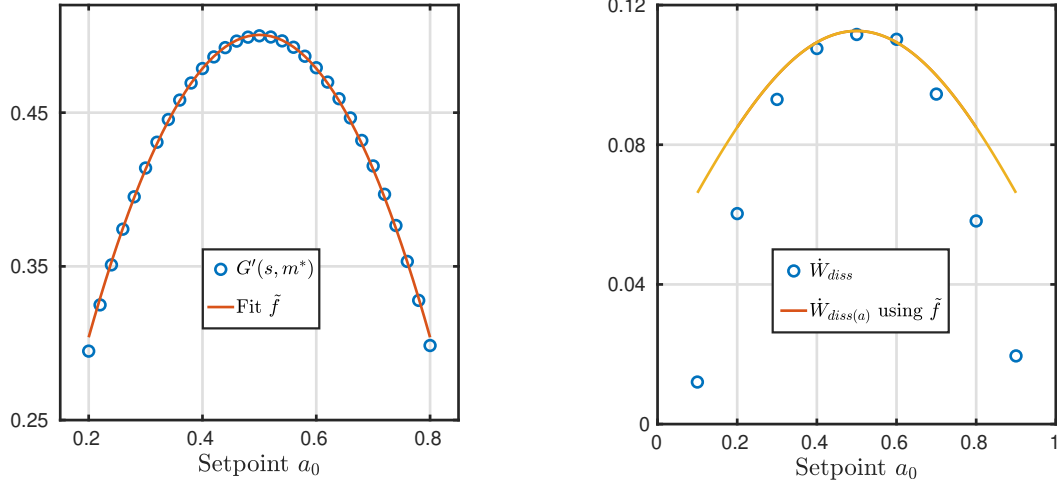
**Figure 7.3: Adaptation accuracy and dissipation in the bulk:** (a) In the bulk (i.e away from boundaries at  $m = 0, m_0$ ) the adaptation accuracy maximizes when  $a_0 \approx 1/2$ . (b) For  $s < s_0$ , both computations for dissipation approximately agree for  $a_0 \in \mathcal{V} := [0.3, 0.7]$ , but disagree when  $a_0 \notin \mathcal{V}$ . Simulation parameters: as in Fig. 6.3.

Moreover, Fig. 7.3a displays the dependence of adaptation error on the setpoint when the input is small. For these values of  $s$ , the effect of the boundary at  $m = m_0$  becomes less prominent. Therefore, the influence of the boundaries along  $a$ -axis becomes prevalent. It is observed that the adaptation error is minimized when  $a_0 \approx 1/2$  and it grows rather symmetrically as  $a_0$  moves away from the centre of its domain  $[0, 1]$ .

### 7.3.2 Dissipation for Weak Stimuli

It was presented in Section 7.2 that the adaptation error of the system seems to be inversely proportional to the dissipation in the system. Therefore, after studying the dependence of adaptation accuracy on the setpoint  $a_0$ , the behavior of dissipation as a function the setpoint can be predicted. That is for  $s < s_0$ , since the error minimizes at  $a_0 \approx 1/2$ , the steady-state dissipation should approach a maximum when the setpoint is near the centre of its domain.

Figure 7.3a illustrates the behavior of adaptation error in the bulk of the phase-space ( $s < s_0$ ). Correspondingly, Fig. 7.3b presents the dissipation as a function of the setpoint with the values of  $s$ , such that the adapted state is localized in the bulk of the phase-space. As a consequence, Figs. 7.3a and 7.3b effectively capture the effect of the phase-space



(a) Fit  $\tilde{f}$  in (7.2) and  $G'(s, m^*)$  as a function of setpoint  $a_0$  at  $s = 100$ . Fit parameters:  $\alpha = 0.045$  and  $r = 2.18$ .

(b) Comparing numerical results with analytical approximation of dissipation using (7.3) at  $s = 100$ .

**Figure 7.4: Fit for  $G'(s, m^*)$ :** (a) The function  $\tilde{f}(a_0)$  almost perfectly reproduces the dependence of  $G'(s, m^*)$  on  $a_0$ . (b) The two plots for dissipation agree for the subset  $\mathcal{V} \subset [0, 1]$ . Outside  $\mathcal{V}$ , the numerical dissipation decreases faster than the analytical approximation. *Simulation parameters: as in Fig. 6.3.*

boundaries at  $a = 0, 1$  on the dynamics of the system.

Figure 7.3b presents numerical and analytical dissipation as a function of the setpoint for different values of  $s$ . It is seen that the dissipation maximizes around  $a_0 \approx 1/2$  (where  $\epsilon$  minimizes) and decreases as the boundaries along the  $a$ -axis come into play. Moreover, since changes in  $s$  do not move the adapted state along the  $a$ -axis, the dissipation in the bulk is independent of  $s$ . This behavior is shown by both (5.26) and (6.19) independently.

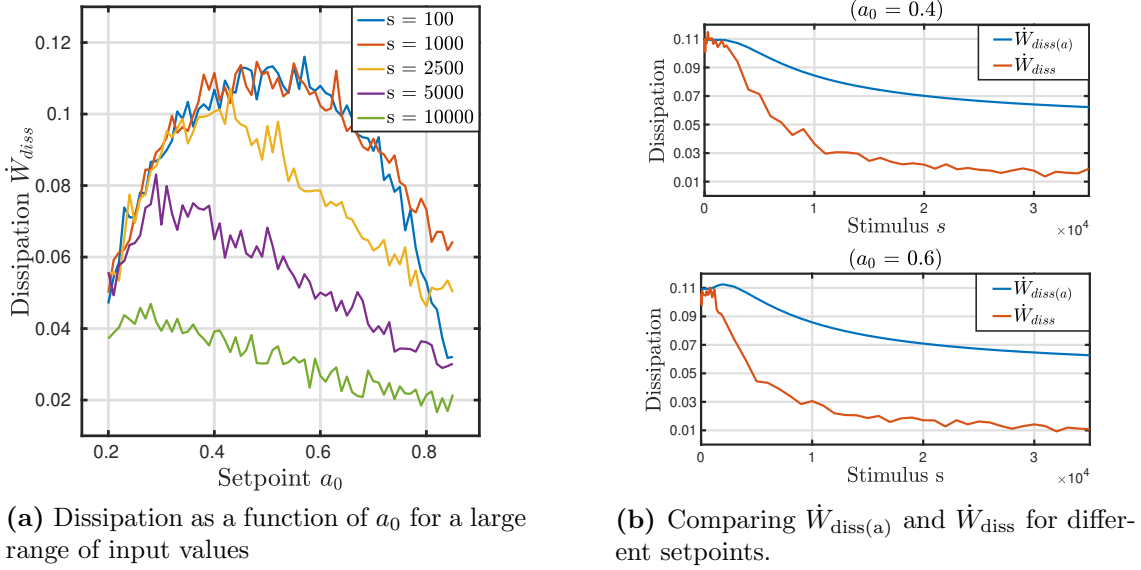
Furthermore, Fig. 7.3b also indicates that in the bulk of the domain,  $\dot{W}_{diss} \approx \dot{W}_{diss(a)}$  for  $a_0 \in \mathcal{V} := [0.3, 0.7]$  i.e., when the effect of boundaries at  $a = 0, 1$  is insignificant. For any  $a_0 \notin \mathcal{V}$ , the two equations for dissipation disagree.

Since the dissipation in the bulk only seems to depend on  $a_0$ , carefully examining (5.26) suggests that this behavior must be due the dependence of  $G'(s, m^*)$  on  $a_0$ . Further numerical investigation reveals that  $G'(s, m^*)$  depends on  $a_0$  via the following logistic-map like fit (see Fig. 7.4a),

$$G'(s, m^*) \approx \tilde{f}(a_0), \quad (7.1)$$

where

$$\tilde{f}(\psi) = \alpha + r\psi(1 - \psi) \quad \text{for } \psi \in [0, 1], \quad (7.2)$$



**Figure 7.5: Dissipation for large input range:** (a) For small values of  $s$ , the dependence of dissipation on  $a_0$  is as in Fig. 7.6. However, for larger values of  $s$  the shape of the function completely changes due the effect of the boundary at  $m = m_0$ . (b) The numerical and the analytical approximation for the dissipation agree with each other when  $s$  is small. But they disagree as  $s$  becomes larger than  $s_0$ .

with  $\alpha$  and  $r$  as real-valued fit parameters. Therefore, the analytically obtained dissipation can be approximated as,

$$\dot{W}_{diss(a)} \approx \omega_m T_{eff} (1 + \tilde{f}(a_0))^2. \quad (7.3)$$

**Remark:** This fit for  $G'(s, m^*)$  holds true for a wide range of Hill-coefficients  $H$  in  $G(s, m)$ , with different fit parameters for each  $H$ .

The results in Fig. 7.4b compare the numerical results with the analytical approximation obtained using (7.3). The results in this figure agree with the ones in Fig. 7.3b. In addition, for  $a_0 \in \mathcal{V}$  and  $s$  such that  $m^*$  lies in the bulk of  $\bar{\Omega}$ , the following approximation holds,

$$\dot{W}_{diss} \approx \dot{W}_{diss(a)} \approx \omega_m T_{eff} (1 + \tilde{f}(a_0))^2. \quad (7.4)$$

This equation indicates that in a given subset of the phase-space, the dissipation rate is independent of  $s$  and is completely determined by the setpoint  $a_0$ .

### 7.3.3 Dissipation Near Boundaries

In the previous section, the behavior of steady-state dissipation for a weak stimulus was studied. In this limit, the dissipation becomes strongly dependent on the choice of setpoint  $a_0$ . However, this functionality changes considerably when the adapted state moves out of

the bulk and the effect of boundary at  $m = m_0$  becomes dominant.

Figure 7.5a presents steady-state dissipation as a function of the setpoint for different stimuli. For small values of  $s$  (i.e.,  $s = 100, 1000$ ), the behavior in Fig. 7.5a is well captured by (7.3). However, for larger stimulus values ( $s > s_0$ ) i.e., when the boundary at  $m = m_0$  comes into play, the numerical and analytical results disagree. Figure 7.5b shows the disagreement between the two equations, by presenting a comparison between  $\dot{W}_{\text{diss}}$  and  $\dot{W}_{\text{diss(a)}}$  for a wide range of input values at  $a_0 = 0.4$  and  $0.6$ .

Based on the results of the previous sections of this chapter and by carefully analyzing the data in Fig. 7.5b it is possible to construct a functional form for the dissipation, the validity of which extends outside the bulk of the phase-space. That is, for  $a_0 \in \mathcal{V}$ ,

$$\dot{W}_{\text{diss}} \approx \begin{cases} \dot{W}_{\text{diss(a)}} & s < s_0 \\ K_0 e^{-\lambda(s-s_0)} + \epsilon_0 (1 - e^{-\lambda(s-s_0)}) & s \geq s_0 \end{cases} \quad (7.5)$$

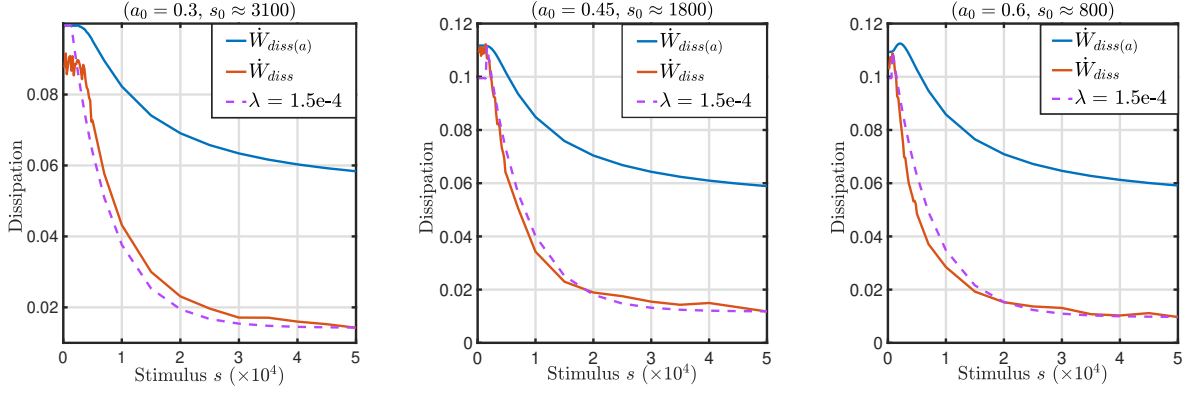
where  $K_0 = \dot{W}_{\text{diss(a)}}|_{s=s_0}$  is the *dissipation coefficient* and  $\epsilon_0 \equiv \min_{s \geq s_0} \dot{W}_{\text{diss}}$ . Figures 7.6a, 7.6b and 7.6c show that (7.5) very well approximates  $\dot{W}_{\text{diss}}$  for all  $s > 0$  and  $a_0 \in \mathcal{V}$  (ignoring the effect of the boundary at  $m = 0$ ). Furthermore, they also indicate that the parameter  $\lambda$  in (7.5) can be approximated to be independent of  $a_0$ , since the same  $\lambda$  is used for different  $a_0$  in Fig. 7.6.

**Remark:** *It can be assumed that in the case where there is no numerical error and when the boundary potential perfectly approximates reflective nature of the domain boundaries, the dissipation at very large  $s$  is vanishingly small. Therefore, it can be assumed that  $\epsilon_0 = 0$ .*

**Remark:** *Since the effective temperature determines the strength of the noise in (6.13), it is very likely that the parameter  $\lambda$  depends on  $T_{\text{eff}}$ .*

Moreover, understanding of the system's behavior in Section 7.3.1 already suggests that for an increasing  $a_0$ , the input threshold  $s_0$  must be lowered. The results in Fig. 7.7b confirm this claim. For Fig. 7.7b,  $s_0$  is chosen to be the value of  $s$  at which the adaptation error  $\epsilon$  becomes larger than 0.75%. This choice of constraint has been made to account for the adaptation error due to  $a_0$  being away from the centre of its domain. Additionally, Fig. 7.7a shows how  $K_0$  changes with the setpoint. This behavior is similar as for  $\dot{W}_{\text{diss(a)}}$  in Fig. 7.3b. However, this similarity is not surprising, since it is a consequence of the definition of  $K_0$ .

**Remark:** *In addition to  $a_0$ , it is expected that the input threshold  $s_0$  must depend on the choice of  $m_0$  as it characterizes the size of the system. This is because  $s_0$  depends on the distance between the location of the adapted state and the boundary at  $m = m_0$ .*

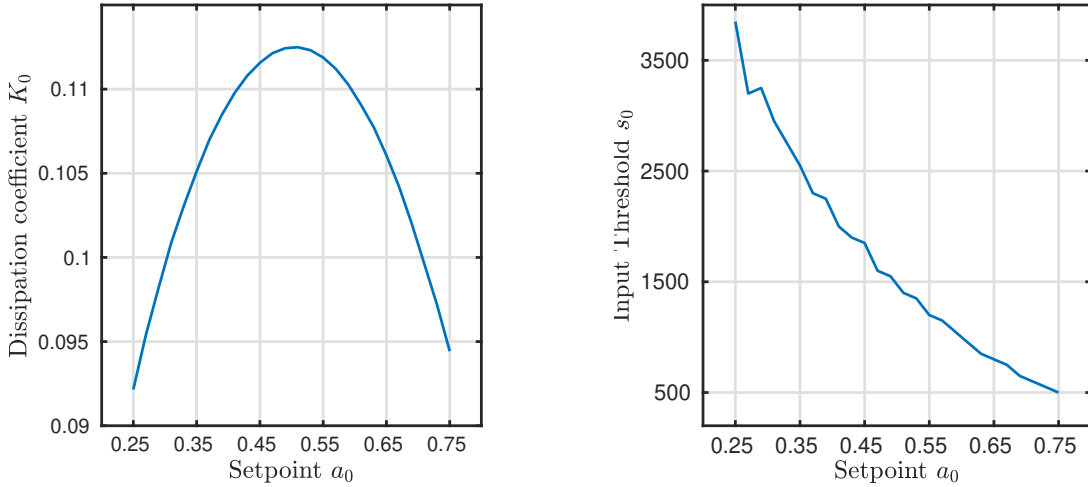


(a)  $\dot{W}_{diss}$ ,  $\dot{W}_{diss(a)}$  and (7.5) at  $\lambda = 1.5 \times 10^{-4}$  for varying stimulus values at  $a_0 = 0.3$ .

(b)  $\dot{W}_{diss}$ ,  $\dot{W}_{diss(a)}$  and (7.5) at  $\lambda = 1.5 \times 10^{-4}$  for varying stimulus values at  $a_0 = 0.45$ .

(c)  $\dot{W}_{diss}$ ,  $\dot{W}_{diss(a)}$  and (7.5) at  $\lambda = 1.5 \times 10^{-4}$  for varying stimulus values at  $a_0 = 0.6$ .

**Figure 7.6: Functional Approximation for  $\dot{W}_{diss}$ :** (a, b, c) The analytical approximation for dissipation (i.e.  $\dot{W}_{diss(a)}$ ) using (5.26) deviates further away from (6.19) as  $s$  becomes larger. However, the functional approximation (7.5) resembles the numerically computed dissipation very closely. The choice of  $s_0$  has been made using a constraint imposed on the error  $\epsilon$  i.e.,  $s_0 = s$  when  $\epsilon$  becomes larger than 0.75%.



(a) Dissipation coefficient  $K_0$  as a function of setpoint  $a_0$ .

(b) Input threshold  $s_0$  as a function of setpoint  $a_0$ .

**Figure 7.7: Dissipation coefficient and input threshold:** For both the figures, the choice of  $s_0$  is the same as in Fig. 7.6. (a) Dissipation coefficient  $K_0$  depends on the setpoint in the same way as indicated by (7.3). (b) The input stimulus  $s_0$  becomes smaller as setpoint is increased. The reason for this behavior is that the adapted state moves along the positive  $m$ -axis as  $a_0$  is increased.



# Chapter 8

## Interacting Systems

### 8.1 Introduction

Chapter 5 presents how Lan and Tu, make use of a three-node feedback network as a minimal model (5.1) to discuss sensory adaptation in a biological system (e.g., in a cell) [15]. In the context of this model, they managed to obtain an energy-speed-accuracy trade-off relation (5.30). However, it is natural to wonder whether such a cost-performance trade-off relation holds true when the system consists of more than one subsystem and they are allowed to interact with each other. In general, one can also question how the dissipation rate and the adaptation error behave for a system of interacting sensory systems (receptors in *E. coli*).

In this thesis two types of interactions are studied:

- *Type 1*: the activities of the two subsystems are coupled via a linear and symmetric interaction,
- *Type 2*: the methylation levels of the two receptors are coupled linearly.

However, before discussing the adaptive properties of interacting systems it is necessary to first investigate how one can define the adaptation error for such a system. Henceforth, defining the notion of overall activity and overall adaptation error.

#### 8.1.1 Overall Adaptation Error

The notion of adaptation error for a single non-interacting system (5.27) is rather simple. However, for a system of interacting subsystems, there is some freedom on how one can define the adaptation error.

When considering a system of two interacting subsystems; at first, there is a notion of individual adaptation error,

$$\epsilon_i \equiv \left| \frac{\langle a_i \rangle - a_0}{a_0} \right|, \quad i \in \{1, 2\} \quad (8.1)$$

where  $\epsilon_i$  and  $a_i$  correspond to the adaptation error and activity of subsystem  $i$ , respectively. However, when considering the system as a whole, the statistical mean of the *overall output activity* ( $a \in [0, 1]$ ) must track the setpoint  $a_0$ . Hence, the *overall adaptation error* of a system can be defined as,

$$\epsilon \equiv \left| \frac{\langle a \rangle - a_0}{a_0} \right|. \quad (8.2)$$

The freedom in the definition of  $\epsilon$  comes from the choice in how the overall activity  $a$  can be defined. One way to define the overall activity is by means of a norm. For instance, consider the two-norm i.e.  $a = \|\mathbf{a}\|_2 = \sqrt{a_1^2 + a_2^2}$  where  $\mathbf{a} = (a_1, a_2)^T$ . However, in the case when  $a_1 = a_2 = 1$ ,  $\|\mathbf{a}\|_2 \notin [0, 1]$ . Therefore, a more reasonable definition is given by the average of the two individual activities,

$$a = \frac{a_1 + a_2}{2}. \quad (8.3)$$

In addition to being simple, this definition also seems physically more appealing as the individual activities are stochastic variables and therefore, the overall activity of the system is the mean activity of the two receptors.

This definition for the overall output activity leads to an interesting consequence, i.e.,

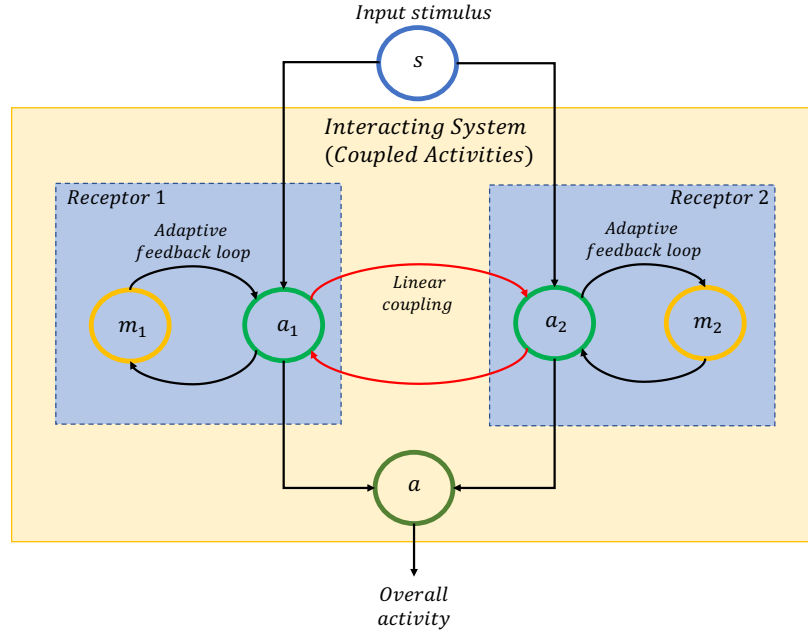
$$\begin{aligned} \epsilon &= \left| \frac{\langle a \rangle - a_0}{a_0} \right| \\ &= \left| \frac{\langle \frac{a_1 + a_2}{2} \rangle - a_0}{a_0} \right| = \left| \frac{\langle a_1 \rangle - a_0}{2a_0} + \frac{\langle a_2 \rangle - a_0}{2a_0} \right| \\ &= \frac{1}{2} \left| \frac{\langle a_1 \rangle - a_0}{a_0} + \frac{\langle a_2 \rangle - a_0}{a_0} \right| \leq \frac{1}{2} \left| \frac{\langle a_1 \rangle - a_0}{a_0} \right| + \frac{1}{2} \left| \frac{\langle a_2 \rangle - a_0}{a_0} \right|, \end{aligned}$$

where the linearity of the expectation value and the triangle inequality of the norm are employed. Then using (8.1) yields,

$$\epsilon \leq \frac{1}{2}(\epsilon_1 + \epsilon_2). \quad (8.4)$$

Equation (8.4) suggests that the mean of the two individual errors places an upper bound on the overall error of a system. Furthermore, (8.4) is independent of the type of interaction between the two subsystems and is purely due to the definition in (8.3).

**Remark:** Equation (8.4) can be trivially extended to a system with any finite number of subsystems (interacting or non-interacting).



**Figure 8.1: Interacting system with coupled activities:** The interacting system consists of two subsystems both consisting of an adaptive feedback loop. The activities of both the receptors are coupled via a linear symmetric coupling. Both the sensory systems are subject to the same stimulus  $s$  which generates output activities  $a_1$  and  $a_2$ , which together determine the overall activity  $a$ .

## 8.2 Coupling Activities

The simplest kind of interaction that can be introduced between two subsystems is the one that is symmetric and linearly couples the activities of the two subsystems (see Fig. 8.1). In this section, the effect of introducing such an interaction on the adaptive performance of the system is explored.

### 8.2.1 Model Introduction

A system consisting of two identical sensory systems is completely determined by four variables i.e.,  $a_1$ ,  $m_1$ ,  $a_2$  and  $m_2$  therefore, the phase-space is a 4-dimensional space –  $\bar{\Omega} = [0, 1]^2 \times [0, m_0]^2$  with  $(a_1, a_2, m_1, m_2)^T \in \bar{\Omega}$ . The system is such that both the receptors have the same setpoint  $a_0$  and their control elements  $m_1$  and  $m_2$  only depend on their corresponding output activities. Hence, the governing set of Langevin equations is,

$$\dot{a}_1(t) = F_{a_1}(a_1, m_1, s) + \eta_{a_1}(t) - \omega_{12}(a_1 - a_2) \quad (8.5a)$$

$$\dot{m}_1(t) = F_{m_1}(a_1, m_1, s) + \eta_{m_1}(t) \quad (8.5b)$$

$$\dot{a}_2(t) = F_{a_2}(a_2, m_2, s) + \eta_{a_2}(t) - \omega_{21}(a_2 - a_1) \quad (8.5c)$$

$$\dot{m}_2(t) = F_{m_2}(a_2, m_2, s) + \eta_{m_2}(t) \quad (8.5d)$$

where  $\omega_{12}, \omega_{21} \geq 0$  are the coupling rates and

$$F_{a_1}(a_1, m_1, s) = -\omega_{a_1}[a_1 - G_1(s, m_1)], \quad (8.6a)$$

$$F_{a_2}(a_2, m_2, s) = -\omega_{a_2}[a_2 - G_2(s, m_2)], \quad (8.6b)$$

with  $\omega_{a_1}, \omega_{a_2} > 0$  and

$$G_1(s, m_1) = \frac{1}{1 + (s/e^{2m_1})^{H_1}} \quad \text{and} \quad G_2(s, m_2) = \frac{1}{1 + (s/e^{2m_2})^{H_2}}, \quad (8.7)$$

where  $H_1, H_2 > 0$ . The Gaussian noise terms in (8.5) obey the following correlation relation,

$$\langle \eta_i(t) \eta_j(t') \rangle = 2\Delta_i \delta_{ij} \delta(t - t') \quad (8.8)$$

where  $\Delta_i \in \mathbb{R}_{\geq 0}$  for  $i, j \in \{a_1, a_2, m_1, m_2\}$ .

**Remark:** For the sake of simplicity, it is chosen that  $H_1 = 1$ . And as the subsystems are identical, it follows that  $H_1 = H_2$ ,  $\omega_{a_1} = \omega_{a_2}$  and  $\Delta_{i_1} = \Delta_{i_2}$  with  $i = a, m$ . Lastly, since the interaction is considered to be symmetric, it implies that  $\omega_{21} = \omega_{12}$ .

### 8.2.2 Detailed Balance Conditions

The Fokker-Planck equation corresponding to the Langevin system (8.5) is,

$$\begin{aligned} \partial_t P = & -\partial_{m_1}(F_{m_1} - \Delta_{m_1} \partial_{m_1} P) - \partial_{m_2}(F_{m_2} - \Delta_{m_2} \partial_{m_2} P) \\ & - \partial_{a_1}(\tilde{F}_{a_1} - \Delta_{a_1} \partial_{a_1} P) - \partial_{a_2}(\tilde{F}_{a_2} - \Delta_{a_2} \partial_{a_2} P) \end{aligned} \quad (8.9)$$

where  $P \equiv P(a_1, a_2, m_1, m_2, t)$  with  $(a_1, a_2, m_1, m_2)^T \in \Omega \equiv \bar{\Omega} \setminus \partial\Omega = (0, 1)^2 \times (0, m_0)^2$  and

$$\tilde{F}_{a_1}(a_1, a_2, m_1, s) \equiv F_{a_1}(a_1, m_1, s) - \omega_{12}(a_1 - a_2) \quad (8.10a)$$

$$\tilde{F}_{a_2}(a_1, a_2, m_2, s) \equiv F_{a_2}(a_2, m_2, s) - \omega_{21}(a_2 - a_1) \quad (8.10b)$$

with,

$$\mathbf{n} \cdot \mathbf{J} = 0, \quad (a_1, a_2, m_1, m_2)^T \in \partial\Omega \quad (8.11)$$

as boundary conditions. In (8.11), the probability current  $\mathbf{J}$  is a four-dimensional vector defined in accordance with (5.11).

As in the non-interacting case, it is possible to derive a set of detailed balance conditions for an interacting system. The derivation for the detailed balance conditions in this case follows the same line of thought as in the non-interacting case (cf. Appendix C.1).

**Derivation:**

Defining the vectors,

$$\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}, \quad \mathbf{m} = \begin{pmatrix} m_1 \\ m_2 \end{pmatrix}. \quad (8.12)$$

A general forward Chapman-Kolmogorov equation with phase space  $\bar{\Omega}$  is of the form,

$$\partial_t p(\mathbf{a}, \mathbf{m}) = \mathcal{L}p(\mathbf{a}, \mathbf{m}) = \iint_{\Omega} d\mathbf{a} d\mathbf{m} \mathcal{Q}(\mathbf{a}, \mathbf{m} | \mathbf{a}', \mathbf{m}') p(\mathbf{a}', \mathbf{m}'), \quad (8.13)$$

where  $\mathcal{L}$  is the forward Chapman-Kolmogorov operator and

$$\mathcal{Q}(\mathbf{a}, \mathbf{m} | \mathbf{a}', \mathbf{m}') = \mathcal{W}(\mathbf{a}, \mathbf{m} | \mathbf{a}', \mathbf{m}') - \delta(\mathbf{a} - \mathbf{a}') \delta(\mathbf{m} - \mathbf{m}') \iint_{\Omega} \mathcal{W}(\mathbf{a}'', \mathbf{m}'' | \mathbf{a}, \mathbf{m}) d\mathbf{a}'' d\mathbf{m}'' \quad (8.14)$$

is the forward transition operator with  $\mathcal{W}$  as the rate function. Then detailed balance entails that

$$\mathcal{W}(\mathbf{a}, \mathbf{m} | \mathbf{a}', \mathbf{m}') p^{eq}(\mathbf{a}', \mathbf{m}') = \mathcal{W}(\mathbf{a}', \mathbf{m}' | \mathbf{a}, \mathbf{m}) p^{eq}(\mathbf{a}, \mathbf{m}) \quad (8.15)$$

must be true, where  $p^{eq}$  is the steady state probability distribution in equilibrium. It can be shown that (8.15) implies that operator  $\mathcal{L}$  must be self-adjoint with respect to the following inner-product (proof in Appendix C.1),

$$(f, g) := \iint_{\Omega} \frac{f(\mathbf{a}, \mathbf{m}) g(\mathbf{a}', \mathbf{m}')}{p^{eq}(\mathbf{a}, \mathbf{m})} d\mathbf{a} d\mathbf{m}, \quad (8.16)$$

i.e., it satisfies the following equality,

$$(\mathcal{L}g, f) = (g, \mathcal{L}f), \quad (8.17)$$

for arbitrary test functions  $f$  and  $g$ , which vanish at  $\partial\Omega$ . Since the inner product is symmetric,

$$(\mathcal{L}g, f) = (\mathcal{L}f, g). \quad (8.18)$$

For the Fokker-Planck equation (8.9) on  $\Omega = (0, 1)^2 \times (0, m_0)^2$ ,

$$\begin{aligned} \mathcal{L} = & -\partial_{m_1}(F_{m_1} - \Delta_{m_1}\partial_{m_1}) - \partial_{m_2}(F_{m_2} - \Delta_{m_2}\partial_{m_2}) \\ & - \partial_{a_1}(\tilde{F}_{a_1} - \Delta_{a_1}\partial_{a_1}) - \partial_{a_2}(\tilde{F}_{a_2} - \Delta_{a_2}\partial_{a_2}). \end{aligned} \quad (8.19)$$

As detailed balance demands  $\mathcal{L}$  to be self-adjoint, it must satisfy (8.17). Then one can evaluate the left side of (8.18) using (8.19) and employing integration by parts. The same can be done to evaluate the right side, which yields

$$\begin{aligned} & \iint_{\Omega} d\mathbf{a} d\mathbf{m} d\mathbf{a}' d\mathbf{m}' \left[ \frac{F_{m_1} f}{p^{eq}} \partial_{m_1} g + \frac{F_{m_2} f}{p^{eq}} \partial_{m_2} g + \frac{\tilde{F}_{a_1} f}{p^{eq}} \partial_{a_1} g + \frac{\tilde{F}_{a_2} f}{p^{eq}} \partial_{a_2} g \right. \\ & \quad \left. + \frac{\Delta_{m_1} g}{(p^{eq})^2} \partial_{m_1} f \partial_{m_1} p^{eq} + \frac{\Delta_{m_2} g}{(p^{eq})^2} \partial_{m_2} f \partial_{m_2} p^{eq} + \frac{\Delta_{a_1} g}{(p^{eq})^2} \partial_{a_1} f \partial_{a_1} p^{eq} + \frac{\Delta_{a_2} g}{(p^{eq})^2} \partial_{a_2} f \partial_{a_2} p^{eq} \right] \\ & = \iint_{\Omega} d\mathbf{a} d\mathbf{m} d\mathbf{a}' d\mathbf{m}' \left[ \frac{F_{m_1} g}{p^{eq}} \partial_{m_1} f + \frac{F_{m_2} g}{p^{eq}} \partial_{m_2} f + \frac{\tilde{F}_{a_1} g}{p^{eq}} \partial_{a_1} f + \frac{\tilde{F}_{a_2} g}{p^{eq}} \partial_{a_2} f \right. \\ & \quad \left. + \frac{\Delta_{m_1} f}{(p^{eq})^2} \partial_{m_1} g \partial_{m_1} p^{eq} + \frac{\Delta_{m_2} f}{(p^{eq})^2} \partial_{m_2} g \partial_{m_2} p^{eq} + \frac{\Delta_{a_1} f}{(p^{eq})^2} \partial_{a_1} g \partial_{a_1} p^{eq} + \frac{\Delta_{a_2} f}{(p^{eq})^2} \partial_{a_2} g \partial_{a_2} p^{eq} \right] \end{aligned} \quad (8.20)$$

Since the integrals on both the sides of (8.20) are over the same domain, the integrands must be equal,

$$\begin{aligned} (F_{m_1} - \frac{\Delta_{m_1}}{p^{eq}} \partial_{m_1} p^{eq}) [f \partial_{m_1} g - g \partial_{m_1} f] + (F_{m_2} - \frac{\Delta_{m_2}}{p^{eq}} \partial_{m_2} p^{eq}) [f \partial_{m_2} g - g \partial_{m_2} f] - \\ (\tilde{F}_{a_1} - \frac{\Delta_{a_1}}{p^{eq}} \partial_{a_1} p^{eq}) [f \partial_{a_1} g - g \partial_{a_1} f] - (\tilde{F}_{a_2} - \frac{\Delta_{a_2}}{p^{eq}} \partial_{a_2} p^{eq}) [f \partial_{a_2} g - g \partial_{a_2} f] = 0. \end{aligned} \quad (8.21)$$

As (8.21) is true for all  $f$  and  $g$ , the following set of equations must hold true,

$$\frac{1}{\Delta_{m_1}} F_{m_1} = \partial_{m_1} \ln p^{eq} \quad (8.22a)$$

$$\frac{1}{\Delta_{a_1}} \tilde{F}_{a_1} = \partial_{a_1} \ln p^{eq} \quad (8.22b)$$

$$\frac{1}{\Delta_{m_2}} F_{m_2} = \partial_{m_2} \ln p^{eq} \quad (8.22c)$$

$$\frac{1}{\Delta_{a_2}} \tilde{F}_{a_2} = \partial_{a_2} \ln p^{eq}. \quad (8.22d)$$

Lastly, the commutativity of derivatives in the (8.22) gives rise to 6 detailed balance conditions,

$$\Delta_{a_1} \partial_{a_1} \tilde{F}_{a_2} = \Delta_{a_2} \partial_{a_2} \tilde{F}_{a_1} \quad (8.23a)$$

$$\Delta_{a_1} \partial_{a_1} F_{m_2} = \Delta_{m_2} \partial_{m_2} \tilde{F}_{a_1} \quad (8.23b)$$

$$\Delta_{a_2} \partial_{a_2} F_{m_1} = \Delta_{m_1} \partial_{m_1} \tilde{F}_{a_2} \quad (8.23c)$$

$$\Delta_{a_1} \partial_{a_1} F_{m_1} = \Delta_{m_1} \partial_{m_1} \tilde{F}_{a_1} \quad (8.23d)$$

$$\Delta_{a_2} \partial_{a_2} F_{m_2} = \Delta_{m_2} \partial_{m_2} \tilde{F}_{a_2} \quad (8.23e)$$

$$\Delta_{m_1} \partial_{m_1} F_{m_2} = \Delta_{m_2} \partial_{m_2} F_{m_1} \quad (8.23f)$$

Even though broken detailed balance is required to achieve adaptation, these equations can help to relate some of the system parameters. Moreover, as for the non-interacting case (8.23d) and (8.23e) can be used to construct the functions  $F_{m_1}$  and  $F_{m_2}$ , respectively.

## Implications

### Condition 1:

$$\Delta_{a_1} \partial_{a_1} \tilde{F}_{a_2} = \Delta_{a_2} \partial_{a_2} \tilde{F}_{a_1} \implies \frac{\Delta_{a_1}}{\Delta_{a_2}} = \frac{\omega_{12}}{\omega_{21}} \quad (8.24)$$

If it is assumed that the effective temperatures of the activities of both systems are the same, then

$$T_{\text{eff}} = \frac{\Delta_{a_1}}{\omega_{a_1}} = \frac{\Delta_{a_2}}{\omega_{a_2}} \implies \frac{\omega_{a_1}}{\omega_{a_2}} = \frac{\omega_{21}}{\omega_{12}} \implies \frac{\omega_{21}}{\omega_{a_2}} = \frac{\omega_{12}}{\omega_{a_1}} = T_{\text{eff}}. \quad (8.25)$$

Hence,

$$\omega_{12} = \omega_{a_1} T_{\text{eff}}, \quad (8.26a)$$

$$\omega_{21} = \omega_{a_2} T_{\text{eff}}. \quad (8.26b)$$

Condition 2 & 3:

As  $\tilde{F}_{a_1}$  is independent of  $m_2$ ,

$$\Delta_{a_1} \partial_{a_1} F_{m_2} = \Delta_{m_2} \partial_{m_2} \tilde{F}_{a_1} \implies \partial_{a_1} F_{m_2} = 0, \quad (8.27)$$

and as  $\tilde{F}_{a_2}$  is independent of  $m_1$ ,

$$\Delta_{a_2} \partial_{a_2} F_{m_1} = \Delta_{m_1} \partial_{m_1} \tilde{F}_{a_2} \implies \partial_{a_2} F_{m_1} = 0. \quad (8.28)$$

Hence, as indicated in (8.5b) and (8.5d),  $F_{m_i}$  does not depend on  $a_j$  for  $i, j = 1, 2$  with  $i \neq j$ .

Condition 4 & 5:

Equations (8.6a) and (8.6b) imply,

$$\Delta_{a_1} \partial_{a_1} F_{m_1} = \Delta_{m_1} \partial_{m_1} \tilde{F}_{a_1} \implies \partial_{a_1} F_{m_1} = \frac{\Delta_{m_1}}{\Delta_{a_1}} \omega_{a_1} G_1(s, m_1), \quad (8.29)$$

and

$$\Delta_{a_2} \partial_{a_2} F_{m_2} = \Delta_{m_2} \partial_{m_2} \tilde{F}_{a_2} \implies \partial_{a_2} F_{m_2} = \frac{\Delta_{m_2}}{\Delta_{a_2}} \omega_{a_2} \partial_{m_2} G_2(s, m_2). \quad (8.30)$$

Then following the discussion in Section (5.1) leads to,

$$F_{m_1} = \omega_{m_1} (a_1 - a_0) [\beta + (1 - \beta) \frac{\Delta_{m_1}}{\Delta_{a_1}} \frac{\omega_{a_1}}{\omega_{m_1}} \partial_{m_1} G_1(s, m_1)] \quad (8.31a)$$

$$F_{m_2} = \omega_{m_2} (a_2 - a_0) [\beta + (1 - \beta) \frac{\Delta_{m_2}}{\Delta_{a_2}} \frac{\omega_{a_2}}{\omega_{m_2}} \partial_{m_2} G_2(s, m_2)] \quad (8.31b)$$

where  $\omega_{m_1}, \omega_{m_2}$  are the adaptation rates and  $\beta \in [0, 1]$  is the parameter that moves the interacting system in and out-of-equilibrium.

Moreover, assuming that the effective temperatures of the control element  $m$  and output  $a$  of a sensory system are equal implies that

$$\frac{\Delta_{m_1}}{\Delta_{a_1}} \frac{\omega_{a_1}}{\omega_{m_1}} = \frac{\Delta_{m_2}}{\Delta_{a_2}} \frac{\omega_{a_2}}{\omega_{m_2}} = 1. \quad (8.32)$$

Condition 6:

This condition is trivially satisfied by the construction of (8.31a) and (8.31b).

### 8.2.3 Boundary Conditions

As in the non-interacting case, there are two ways to impose the reflective boundary conditions (reflection operators and boundary potential) for interacting systems. When computing the energetics of the interacting system, the boundary potential  $\tilde{V} : \Omega \rightarrow \mathbb{R}$  of the form,

$$\begin{aligned} \tilde{V}(a_1, a_2, m_1, m_2) = & \frac{1}{\omega_{a_1}} V_b(a_1, a_{1\max}) + \frac{1}{\omega_{a_2}} V_b(a_2, a_{2\max}) + \\ & \frac{1}{\omega_{m_1}} V_b(m_1, m_{1\max}) + \frac{1}{\omega_{m_2}} V_b(m_2, m_{2\max}), \end{aligned} \quad (8.33)$$

is used. In (8.33),  $V_b$  is as in (6.4) and  $a_{1\max} = a_{2\max} = 1$  and  $m_{1\max} = m_{2\max} = m_0$ .

### 8.2.4 Energetics & Adaptation Accuracy

A design choice that one can make for this interacting system is to set the scale of interaction by introducing a coupling constant  $k$  in equations (8.26a) and (8.26b),

$$\omega_{12} = k\omega_{a_1}T_{\text{eff}}, \quad (8.34a)$$

$$\omega_{21} = k\omega_{a_2}T_{\text{eff}}. \quad (8.34b)$$

The dissipation for the interacting system can be computed in an analogous way as for the non-interacting case. That is, by employing (3.22), which yields,

$$\dot{W}_{\text{diss}} = T \iint_{\Omega} \left[ \frac{J_{a_1} \tilde{F}_{a_1}}{\Delta_{a_1}} + \frac{J_{a_2} \tilde{F}_{a_2}}{\Delta_{a_2}} + \frac{J_{m_1} F_{m_1}}{\Delta_{m_1}} + \frac{J_{m_2} F_{m_2}}{\Delta_{m_2}} \right] d\mathbf{a} d\mathbf{m}. \quad (8.35)$$

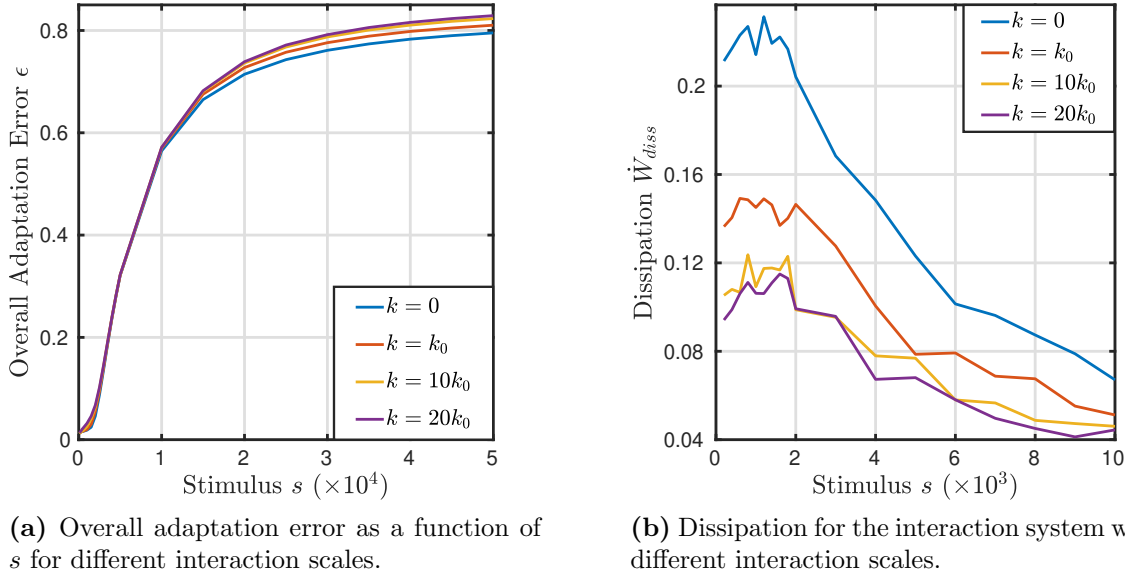
where  $T = T_{\text{eff}}$ . Since (5.23) is equivalent to (6.19) on a trajectory level, (8.35) for a single trajectory can be written as (including the boundary potential),

$$\begin{aligned} \dot{W}_{\text{diss}} = & \frac{T}{\Delta_{a_1}} \frac{da_1}{dt} \circ (\tilde{F}_{a_1} + f_{a_1}) + \frac{T}{\Delta_{a_2}} \frac{da_2}{dt} \circ (\tilde{F}_{a_2} + f_{a_2}) \\ & + \frac{T}{\Delta_{m_1}} \frac{dm_1}{dt} \circ (F_{m_1} + f_{m_1}) + \frac{T}{\Delta_{m_2}} \frac{dm_2}{dt} \circ (F_{m_2} + f_{m_2}). \end{aligned} \quad (8.36)$$

where  $f_i \equiv -\omega_i \partial_i \tilde{V}$  for  $i = a_1, a_2, m_1, m_2$  – are the forces due to the boundary potential.

Figure 8.2 presents the simulation results for this interacting system. Figure 8.2a exhibits how the overall adaptation error changes with the stimulus level for different coupling constants. Whereas, Fig. 8.2b shows the behavior of dissipation for different values of coupling constant  $k$ . The simulation results exhibit that for an interacting system with linearly coupled activities, stronger interaction (larger  $k$ ) results in a higher overall adaptation error, when compared to weakly or non-interacting systems. This behavior appears to be true for all  $s > 0$ . Moreover, as seen for the non-interacting case, the adaptation error increases





**Figure 8.2: Adaptation Error and Dissipation for Interacting System with Coupled Activities:** (a, b) It is seen that as the interaction strength  $k$  is increased, the overall adaptation error also increases, whereas the energy cost associated to it decreases. Furthermore, as seen for a non-interacting system, for a fixed  $k$  as  $s$  becomes larger,  $\epsilon$  increases and the dissipation decreases. *Simulation Parameters:* As in Figure 6.3 with  $k_0 = 100$  and  $\omega_{12/21} = k\omega_{a_1/a_2}T_{eff}$ .

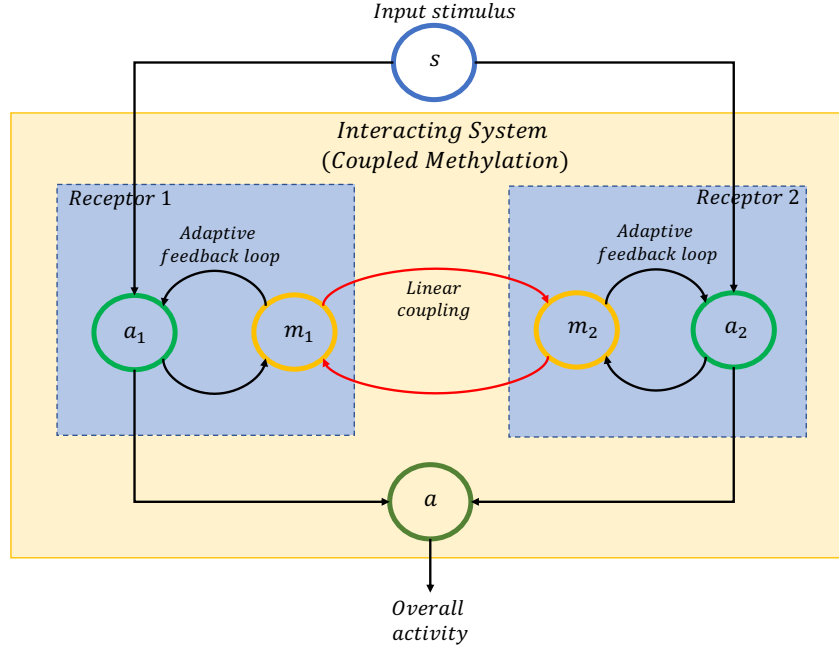
as  $s$  increases. This functionality is true irrespective of the interaction strength of the system and appears to be due to the effect of the boundary as for the non-interacting systems.

Furthermore, when the coupling constant  $k$  increases, the adaptation accuracy is lowered and in consequence, the energy dissipation is also reduced (see Fig. 8.2b). This behavior agrees with what was observed for the non-interacting system as the energy cost seems to be inversely proportional to the adaptation error.

To summarize, coupling the output activities of the two subsystems seems to enforce a trade-off between the energy cost and the adaptive performance of the overall system. That is, for a fixed input, a highly coupled system has lower energy consumption but also has least adaptation accuracy. Therefore, it can be concluded that linearly coupling the activities increases the maximum adaptation error of the system but reduces the maximum dissipation.

### 8.3 Coupled Methylation

In the previous section, the behavior of an interacting system in which the output activities of the two sensory systems were coupled linearly was investigated. Here, the consequences



**Figure 8.3: Interacting system with coupled methylation:** The details of this system are the same as in Fig. 8.1, with a different interaction between the receptors (sensory systems). In this case, the linear interaction couples the control elements of the two subsystems.

of coupling the control elements of the two subsystems instead of their outputs are studied (see Fig. 8.3).

### 8.3.1 Model Introduction

The phase space of this interacting system is a four-dimensional space exactly as in Section 8.2.1. Moreover, the governing SDE's are the same as in (8.5), except that the interaction terms couple the methylation levels instead of the activities. The Langevin equations that govern this system are,

$$\dot{a}_1(t) = F_{a_1}(a_1, m_1, s) + \eta_{a_1}(t) \quad (8.37a)$$

$$\dot{m}_1(t) = F_{m_1}(a_1, m_1, s) + \eta_{m_1}(t) - \omega_{12}(m_1 - m_2) \quad (8.37b)$$

$$\dot{a}_2(t) = F_{a_2}(a_2, m_2, s) + \eta_{a_2}(t) \quad (8.37c)$$

$$\dot{m}_2(t) = F_{m_2}(a_2, m_2, s) + \eta_{m_2}(t) - \omega_{21}(m_2 - m_1) \quad (8.37d)$$

where  $F_{a_1}$ ,  $F_{a_2}$ ,  $F_{m_1}$  and  $F_{m_2}$  are as in (8.6) and (8.31), respectively. The noise terms in (8.37) also obey the correlations in (8.8).

The Fokker-Planck equation for this system of Langevin equations is,

$$\begin{aligned} \partial_t P = & -\partial_{m_1}(\tilde{F}_{m_1} - \Delta_{m_1} \partial_{m_1} P) - \partial_{m_2}(\tilde{F}_{m_2} - \Delta_{m_2} \partial_{m_2} P) \\ & - \partial_{a_1}(F_{a_1} - \Delta_{a_1} \partial_{a_1} P) - \partial_{a_2}(F_{a_2} - \Delta_{a_2} \partial_{a_2} P), \end{aligned} \quad (8.38)$$

which is defined in the interior  $\Omega$ , where

$$\tilde{F}_{m_1}(a_1, m_1, m_2, s) \equiv F_{m_1}(a_1, m_1, s) - \omega_{12}(m_1 - m_2), \quad (8.39a)$$

$$\tilde{F}_{m_2}(a_1, m_1, m_2, s) \equiv F_{m_2}(a_2, m_2, s) - \omega_{21}(m_2 - m_1). \quad (8.39b)$$

The boundary conditions for the FPE (8.38) are as in the first case i.e., (8.11).

As in the previous case, the detailed balance conditions can be used (specifically 8.23f) to obtain the interaction rates in terms of adaptation rates and effective temperatures,

$$\omega_{12} = \omega_{m_1} T_{\text{eff}}, \quad (8.40a)$$

$$\omega_{21} = \omega_{m_2} T_{\text{eff}}. \quad (8.40b)$$

Also, a coupling constant  $k$  is introduced to tune the interaction rates,

$$\omega_{12} = k\omega_{m_1} T_{\text{eff}}, \quad (8.41a)$$

$$\omega_{21} = k\omega_{m_2} T_{\text{eff}}. \quad (8.41b)$$

Lastly, for the implementation of the phase-space boundaries, the boundary potential (8.33) is employed.

### 8.3.2 Adaptation Error & Dissipation

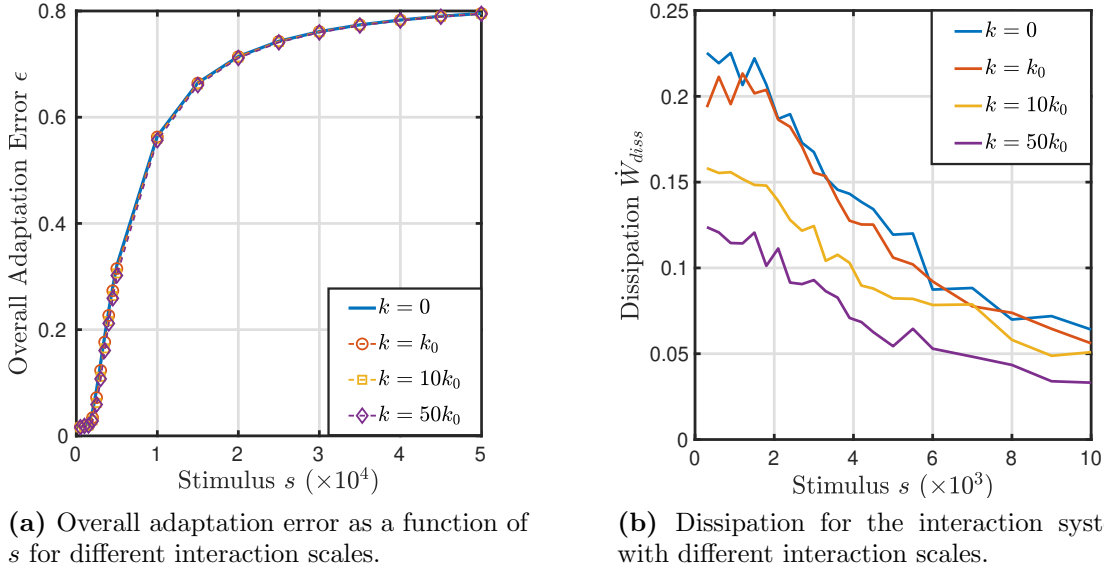
The dissipation for this interacting system can be computed in a similar manner as for the first interaction type. The resulting equation for the dissipation is,

$$\begin{aligned} \dot{W}_{\text{diss}} = & \frac{T}{\Delta_{a_1}} \frac{da_1}{dt} \circ (F_{a_1} + f_{a_1}) + \frac{T}{\Delta_{a_2}} \frac{da_2}{dt} \circ (F_{a_2} + f_{a_2}) \\ & + \frac{T}{\Delta_{m_1}} \frac{dm_1}{dt} \circ (\tilde{F}_{m_1} + f_{m_1}) + \frac{T}{\Delta_{m_2}} \frac{dm_2}{dt} \circ (\tilde{F}_{m_2} + f_{m_2}). \end{aligned} \quad (8.42)$$

where  $\circ$  entails that the product of the stochastic terms is to be taken in the Stratonovich sense.

The results of the numerical simulation for this interacting system are presented in Fig. 8.4. The results presented in Fig. 8.4a compare the adaptation accuracy of the interacting system ( $k \neq 0$ ) with that of the non-interacting system ( $k = 0$ ).

As for an individual system, the overall adaptation error of the system increases with increasing input irrespective of the interaction. More importantly, it is observed that linearly coupling the methylation levels does not change the adaptation accuracy of the system. Furthermore, Fig. 8.4b shows that, as the input is increased, the dissipation for a fixed  $k$  decreases (effect of the boundary). Additionally, it is observed that for a fixed  $s$ , increasing the interaction strength reduces the energy cost associated with adaptation.



**Figure 8.4: Adaptation Error and Dissipation for Interacting System with Coupled Activities:** (a) The overall adaptation error increases as  $s$  gets larger. However, the adaptive performance appears to be independent of the interaction strength. (b) The energy consumption for an interacting system with coupled control elements is lower than that for a system with two non-interacting subsystems. That is for an interacting system consumes less energy for the same adaptive performance. *Simulation Parameters:* As in Fig. 8.2 with  $\omega_{12/21} = k\omega_{m_1/m_2}T_{eff}$ .

In conclusion, in an interacting system with linearly coupled methylation levels, the energy cost associated with the same adaptation accuracy is lowered as the interaction between the two subsystems becomes stronger. Therefore, from an energy cost perspective, coupling the methylation levels (control elements) improves the adaptive performance of the system. However, it does not change the overall adaptation accuracy of the system. That is, two nearby receptors in a cell require less energy to achieve the same adaptation accuracy if their methylation levels are linearly related on each other. However, such interaction between the two receptors does not improve the adaptation accuracy.

# Concluding Remarks & Outlook

Adaptation is an essential process that is observed across all biological scales, from populations of species over an evolutionary period to processes in a cell [56]. It is the ability of a system to adjust to changes in its surroundings. Adaptation in biological sensory systems can be characterized by a fast response in the output activity due to an external input and a slowly varying control element, which counters the effect of the input and brings the output to a predefined setpoint [15]. The work in this thesis investigates the physical trade-offs between the adaptation capabilities and the energetics of noisy biological systems.

In the first part of the thesis i.e., Chapters 1-4, I presented the mathematical framework essential for this topic. In Chapter 5, I discussed some properties of the model for sensory adaptation (5.1) suggested in Ref. [15]. By presenting the phase-space dynamics of the system, it is first discussed how the system accomplishes adaptation. Moreover, by employing the steepest descent approximation for integrals, an analytical expression for the dissipation rate is obtained. Furthermore, the results obtained above are then used to construct an energy-speed-accuracy (ESA) relation, as in Ref. [15].

Subsequently, in Chapter 6, a framework for stochastic energetics is employed to obtain an expression for the steady-state dissipation in the system. Then this expression for dissipation and the numerical solution to the Langevin system are used to computationally evaluate the energetics of the model.

The analytical results for the adaptation error in Chapter 5 imply that the adaptation error for a fully adaptive system is vanishingly small in the bulk of the phase-space and is non-zero only near the domain-boundaries. This result is confirmed by the numerical results presented in Chapter 7. Additionally, simulation results in Chapter 7 show that phase-space boundaries constrain the trajectories of the system, which deteriorates the ability of the system to adapt accurately. Also, it is observed that the energy dissipation of the system declines as the adaptation error increases. This observation supports the assertion that the adaptation accuracy is proportional to the dissipation even in the vicinity of phase-space boundaries.

Examination of the numerical results in Chapter 7 revealed that the energy dissipation of the system in the bulk of the phase-space only depends on the pre-defined setpoint.

However, when the input level exceeds a threshold, the two approximations of dissipation i.e., numerical and analytical tend to disagree as they depend on the input differently. Based on the evaluation of the data, an empirical expression for the dissipation is conjectured, which describes the dependence of the dissipation on the input near the boundaries.

After exploring the role of the phase-space boundaries in a non-interacting system, a system of two interacting subsystems is studied in Chapter 8. To quantify the adaptation accuracy of an interacting system, the notion of overall output activity and the notion of overall adaptation error is developed. Here, the overall activity is defined to be the mean of the two individual activities. Two types of interactions between the sensory systems are investigated. In the first case, the output activities of the two subsystems are coupled linearly via a conservative and symmetric force. And for the second case, instead of coupling the activities, the control elements are coupled. Numerical simulations for the interacting systems show that coupling the activities hinders the adaptation capabilities of the system, but also reduces the energy cost associated with it. As a result, such an interaction seems to reduce both the maximum adaptation accuracy as well as the maximum dissipation in the system. However, coupling the control elements turns out to be an energetically favorable interaction as the interacting system seems to dissipate less than the non-interacting system without compromising on the adaptation accuracy.

This thesis presents a first step into exploring the significance of the phase-space limitations for sensory adaptation. The next stages in this research could involve considering the dependence of the dissipation and adaptive performance on the effective temperatures. In addition, an analytical approximation of the dissipation is required that holds true even at the boundaries of the phase-space. Furthermore, the validity of the ESA suggested in Ref. [15] in this setting ( $C = 1$ ) appears questionable, which is to be understood.

For interacting systems, the first general results have been presented, which, however, only scratches the surface of a complex topic. Further numerical and analytical work is required to fully understand the cost-performance relationship for interacting systems. Additionally, to understand the collective behavior of such systems, one could further extend to  $n$ -interacting sensory systems.

The answers to such questions will provide insights into generic relationships that connect dissipation, adaptive performance and phase-space limitations in various biological systems such as molecular motor systems [51] and cellular processes such as kinetic proof-reading [22]. The optimization principles emerging from such relationships can provide a deeper understanding of the design principles that underlie the structure of biochemical networks [15].

# Appendix A

## Markov Processes and Chapman-Kolmogorov Equation

### A.1 Markov Processes

Lets  $\Psi(t)$  be a random variable with  $t \in \mathbb{R}_{\geq 0}$ . If  $\Psi(t)$  represents a discrete random variable then  $\Psi(t) \in \mathbb{Z}$ , alternatively, if it is a continuous real-valued random process then  $\Psi(t) \in \Omega \subseteq \mathbb{R}$ , where  $\Omega$  is the sample space.

The statistical properties of  $\Psi(t)$  for a single measurement is completely determined by  $p(\psi, t)$ , which is the probability density that  $\Psi(t)$  takes a value in the interval  $(\psi, \psi + d\psi)$  at time  $t$ .

Let  $t_1 \leq t_2 \leq \dots \leq t_n$  be the time instances for a sequence of  $n$  measurements then one needs a set of  $n$  probability densities [27],

$$p(\psi_1, t_1), p(\psi_2, t_2; \psi_1, t_1), \dots, p(\psi_n, t_n; \psi_{n-1}, t_{n-1}; \dots; \psi_1, t_1)$$

to completely specify the statistical properties of  $\psi(t)$ . Here, the last term i.e.,

$$p(\psi_n, t_n; \psi_{n-1}, t_{n-1}; \dots; \psi_1, t_1),$$

is an  $n$ -time joint probability distribution function (PDF), which entails that

$$p(\psi_n, t_n; \dots; \psi_1, t_1) d\psi_1 d\psi_2 \dots d\psi_n$$

is the probability that the stochastic variable  $\Psi$  lies in interval  $(\psi_k, \psi_k + d\psi_k)$  at time  $t = t_k$ .

The  $n$ -time joint probability distribution can be related to  $(n - 1)$ -time PDF via a  $n$ -time conditional probability distribution,

$$p(\psi_n, t_n | \psi_{n-1}, t_{n-1}; \dots; \psi_1, t_1) = \frac{p(\psi_n, t_n; \psi_{n-1}, t_{n-1}; \dots; \psi_1, t_1)}{p(\psi_{n-1}, t_{n-1}; \dots; \psi_1, t_1)}. \quad (\text{A.1})$$

If the random process  $\Psi(t)$  is endowed with the Markov property, then the following equivalence holds true [27],

$$p(\psi_n, t_n | \psi_{n-1}, t_{n-1}; \dots; \psi_1, t_1) = p(\psi_n, t_n | \psi_{n-1}, t_{n-1}). \quad (\text{A.2})$$

This means that Markovian property imposes a restriction that the conditional joint probability of the  $n^{\text{th}}$ -time step (next) only depends on the  $(n-1)^{\text{th}}$ -step (current) and the memory of the rest of previous steps is forgotten or irrelevant. A stochastic process equipped with the Markovian property is called a *Markov Process*.

Rewriting (A.1) and employing Markov property n-times yields [27],

$$\begin{aligned} p(\psi_n, t_n; \dots; \psi_1, t_1) &= p(\psi_n, t_n | \psi_{n-1}, t_{n-1}) p(\psi_{n-1}, t_{n-1}; \dots; \psi_1, t_1) \\ &= p(\psi_n, t_n | \psi_{n-1}, t_{n-1}) p(\psi_{n-1}, t_{n-1} | \psi_{n-2}, t_{n-2}) \\ &\quad p(\psi_{n-2}, t_{n-2}; \dots; \psi_1, t_1) \\ &= p(\psi_n, t_n | \psi_{n-1}, t_{n-1}) p(\psi_{n-1}, t_{n-1} | \psi_{n-2}, t_{n-2}) \\ &\quad \dots p(\psi_2, t_2 | \psi_1, t_1) p(\psi_1, t_1) \end{aligned} \quad (\text{A.3})$$

Equation (A.3) is an important consequence of the Markov property, which is that the  $n$ -time joint probability distribution can be obtained using 2-time conditional probability distributions and a single time PDF  $p_1(\psi_1, t_1)$ .

## A.2 Chapman-Kolmogorov Equation

Consider two random variables  $X$  and  $Y$  and let  $\Omega$  be their sample space. Then it is known from basic probability theory that the joint probability distribution  $p(x, y)$  of the two random variables is associated to their marginal probabilities by the following equations,

$$p(x) = \int_{\Omega} dy p(x, y) \quad \text{and} \quad p(y) = \int_{\Omega} dx p(x, y). \quad (\text{A.4})$$

Using this concept for the stochastic process  $\Psi(t)$  (for  $n = 3$ ) yields

$$p(\psi_3, t_3; \psi_1, t_1) = \int_{\Omega} d\psi_2 p(\psi_3, t_3; \psi_2, t_2; \psi_1, t_1). \quad (\text{A.5})$$

Employing (A.1) in (A.5),

$$\begin{aligned} p(\psi_3, t_3 | \psi_1, t_1) p(\psi_2, t_2) &= \int_{\Omega} d\psi_2 p(\psi_3, t_3 | \psi_2, t_2; \psi_1, t_1) p(\psi_2, t_2; \psi_1, t_1) \\ &= \int_{\Omega} d\psi_2 p(\psi_3, t_3 | \psi_2, t_2; \psi_1, t_1) p(\psi_2, t_2 | \psi_1, t_1) p(\psi_1, t_1). \end{aligned} \quad (\text{A.6})$$

Since  $\Psi(t)$  is a Markov process, (A.6) becomes

$$p(\psi_3, t_3 | \psi_1, t_1) = \int_{\Omega} d\psi_2 p(\psi_3, t_3 | \psi_2, t_2) p(\psi_2, t_2 | \psi_1, t_1). \quad (\text{A.7})$$

Equation (A.7) is the *Chapman-Kolmogorov equation*. It is a functional equation that relates conditional probabilities and only holds true for Markov processes [36].



# Appendix B

## Itô-Taylor Expansion

**Definition (Linear Growth Bound)** A function  $\phi(t, \mathbf{x}) : \mathbb{R}_+ \times \mathbb{R}^d \rightarrow \mathbb{R}^d$  satisfies the linear growth condition, if there exists a  $K > 0$ , such that

$$\|\phi(t, \mathbf{x})\|^2 \leq K(1 + \|\mathbf{x}\|^2)$$

for all  $t > 0$  and  $\mathbf{x} \in \mathbb{R}^d$  [39].

Consider the SDE in (4.1), where it is assumed that  $a(X, t)$  and  $b(X, t)$  are sufficiently smooth and satisfy the linear-growth condition [53]. Additionally, it is assumed that  $a$  and  $b$  have non-explicit dependence on  $t$ . That is,  $a(X, t) = a[X(t)]$  and  $b(X, t) = b[X(t)]$ .

Let  $Y = f[X(t)]$ , then the differential of  $Y$  in the limit  $dt \rightarrow 0$  is given by Itô's formula,

$$dY = \left[ a[X(t)] \frac{\partial f}{\partial X} + \frac{1}{2} b^2[X(t)] \frac{\partial^2 f}{\partial X^2} \right] dt + b[X(t)] \frac{\partial f}{\partial X} dW_t. \quad (\text{B.1})$$

From this one can define the following operators,

$$L_0 := a[X(t)] \frac{\partial}{\partial X} + \frac{1}{2} b^2[X(t)] \frac{\partial^2}{\partial X^2}; \quad L_1 := b[X(t)] \frac{\partial}{\partial X}, \quad (\text{B.2})$$

hence, (4.1) becomes

$$dY = L_0[f[X(t)]] dt + L_1[f[X(t)]] dW_t. \quad (\text{B.3})$$

This equation is equivalent to the following integral equation,

$$Y[X(t)] = Y[X(t_0)] + \int_{t_0}^t L_0[f[X(\tau)]] d\tau + \int_{t_0}^t L_1[f[X(\tau)]] dW_\tau, \quad (\text{B.4})$$

where  $t_0 < t$  is some initial time point. If  $f[X(t)] = X(t)$ , then (B.4) becomes,

$$X(t) = X(t_0) + \int_{t_0}^t a[X(\tau)] d\tau + \int_{t_0}^t b[X(\tau)] dW_\tau, \quad (\text{B.5})$$

which is the integral form of (4.1). And similarly, if  $f[X(t)] = a[X(t)]$  and  $f[X(t)] = b[X(t)]$  then,

$$a[X(t)] = a[X(t_0)] + \int_{t_0}^t L_0[a[X(\tau)]] d\tau + \int_{t_0}^t L_1[a[X(\tau)]] dW_\tau, \quad (\text{B.6a})$$

$$b[X(t)] = b[X(t_0)] + \int_{t_0}^t L_0[b[X(\tau)]] d\tau + \int_{t_0}^t L_1[b[X(\tau)]] dW_\tau. \quad (\text{B.6b})$$

Plugging (B.6) in (B.5) yields,

$$\begin{aligned} X(t) &= X(t_0) + \int_{t_0}^t \left( a[X(t_0)] + \int_{t_0}^{\tau'} L_0[a[X(\tau)]] d\tau + \int_{t_0}^{\tau'} L_1[a[X(\tau)]] dW_\tau \right) d\tau' \\ &\quad + \int_{t_0}^t \left( b[X(t_0)] + \int_{t_0}^{\tau'} L_0[b[X(\tau)]] d\tau + \int_{t_0}^{\tau'} L_1[b[X(\tau)]] dW_\tau \right) dW_{\tau'} \quad (\text{B.7}) \\ &= X(t_0) + \int_{t_0}^t a[X(t_0)] d\tau' + \int_{t_0}^t b[X(t_0)] dW_{\tau'} + R, \end{aligned}$$

where,

$$\begin{aligned} R &= \int_{t_0}^t \int_{t_0}^{\tau'} L_0[a[X(\tau)]] d\tau d\tau' + \int_{t_0}^t \int_{t_0}^{\tau'} L_1[a[X(\tau)]] d\tau d\tau' \\ &\quad + \int_{t_0}^t \int_{t_0}^{\tau'} L_0[b[X(\tau)]] dW_\tau dW_{\tau'} + \int_{t_0}^t \int_{t_0}^{\tau'} L_1[b[X(\tau)]] dW_\tau dW_{\tau'}. \end{aligned} \quad (\text{B.8})$$

Equation (B.7) is the *Itô-Taylor expansion* [53] and is a necessary result to derive the numerical methods discussed in Chapter 4.

Using (B.4) in the last term of (B.8) for  $Y = L_1 b[X(t)] = b[X(t)]b'[X(t)]$  and expanding to higher order yields,

$$\begin{aligned} X(t) &= X(t_0) + \int_{t_0}^t a[X(t_0)] d\tau' + \int_{t_0}^t b[X(t_0)] dW_{\tau'} \\ &\quad + b[X(t_0)]b'[X(t_0)] \int_{t_0}^t \int_{t_0}^{\tau} dW_\tau dW_{\tau'} + \tilde{R}, \end{aligned} \quad (\text{B.9})$$

where,

$$\begin{aligned} \tilde{R} &= \int_{t_0}^t \int_{t_0}^{\tau'} L_0[a[X(\tau)]] d\tau d\tau' + \int_{t_0}^t \int_{t_0}^{\tau'} L_1[a[X(\tau)]] d\tau d\tau' + \int_{t_0}^t \int_{t_0}^{\tau'} L_0[b[X(\tau)]] dW_\tau dW_{\tau'} \\ &\quad + \int_{t_0}^t \int_{t_0}^{\tau'} \int_{t_0}^{\tau''} L_0 L_1 g[X(\tau'')] d\tau'' dW_{\tau'} dW_\tau + \int_{t_0}^t \int_{t_0}^{\tau'} \int_{t_0}^{\tau''} L_1 L_1 g[X(\tau'')] dW_{\tau''} dW_{\tau'} dW_\tau. \end{aligned} \quad (\text{B.10})$$

# Appendix C

## Calculations for Chapter 5

### C.1 Detailed Balance Condition

The detailed balance condition discussed in this section is a very general one and is true for all such feedback systems [15].

A general Chapman-Kolmogorov equation with rate  $\mathcal{W}$  can be written as,

$$\partial_t p(a, m) = \iint (\mathcal{W}(a, m|a', m')p(a', m') - \mathcal{W}(a', m'|a, m)p(a, m)) da' dm', \quad (\text{C.1})$$

where (all) the double integrals are over  $\Omega \equiv \bar{\Omega}/\partial\Omega$ .

In terms of a forward transition operator:

$$\mathcal{Q}(a, m|a', m') = \mathcal{W}(a, m|a', m') - \delta(a - a')\delta(m - m') \iint \mathcal{W}(a'', m''|a, m) dm'' da''$$

equation (C.1) becomes,

$$\partial_t p(a, m) = \iint \mathcal{Q}(a, m|a', m')p(a', m') da' dm' \iff \partial_t p = \mathcal{L}p, \quad (\text{C.2})$$

where  $\mathcal{L}$  is the forward Chapman-Kolmogorov operator.

Then the definition of detailed balance dictates that:

$$\frac{\mathcal{W}(a, m|a', m')}{p^{eq}(a, m)} = \frac{\mathcal{W}(a', m'|a, m)}{p^{eq}(a', m')}. \quad (\text{C.3})$$

Then for test functions  $f$  and  $g$ , (C.3) yields the following,

$$\underbrace{\iint \iint da \, dm \, da' \, dm' f(a, m) g(a', m') \frac{\mathcal{W}(a, m|a', m')}{p^{eq}(a, m)}}_I = \underbrace{\iint \iint da \, dm \, da' \, dm' \frac{\mathcal{W}(a', m'|a, m)}{p^{eq}(a', m')} f(a, m) g(a', m')}_I. \quad (C.4)$$

First looking at  $I$ :

$$I = \iint da \, dm \frac{f(a, m)}{p^{eq}(a, m)} \iint da' \, dm' \mathcal{W}(a, m|a', m') g(a', m'),$$

and using the forward transition operator gives,

$$\begin{aligned} &= \iint da \, dm \frac{f(a, m)}{p^{eq}(a, m)} \iint da' \, dm' \left[ \mathcal{Q}(a, m|a', m') + \right. \\ &\quad \left. \delta(a - a') \delta(m - m') \iint da'' \, dm'' \mathcal{W}(a'', m''|a, m) \right] g(a', m') \\ &= \iint da \, dm \frac{f(a, m)}{p^{eq}(a, m)} \iint da' \, dm' \mathcal{Q}(a, m|a', m') g(a', m') + \\ &\quad \iint da \, dm \frac{f(a, m)}{p^{eq}(a, m)} g(a, m) \iint dm'' da'' \mathcal{W}(a'', m''|a, m). \end{aligned}$$

Defining  $C(a, m) = \iint dm'' da'' \mathcal{W}(a'', m''|a, m)$ :

$$\begin{aligned} I &= \iint da \, dm \frac{f(a, m)}{p^{eq}(a, m)} \iint da' \, dm' \mathcal{Q}(a, m|a', m') g(a', m') \\ &\quad + \underbrace{\iint da \, dm \frac{f(a, m)}{p^{eq}(a, m)} g(a, m) C(a, m)}_{\text{const}=c}. \end{aligned}$$

Then looking at  $II$ :

$$II = \iint da' \, dm' \frac{g(a', m')}{p^{eq}(a', m')} \iint da \, dm \mathcal{W}(a', m'|a, m) f(a, m),$$

using the backward transition operator

$$\mathcal{Q}(a', m'|a, m) = \mathcal{W}(a', m'|a, m) - \delta(a - a') \delta(m - m') \iint \mathcal{W}(a'', m''|a, m) da'' dm''$$

yields,

$$\begin{aligned}
&= \iint da' dm' \frac{g(a', m')}{p^{eq}(a', m')} \iint da dm f(a, m) \left[ \mathcal{Q}(a', m'|a, m) \right. \\
&\quad \left. + \delta(a - a')\delta(m - m') \iint \mathcal{W}(a'', m''|a, m) da'' dm'' \right] \\
&= \iint da' dm' \frac{g(a', m')}{p^{eq}(a', m')} \iint da dm f(a, m) \mathcal{Q}(a', m'|a, m) \\
&\quad + \underbrace{\iint da' dm' \frac{g(a', m')}{p^{eq}(a', m')} f(a', m') \iint \mathcal{W}(a'', m''|a', m') da'' dm''}_{C(a', m')} \\
&\quad \underbrace{\hspace{10em}}_c
\end{aligned}$$

As  $I = II$ ,

$$\begin{aligned}
&\iint da dm \frac{f(a, m)}{p^{eq}(a, m)} \iint da' dm' \mathcal{Q}(a, m|a', m') g(a', m') \\
&= \iint da' dm' \frac{g(a', m')}{p^{eq}(a', m')} \iint da dm f(a, m) \mathcal{Q}(a', m'|a, m).
\end{aligned} \tag{C.5}$$

This equation can be written in terms of a scalar product [63],

$$(f, \mathcal{L}g) = (g, \mathcal{L}f). \tag{C.6}$$

where  $(f, g) = \iint_{\Omega} \frac{f(a, m)g(a, m)}{p^{eq}} da dm$ .

**Remark:** It is trivially true that  $(\cdot, \cdot)$  satisfies all the properties of a scalar product (linearity, symmetry and positive definiteness) [64].

The Fokker-Planck equation of the system is (5.3),

$$\partial_t P = -\frac{\partial}{\partial a}(F_a P) - \frac{\partial}{\partial m}(F_m P) + \Delta_m \frac{\partial^2 P}{\partial m^2} + \Delta_a \frac{\partial^2 P}{\partial a^2}.$$

with  $(a, m)^T \in \Omega = (0, 1) \times (0, m_0)$ . This FPE can also be written in terms of an operator  $\mathcal{L}$  on the right hand side, where

$$\mathcal{L} = \partial_a(F_a - \Delta_a \partial_a) - \partial_m(F_m - \Delta_m \partial_m) \tag{C.7}$$

Using symmetry property of a scalar product, the right side of (C.6) is,

$$\begin{aligned}
(g, \mathcal{L}f) &= (\mathcal{L}f, g) = \iint da dm \frac{(\mathcal{L}f)(a, m)g(a, m)}{p^{eq}} \\
&= \iint da dm \left\{ \left[ -\frac{\partial}{\partial a}(F_a f - \Delta_a \frac{\partial}{\partial a} f) \right] \frac{g}{p^{eq}} + \left[ -\frac{\partial}{\partial m}(F_m f - \Delta_m \frac{\partial}{\partial m} f) \right] \frac{g}{p^{eq}} \right\},
\end{aligned}$$

where  $p^{eq}$  is the steady state probability distribution in equilibrium.

**Remark:** For the sake of completeness one can specify the domain of operator  $\mathcal{L}$  in (C.7). In this case, it is safe to assume the domain of  $\mathcal{L}$  to be  $D(\mathcal{L}) = H^2(\Omega) \cap H_0^1(\Omega)$ . For a positive integer  $p$ ,  $H_0^p$  is the Sobolev space compactly supported on  $\Omega$ , where every  $u \in H_0^p$  has a weak derivative  $D^\alpha u \in L^2(\Omega)$  with  $|\alpha| \leq p$  [65].

Then using integration by parts, assuming that  $f$  and  $g$  vanish at the boundaries,

$$\begin{aligned} &= \iint da dm \left\{ (F_a f - \Delta_a \frac{\partial}{\partial a} f) \frac{\partial}{\partial a} \left( \frac{g}{p^{eq}} \right) + (F_m f - \Delta_m \frac{\partial}{\partial m} f) \frac{\partial}{\partial m} \left( \frac{g}{p^{eq}} \right) \right\} \\ &= \iint da dm \left\{ F_a f \frac{\partial_a g}{p^{eq}} - F_a f g \frac{\partial_a p^{eq}}{(p^{eq})^2} - \Delta_a \frac{\partial_a f \partial_a g}{p^{eq}} + \Delta_a g \frac{\partial_a f \partial_a p^{eq}}{(p^{eq})^2} \right. \\ &\quad \left. + F_m f \frac{\partial_m g}{p^{eq}} - F_m f g \frac{\partial_m p^{eq}}{(p^{eq})^2} - \Delta_m \frac{\partial_m f \partial_m g}{p^{eq}} + \Delta_m g \frac{\partial_m f \partial_m p^{eq}}{(p^{eq})^2} \right\}. \end{aligned}$$

The other side of (C.6) is similarly obtained,

$$\begin{aligned} (\mathcal{L}g, f) &= \iint da dm \left\{ \left[ -\frac{\partial}{\partial a} (F_a g - \Delta_a \frac{\partial}{\partial a} g) \right] \frac{f}{p^{eq}} + \left[ -\frac{\partial}{\partial m} (F_m g - \Delta_m \frac{\partial}{\partial m} g) \right] \frac{f}{p^{eq}} \right\} \\ &= \iint da dm \left\{ F_a g \frac{\partial_a g}{p^{eq}} - F_a f g \frac{\partial_a p^{eq}}{(p^{eq})^2} - \Delta_a \frac{\partial_a f \partial_a g}{p^{eq}} + \Delta_a f \frac{\partial_a g \partial_a p^{eq}}{(p^{eq})^2} \right. \\ &\quad \left. + F_m g \frac{\partial_m f}{p^{eq}} - F_m f g \frac{\partial_m p^{eq}}{(p^{eq})^2} - \Delta_m \frac{\partial_m f \partial_m g}{p^{eq}} + \Delta_m f \frac{\partial_m g \partial_m p^{eq}}{(p^{eq})^2} \right\}. \end{aligned}$$

Equating the terms, leads to cancellation of many terms. As  $f, g$  are arbitrary, and the integrals are over the same domain,

$$\left( F_a - \Delta_a \frac{\partial_a p^{eq}}{p^{eq}} \right) (g \partial_a f - f \partial_a g) + \left( F_m - \Delta_m \frac{\partial_m p^{eq}}{p^{eq}} \right) (g \partial_m f - f \partial_m g) = 0. \quad (C.8)$$

Since (C.8) is true for all  $f$  and  $g$ , it must be true that,

$$\begin{aligned} F_a - \Delta_a \frac{\partial_a p^{eq}}{p^{eq}} &= 0 \quad \text{and} \quad F_m - \Delta_m \frac{\partial_m p^{eq}}{p^{eq}} = 0, \\ \implies \frac{F_a}{\Delta_a} &= \partial_a \ln p^{eq} \quad \text{and} \quad \frac{F_m}{\Delta_m} = \partial_m \ln p^{eq} \end{aligned}$$

and by Schwarz's Theorem, the derivatives commute,

$$\frac{1}{\Delta_a} \partial_m F_a = \frac{1}{\Delta_m} \partial_a F_m = \partial_m \partial_a \ln p^{eq}. \quad (C.9)$$

## C.2 Dissipation Approximation

Since  $\omega_a \gg \omega_m$ , adiabatic approximation implies,

$$\dot{W} \approx \int_0^1 da \int_0^{m_0} dm \frac{J_m^2}{\Delta_m P} = \int_0^1 da \int_0^{m_0} dm \frac{1}{\Delta_m P} \left[ F_m P - \Delta_m \frac{\partial P}{\partial m} \right]^2 \quad (\text{C.10})$$

$$= \int_0^1 da \int_0^{m_0} dm \frac{P}{\Delta_m} \left[ -\omega_m(a - a_0)\beta + \omega_m(1 - \beta) \frac{\Delta_m/\omega_m}{\Delta_a/\omega_a} (a - a_0) G' \right. \\ \left. + \Delta_m \partial_m (F + H + \log Z) \right]^2 \quad (\text{C.11})$$

$$= \int_0^1 da \int_0^{m_0} dm \frac{P}{\Delta_m} \left[ -\omega_m(a - a_0)\beta + \omega_m(1 - \beta) \frac{\Delta_m/\omega_m}{\Delta_a/\omega_a} (a - a_0) G' \right. \\ \left. - \Delta_m \frac{\omega_a}{\Delta_a} G' (a - G) + \Delta_m (G - a_0) [\beta - (1 - \beta) \frac{\Delta_m/\omega_m}{\Delta_a/\omega_a} G'] \right]^2 \quad (\text{C.12})$$

$$= \frac{1}{Z} \int_0^1 da e^{-F} \int_0^{m_0} dm \frac{e^{-H}}{\Delta_m} \left[ -\omega_m(a - a_0)\beta + \omega_m(1 - \beta) \frac{\Delta_m/\omega_m}{\Delta_a/\omega_a} (a - a_0) G' \right. \\ \left. - \Delta_m \frac{\omega_a}{\Delta_a} G' (a - G) + \Delta_m (G - a_0) [\beta - (1 - \beta) \frac{\Delta_m/\omega_m}{\Delta_a/\omega_a} G'] \right]^2 \quad (\text{C.13})$$

$$\approx \frac{1}{Z} \sqrt{\frac{2\pi}{H''(s, m^*)}} \frac{e^{-H(s, m^*)}}{\Delta_m} \int_0^1 da e^{-F|_{m=m^*}} \left[ \omega_m(1 - \beta) \frac{\Delta_m/\omega_m}{\Delta_a/\omega_a} (a - a_0) G' \right. \\ \left. - \omega_m(a - a_0)\beta - \Delta_m \frac{\omega_a}{\Delta_a} G' (a - G) + \Delta_m (G - a_0) [\beta - (1 - \beta) \frac{\Delta_m/\omega_m}{\Delta_a/\omega_a} G'] \right]_{m=m^*}^2 \quad (\text{C.14})$$

$$= \frac{1}{Z} \sqrt{\frac{2\pi}{H''(s, m^*)}} \frac{e^{-H(s, m^*)}}{\Delta_m} \int_0^1 da e^{-F|_{m=m^*}} \left[ -\omega_m(a - a_0)\beta \right. \\ \left. + \omega_m(1 - \beta) \frac{\Delta_m/\omega_m}{\Delta_a/\omega_a} (a - a_0) G' - \Delta_m \frac{\omega_a}{\Delta_a} G' (a - a_0) \right]^2 \quad (\text{C.15})$$

$$\approx \frac{1}{\Delta_m} \int_0^1 da e^{-F|_{m^*}} \left[ -\omega_m(a - a_0)\beta + \omega_m(1 - \beta) \frac{\Delta_m/\omega_m}{\Delta_a/\omega_a} (a - a_0) G' \right. \\ \left. - \omega_m \frac{\Delta_m/\omega_m}{\Delta_a/\omega_a} G' (a - a_0) \right]^2 \quad (\text{C.16})$$

$$= \frac{1}{\Delta_m} \int_0^1 da \exp \left[ -\frac{\omega_a}{2\Delta_a} (a - a_0)^2 \right] \left[ -\omega_m(a - a_0)\beta - \omega_m \beta \frac{\Delta_m/\omega_m}{\Delta_a/\omega_a} (a - a_0) G' \right]^2 \quad (\text{C.17})$$

$$= \frac{\omega_m^2 \beta^2}{\Delta_m} \left[ 1 + \beta \frac{\Delta_m/\omega_m}{\Delta_a/\omega_a} G'(s, m^*) \right]^2 \int_0^1 da (a - a_0)^2 \exp \left[ -\frac{\omega_a}{2\Delta_a} (a - a_0)^2 \right] \quad (\text{C.18})$$

$$\begin{aligned}
&= \frac{\omega_m^2 \beta^2}{\Delta_m} \left[ 1 + \beta \frac{\Delta_m/\omega_m}{\Delta_a/\omega_a} G'(s, m^*) \right]^2 \int_{-\infty}^{\infty} da (a - a_0)^2 \exp \left[ -\frac{\omega_a}{2\Delta_a} (a - a_0)^2 \right] \quad (\text{C.19}) \\
&= \omega_m \beta^2 \frac{\Delta_a/\omega_a}{\Delta_m/\omega_m} \left( 1 + \frac{\Delta_m/\omega_m}{\Delta_a/\omega_a} G'(s, m^*) \right)^2.
\end{aligned}$$

For the above calculation, (C.11) is obtained using that  $P(a, m) = \frac{1}{Z} e^{-F-H}$  and it is used that  $Z$  (partition function) is a constant in  $a$  and  $m$  to obtain (C.12), where  $F \equiv \frac{\omega_a}{2\Delta_a} (a - G(s, m))^2$  and  $H$  is the effective potential as in Section 5.1.

Then Laplace's approximation for integrals (steepest descent) [66] is used to integrate along  $m$  to obtain (C.14). Also, it is used that  $G(s, m^*) = a_0$ .

Using steepest descent method, it is possible to evaluate the partition function  $Z$ , which cancels out most of the terms that were outside the integral.

Then, it is used that the probability distribution vanishes when  $a \notin [0, 1]$  to change the integration limits in (C.18).

And lastly, for the very last step the following Gaussian integral result was used [67],

$$\int_{-\infty}^{\infty} x^{2n} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}} \frac{(2n-1)!!}{(2a)^n} \quad (\text{C.20})$$

where  $!!$  is the double factorial and  $n \in \mathbb{Z}_{\geq 0}$ .

### C.3 Calculations for ESA

To derive (5.33), start by considering (5.29c) i.e.,

$$\epsilon_3 = \frac{1-\beta}{a_0\beta} \int_0^1 da \int_0^{m_0} dm \frac{\Delta_m/\omega_m}{\Delta_a/\omega_a} (a - a_0) G'(s, m) P(a, m).$$

Using (5.12) and the adiabatic approximation for integration in  $a$  gives,

$$\epsilon_3 \approx \frac{1-\beta}{a_0\beta} \frac{\Delta_m/\omega_m}{\Delta_a/\omega_a} \sqrt{\frac{2\pi\Delta_a}{\omega_a}} \int_0^{m_0} dm (G(s, m) - a_0) G'(s, m) P_m(m). \quad (\text{C.21})$$

Performing Taylor expansion for  $(G(s, m) - a_0)G'(s, m)$  around  $m = m^*$  yields,

$$\begin{aligned}
\epsilon_3 \approx \frac{1-\beta}{a_0\beta} \frac{\Delta_m/\omega_m}{\Delta_a/\omega_a} \sqrt{\frac{2\pi\Delta_a}{\omega_a}} \int_0^{m_0} dm P_m(m) & \left( [(G(s, m) - a_0)G'(s, m)]|_{m=m^*} \right. \\
& + (m - m^*) [(G(s, m) - a_0)G'(s, m)]'|_{m=m^*} \\
& \left. + \frac{1}{2} (m - m^*)^2 [(G(s, m) - a_0)G'(s, m)]''|_{m=m^*} \right).
\end{aligned}$$



Using (5.20) results in the following equation,

$$\begin{aligned}
&= \frac{1-\beta}{a_0\beta} \frac{\Delta_m/\omega_m}{\Delta_a/\omega_a} \sqrt{\frac{2\pi\Delta_a}{\omega_a}} \frac{1}{Z} \int_0^{m_0} dm e^{-H(s,m)} \left( (m-m^*)[(G(s,m)-a_0)G''(s,m) \right. \\
&\quad \left. + (G'(s,m))^2] \Big|_{m=m^*} + \frac{1}{2}(m-m^*)^2[2G'(s,m)G''(s,m) \right. \\
&\quad \left. + (G(s,m)-a_0)G'''(s,m) + G'(s,m)G''(s,m)] \Big|_{m=m^*} \right). \tag{C.22}
\end{aligned}$$

Then using the steepest descent to evaluate the partition function and to integrate (C.22) along  $m$ ,

$$\begin{aligned}
&= \frac{1-\beta}{a_0\beta} \frac{\Delta_m/\omega_m}{\Delta_a/\omega_a} \sqrt{\frac{H''(s,m^*)}{2\pi}} e^{H(s,m^*)} \left[ (G'(s,m))^2 \Big|_{m=m^*} \right. \\
&\quad \left. \int_0^{m_0} dm e^{-H(s,m)} (m-m^*) + \frac{3}{2} G'(s,m^*) G'''(s,m^*) \int_0^{m_0} dm e^{-H(s,m)} (m-m^*)^2 \right]. \tag{C.23}
\end{aligned}$$

Using the following Taylor expansion around  $m = m^*$ ,

$$e^{-H(s,m)} \approx \exp[-H(s,m^*) - \underbrace{(m-m^*)H'(s,m^*)}_{=0} - \frac{1}{2}(m-m^*)^2 H''(s,m^*)],$$

in (C.23) yields,

$$\begin{aligned}
\epsilon_3 \approx & \frac{1-\beta}{a_0\beta} \frac{\Delta_m/\omega_m}{\Delta_a/\omega_a} \sqrt{\frac{H''(s,m^*)}{2\pi}} \frac{3}{2} G'(s,m^*) G'''(s,m^*) \\
& \underbrace{\int_0^{m_0} dm e^{-\frac{1}{2}H''(s,m^*)(m-m^*)^2} (m-m^*)^2}_{\sqrt{\frac{2\pi}{H''(s,m^*)}} (H''(s,m^*))^{-1}}, \tag{C.24}
\end{aligned}$$

where the first term vanished because  $\int_{-\infty}^{\infty} x e^{-\alpha x^2} dx = 0$ . Lastly, substituting

$$H''(s,m^*) = \frac{1}{\Delta_m/\omega_m} G'(s,m^*) [\beta - (1-\beta) \frac{\Delta_a/\omega_a}{\Delta_m/\omega_m} G'(s,m^*)],$$

in (C.24) yields,

$$\epsilon_3 \approx \frac{1-\beta}{\beta} \frac{(\Delta_m/\omega_m)^2}{\Delta_a/\omega_a} \frac{3G''(s,m^*)}{2a_0} (\beta - (1-\beta) \frac{\Delta_m/\omega_m}{\Delta_a/\omega_a} G'(s,m^*))^{-1}. \tag{C.25}$$



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# Declaration

I hereby declare that this thesis is my own work, and that I have not used any sources and aids other than those stated in this thesis.

München, 23/08/2022

Author's signature: